Sparse multivariate modeling for stock returns predictability*

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Abstract

We develop a new variational Bayes estimation method for large-scale multivariate predictive regressions. Our approach allows to elicit permutation-invariant shrinkage priors directly on the regression coefficients matrix rather than a Cholesky-based linear transformation, as typically implemented in existing MCMC and variational Bayes approaches. Both a simulation and an empirical study on the cross-industry predictability of equity risk premiums in the US, show that by directly shrinking weak industry inter-dependencies one can substantially improve both the statistical and economic out-of-sample performance of multivariate regression models compared to a naive recursive mean forecast. This holds across alternative continuous shrinkage priors, such as the adaptive Bayesian lasso, adaptive normal-gamma and the horseshoe.

Keywords: mean-field approximation, variational Bayes, shrinkage prior, high-dimensional multivariate regression, returns predictability.

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1 Introduction

While many problems in economics rely on the identification of primitive underlying shocks and structural parameters, measuring the dynamics of expected returns is intimately related to out-of-sample forecasting. This practical view complements the theory-driven approach which often provides the building blocks for the empirical analysis of financial markets. However, expected returns are notoriously difficult to measure, that is the signal provided by commonly used predictors is buried in the noise of realised returns. One of the most common approaches to increase the signal-to-noise ratio is to search for additional predictors (see, e.g., Giannone et al., 2021). Although theoretical models often offer some guidance in understanding which predictors should be searched for and why, they are often too stylized to explicitly describe all sources of time-series and cross-sectional variation in financial returns. Therefore, investors and researchers, alike, typically face a trade-off: they could pre-select simple, perhaps univariate, models to predict excess returns by appealing to economic theories and existing empirical literature, with the risk of omitting important features, or, alternatively, they could use all information available and exploit the flexibility of large-scale multivariate predictive models.

Increasing the scale and flexibility of predictive systems comes at a cost, however: standard estimation techniques, such as ordinary least squares, maximum likelihood and Bayesian inference with non-informative priors tend to fail out-of-sample as the model dimensionality increases. This is the widely known curse of dimensionality whereby large-scale forecasting models tend to have a low bias but a high variance, and therefore overfit in-sample. As a result, the suitability of regularization and variable selection techniques to discipline the parameter space in large-dimensional models makes them particularly attractive in the context of financial returns predictability.

Within a Bayesian multivariate regression context, regularization is often imposed via continuous shrinkage priors like the Bayesian lasso (Park and Casella, 2008), the normal-gamma (Griffin and Brown, 2010) or the horseshoe (Carvalho et al., 2010). Common estimation approaches typically rely on a modified Cholesky factorization of the precision matrix. While this greatly simplify inference, it also prevents to explicitly elicit continuous, global-local, shrinkage priors directly on the regression coefficients. In addition, by breaking down a multivariate system into a sequence of smaller univariate regressions via Cholesky factorization, the posterior estimates are dependent on the ordering implied by the decomposition. Such order is typically arbitrary; yet, different orders would lead to different posterior estimates (see Section 2 for a discussion).

We address this issue by proposing a novel variational Bayes estimation algorithm which allows for fast and accurate estimates of the model parameters in large-scale multivariate predictive regressions. Unlike existing estimation algorithms, our approach allows to explicitly use shrinkage priors on the coefficients of regression matrix without leveraging on standard order-dependent Cholesky-based reparameterizations.

Empirically, we investigate both the statistical and economic significance of our modeling approach within the context of a typical investment decision problem whereby a representative investor faces the choice of investing in a large set of different portfolios. Although the model is general and can be applied to any type of asset returns, as far as data are stationary, our focus on industry-sorted portfolios is motivated by keen interests from researchers (see, e.g., Fama and French, 1997) and practitioners (for example, the increasing popularity of industry ETFs). While there is a vast literature examining the out-of-sample performance of the aggregate or individual stock excess returns, the question of whether industry portfolios can be predicted has received relatively little attention so far. Early

exceptions are Ferson and Harvey (1991), Ferson and Korajczyk (1995) and Ferson and Harvey (1999). Yet, the implications of industry returns predictability are far from trivial. If all industries are unpredictable, then the market return, which is a weighted average of the industry portfolios, should also be unpredictable; the evidence of aggregate market return predictability, thus, implies that at least some industry portfolio returns is predictable. In addition, understanding the dynamics of risk premiums across different industries could have important implications for dynamic asset pricing models, since they ultimately affect the efficient allocation of capital across sectors and over time.

Our primary contribution is two-fold. First, we provide a new benchmark estimation procedure for sparse multivariate regression models within the context of stock returns predictability. This is summarised by a better out-of-sample predictive performance, as well as economic significance, compared to standard linear dense and sparse multivariate models. More precisely, while alternative estimation methods provide negative out-of-sample R^2_{oos} and certainty equivalent returns for the vast majority of industry portfolios when compared against a naive rolling mean forecast, our estimation strategy is often able to generate a positive and significant out-of-sample statistical and economic performance. Second we provide robust evidence, both in simulation and on a real data set, that by eliciting shrinkage priors directly on the regression coefficients matrix, one can substantially outperform state-of-the art estimation methods which are based on a modified Cholesky factorization. In this respect, our view is that by a more accurate identification of the conditional information which affect the dynamics of risk premiums, which is arguably sparse in nature given the weak cross-sectional correlations, one can significantly improve both statistically and economically the out-of-sample performance of large-scale multivariate predictive models compared to benchmark rolling mean forecasts (see, e.g., Campbell and Thompson, 2007).

2 Motivation

Let $\mathbf{y}_t = (y_{1,t}, \ldots, y_{d,t})^\intercal \in \mathbb{R}^d$ be a realization of a multivariate Gaussian random variable and let $\mathbf{x}_t = (1, x_{1,t}, \ldots, x_{p,t})^\intercal \in \mathbb{R}^{(p+1)}$ be a vector of exogenous covariates observed at time t. A multivariate predictive regression model with constant volatility and both endogenous and exogenous variables is defined as follows:

$$\mathbf{y}_t = \boldsymbol{\Theta}\mathbf{z}_{t-1} + \mathbf{u}_t, \qquad \mathbf{u}_t \sim \mathsf{N}_d\left(\mathbf{0}_d, \boldsymbol{\Omega}^{-1}\right), \qquad t = 2, 3, \ldots, \tag{1}$$

with $\boldsymbol{\Theta} = (\boldsymbol{\Gamma}, \boldsymbol{\Phi})$ where $\boldsymbol{\Gamma} \in \mathbb{R}^{d\times(p+1)}$ is the matrix of regression parameters for the exogenous predictors and $\boldsymbol{\Phi} \in \mathbb{R}^{d\times d}$ is the transition matrix containing the autoregressive parameters, $\mathbf{z}_{t-1} = (\mathbf{x}_{t-1}^\intercal, \mathbf{y}_{t-1}^\intercal)^\intercal$, and $\mathbf{u}_t \in \mathbb{R}^d$ is a sequence of uncorrelated stochastic innovation terms such that $\mathbf{u}_{t-k} \perp \mathbf{u}_{t-j}$ for $k \neq j$ and $k, j = \pm 1, \pm 2, \ldots$ and covariance matrix equal to $\boldsymbol{\Omega}^{-1}$, with $\boldsymbol{\Omega} \in \mathbb{S}_{++}^d$ being a symmetric and positive definite precision matrix.

The modified Cholesky factorization of the precision matrix $\boldsymbol{\Omega}$ can be conveniently exploited to re-write the model in (1) with orthogonal innovations, (see, e.g. Rothman et al., 2010; Bitto and Frühwirth-Schnatter, 2019). Let $\boldsymbol{\Omega} = \mathbf{L}^\intercal\mathbf{V}\mathbf{L}$, where $\mathbf{L} \in \mathbb{R}^{d\times d}$ is uni-lower-triangual and $\mathbf{V} \in \mathbb{S}_{++}^d$ is diagonal, then

$$\mathbf{L}\mathbf{y}_t = \mathbf{L}\boldsymbol{\Theta}\mathbf{z}_{t-1} + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \mathsf{N}_d(\mathbf{0}_d, \mathbf{V}^{-1}), \qquad t = 2, 3, \ldots, \tag{2}$$

which corresponds to $\mathbf{y}_t = \mathbf{B}\mathbf{y}_t + \mathbf{A}\mathbf{z}_{t-1} + \boldsymbol{\varepsilon}_t$, for $t = 2, 3, \ldots$, where $\mathbf{A} = \mathbf{L}\boldsymbol{\Theta}$ and the matrix $\mathbf{B} = \mathbf{I}_d - \mathbf{L}$ has a strict-lower-triangular structure with elements $\beta_{j,k} = -l_{j,k}$ for $j = 2, \ldots, d$ and $k = 1, \ldots, j-1$. The distinguishing feature of the regression model in (2) is the equation-by-equation representation in which each variable $y_{j,t}$, $j = 1, \ldots, d$ depends

on the past history defined by the filtration $\{\mathbf{y}_s, s < t\}$, on the exogenous covariates and on the present values of its predecessors $\{y_{k,t}, k < j\}$. More precisely, the univariate regression for the j-th component of $\mathbf{y}_t$ becomes:

$$y_{j,t} = \boldsymbol{\beta}_j \mathbf{r}_{j,t} + \boldsymbol{\vartheta}_j \mathbf{z}_{t-1} + \varepsilon_{j,t}, \quad \varepsilon_{j,t} \sim \mathsf{N}_1(0, 1/\nu_j), \tag{3}$$

for all $j = 1, \ldots, d$ and $t = 2, 3, \ldots$, where $\boldsymbol{\beta}_j \in \mathbb{R}^{j-1}$ is a row vector containing the non-null elements in the j-th row of $\mathbf{B}$ and $\boldsymbol{\vartheta}_j$ denotes the j-th row of $\boldsymbol{\Theta}$. Moreover, for any $j = 1, \ldots, d$, let $\mathbf{r}_{j,t} = \mathbf{y}_t^j - \boldsymbol{\Theta}^j \mathbf{z}_{t-1}$ denotes the the vector of residuals up to the $(j-1)$-th regression, with $\mathbf{y}_t^j = (y_{1,t}, \ldots, y_{j-1,t})^\intercal \in \mathbb{R}^{j-1}$ being the sub-vector of $\mathbf{y}_t$ collecting the variables up to the $(j-1)$-th and $\boldsymbol{\Theta}^j \in \mathbb{R}^{(j-1)\times d}$ is the sub-matrix containing the first $j-1$ rows of $\boldsymbol{\Theta}$. Therefore, the parameter space for the vector $\boldsymbol{\xi} = (\boldsymbol{\nu}^\intercal, \boldsymbol{\beta}^\intercal, \boldsymbol{\vartheta}^\intercal)^\intercal$ is given by the product of the space of each component: $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_d)^\intercal \in \mathbb{R}_+^d$, $\boldsymbol{\beta} = (\boldsymbol{\beta}_2^\intercal, \ldots, \boldsymbol{\beta}_d^\intercal)^\intercal \in \mathbb{R}^{d(d-1)/2}$, $\boldsymbol{\vartheta} = \mathrm{vec}(\boldsymbol{\Theta}^\intercal)^\intercal \in \mathbb{R}^{d(d+p+1)}$ and it is equal to $\boldsymbol{\Xi} = \mathbb{R}_+^d \times \mathbb{R}^{d(d-1)/2} \times \mathbb{R}^{d(d+p+1)}$, where $\mathrm{vec}\,(\mathbf{x})$ denotes the vectorization operator that returns a $mn \times 1$ vector obtained by stacking the columns of the $(m \times n)$ matrix $\mathbf{x}$ on top of one another.

Two comments are in order. First, although not strictly needed for the development of the variational approximation, we force the data generating process to be weakly stationary and ergodic, by restricting the eigenvalues of the autoregressive matrix $\boldsymbol{\Phi}$ to be in modulus strictly less than one. Second, since we are primarily interested in the identification of the regression matrix $\boldsymbol{\Theta}$, we assume for simplicity that each of the elements in $\boldsymbol{\nu}$ are time invariant. This is consistent with some of the recent empirical finance literature (see, e.g., Stambaugh, 1999; Goyal and Welch, 2008; Dangl and Halling, 2012; Smith and Timmermann, 2021). This assumption can be relaxed by assuming each $\nu_j^{-1}, j = 1, \ldots, d$ as a latent process and leverage standard stochastic volatility modeling frameworks (see, e.g.,

Clark, 2011; Chan and Eisenstat, 2018; Carriero et al., 2019). We leave such development for future research.

Common approaches to inference for high-dimensional models that rely either on simulation-based and variational Bayes methods usually consider the elements in $\mathbf{A} = \mathbf{L}\boldsymbol{\Theta}$ as the parameters of interest (see, e.g. Chan and Eisenstat, 2018; Gefang et al., 2019; Chan and Yu, 2020). This has the merit of simplifying the estimation procedure making feasible the efficient implementation of standard `MCMC` (see, e.g., Chan and Eisenstat, 2018) and standard variational Bayes (`LVB`) algorithms (see, e.g., Chan and Yu, 2020). However, one should notice that the efficiency gain comes at the cost of preventing to elicit any shrinkage prior directly on the regression parameters matrix $\boldsymbol{\Theta}$. Perhaps even more importantly, the posterior estimates are not permutation invariant since the matrix $\mathbf{L}$ mechanically depends on the order of the vector $\mathbf{y}_t$. This might lead to a misleading characterisation of the sparse structure in the coefficients matrix. Proposition 2.1 provides a theoretical justification of these claims.

Proposition 2.1. *Under the parametrization* $\mathbf{A} = \mathbf{L}\boldsymbol{\Theta}$*, the estimate* $\boldsymbol{\Theta} = \mathbf{L}^{-1}\mathbf{A}$ *has the following characteristics: i) does not preserve the true structure of* $\boldsymbol{\Theta}$*, i.e.* $a_{i,j} = 0$ *does not imply* $\vartheta_{i,j} = 0$ *where* $a_{i,j}$ *and* $\vartheta_{i,j}$ *denotes the* (i,j)*-entry of* $\mathbf{A}$ *and* $\boldsymbol{\Theta}$ *respectively; ii)* $(a_{i,1}, \ldots, a_{i,j}) = \mathbf{0}_j$ *represents a sufficient condition to preserve a sparse pattern* $(\vartheta_{i,1}, \ldots, \vartheta_{i,j}) = \mathbf{0}_j$*; iii) is not permutation invariant.*

Proof. The element $\vartheta_{i,j}$ is computed as the linear combination $\vartheta_{i,j} = a_{i,j} + \sum_{k=1}^{i-1} c_{i,k} a_{k,j}$, where $c_{i,j}$ denotes the (i,j)-entry of $\mathbf{L}^{-1}$. From the latter expression, statements *i)* and *ii)* are immediately verified. As a consequence of *ii)*, the ordering plays a central role to estimate the sparse structure, which validate *iii)*. □

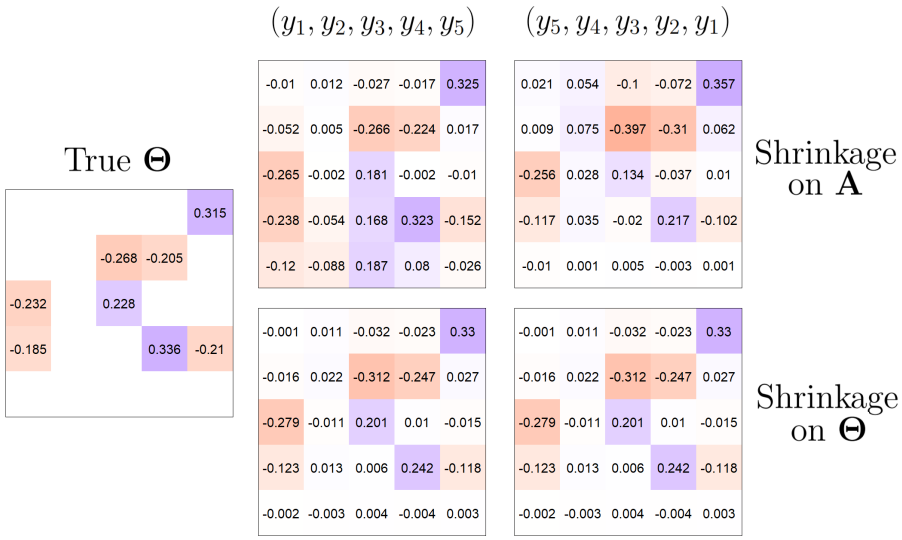


Figure 1: Comparison between the posterior inference for the linear representation $\mathbf{A} = \mathbf{L\Theta}$ (first row) and the original parametrization $\mathbf{\Theta}$ (second row), for two different permutations of $\mathbf{y}_t$.

Figure 1 provides a numerical representation of the results in Proposition 2.1. We compare the posterior estimates obtained by using the linear representation $\mathbf{A} = \mathbf{L\Theta}$ and the original parametrization $\mathbf{\Theta}$, for two different permutations of $\mathbf{y}_t$. The numerical results confirm that the posterior estimates are permutation dependent (see different columns) in the first case, and, therefore, the sparsity pattern is not preserved.

3 Variational Bayes inference

The variational Bayes approach to inference requires to minimize the Kullback-Leibler (KL) (Kullback and Leibler, 1951) divergence between a variational density $q(\boldsymbol{\psi})$ and the true posterior density $p(\boldsymbol{\psi}|\mathbf{y})$, (see, e.g. Bishop, 2006; Ormerod and Wand, 2010; Blei et al., 2017; Wand, 2017). The KL divergence cannot be directly minimized with respect to $\boldsymbol{\psi}$ because it involves the expectation with respect to the unknown true posterior distribution. Ormerod and Wand (2010) show that the problem of minimizing the KL divergence can be equivalently stated as the maximization of the variational lower bound (ELBO), here

denoted by $\underline{p}\,(\mathbf{y}; q)$:

$$q^*(\boldsymbol{\psi}) = \arg \min_{q(\boldsymbol{\psi}) \in \mathcal{Q}} \log \underline{p}\,(\mathbf{y}; q)\,, \quad \underline{p}\,(\mathbf{y}; q) = \int q(\boldsymbol{\psi}) \log \left\{ \frac{p(\mathbf{y}, \boldsymbol{\psi})}{q(\boldsymbol{\psi})} \right\} d\boldsymbol{\psi}, \tag{4}$$

where $q^*(\boldsymbol{\psi}) \in \mathcal{Q}$ denotes the optimal variational density and $\mathcal{Q}$ is a space of functions. According to the assumption we make on the space $\mathcal{Q}$, we fall into different variational paradigms (see Bishop, 2006, for a complete discussion). The mean-field variational Bayes (`MFVB`, hereafter) approach only assumes a non-parametric restriction for the variational density, i.e. $q(\boldsymbol{\psi}) = \prod_{i=1}^{p} q_i(\boldsymbol{\psi}_i)$ for a convenient partition $\{\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_p\}$ of the parameter vector $\boldsymbol{\psi}$. Under the `MFVB` restriction, an established result provides a closed form expression for the optimal variational density of each component $q(\boldsymbol{\psi}_j)$:

$$q^*(\boldsymbol{\psi}_j) \propto \exp \left\{ \mathbb{E}_{q^\star(\boldsymbol{\psi} \setminus \boldsymbol{\psi}_j)} \Big[\log p(\mathbf{y}, \boldsymbol{\psi}) \Big] \right\}, \qquad q^\star(\boldsymbol{\psi} \setminus \boldsymbol{\psi}_j) = \prod_{\substack{i=1 \\ i \neq j}}^{p} q_i(\boldsymbol{\psi}_i), \tag{5}$$

where the expectation is taken with respect to the joint approximating density with the j-th element of the partition removed $q^\star(\boldsymbol{\psi} \setminus \boldsymbol{\psi}_j)$. The latter result provides an easy way to implement an iterative algorithm to estimate the optimal density $q^*(\boldsymbol{\psi})$.

From the previous discussion it emerges that the most relevant issue in developing the `MFVB` algorithm consists to properly define the factorization of $q(\boldsymbol{\psi})$ which plays a central role within this approximation scheme. Next paragraphs present the factorization of the variational density for a non-informative prior, as well as the three alternative shrinkage priors for $\boldsymbol{\Theta}$. In the main text we will summarize the optimal approximating density $q^\star$, and leave the full derivation of the optimal variational densities together with the analytical form of the lower bound and the correspondent iterative algorithms in Appendix B.

3.1 Prior and mean-field factorization

As a default prior specification, we assume a non-shrinkage independent and non-informative diffuse priors: $\nu_j \sim \mathsf{Ga}(a_\nu, b_\nu)$, $j = 1, \ldots, d$, $\beta_{j,k} \sim \mathsf{N}(0, \tau)$, $j = 2, \ldots, d, \quad k = 1, \ldots, j-1$ and $\vartheta_{j,k} \sim \mathsf{N}(0, \upsilon)$ for $j = 1, \ldots, d$ and $k = 1, \ldots, d+p+1$, where $\mathsf{Ga}(\cdot, \cdot)$ denotes the gamma distribution, and $a_\nu > 0$, $b_\nu > 0$, $\tau \gg 0$ and $\upsilon \gg 0$ are fixed hyper-parameters. The variational density $q(\boldsymbol{\xi})$ can be factorised as follows:

$$q(\boldsymbol{\xi}) = q(\boldsymbol{\nu})q(\boldsymbol{\beta})q(\boldsymbol{\vartheta}), \quad q(\boldsymbol{\nu}) = \prod_{j=1}^{d} q(\nu_j), \quad q(\boldsymbol{\beta}) = \prod_{j=2}^{d} q(\boldsymbol{\beta}_j). \tag{6}$$

Proposition 3.1 provides the optimal variational density for this default prior specification. The proof and analytical derivations are available in Appendix B.1.

Proposition 3.1. *Under the default normal prior, the optimal variational densities for* $(\boldsymbol{\nu}^\intercal, \boldsymbol{\beta}^\intercal, \boldsymbol{\vartheta}^\intercal)^\intercal$ *in* (6) *are:* $q^*(\nu_j) \equiv \mathsf{Ga}(a_{q(\nu_j)}, b_{q(\nu_j)})$, $j = 1, \ldots, d$ *for the inverse of variance parameters,* $q^*(\boldsymbol{\beta}_j) \equiv \mathsf{N}_{j-1}(\boldsymbol{\mu}_{q(\beta_j)}, \boldsymbol{\Sigma}_{q(\beta_j)})$, $j = 2, \ldots, d$ *for the rows of the Cholesky factor and* $q^*(\boldsymbol{\vartheta}) \equiv \mathsf{N}_{d(d+p+1)}(\boldsymbol{\mu}_{q(\vartheta)}, \boldsymbol{\Sigma}_{q(\vartheta)})$ *for the regression parameters.*

The assumption of a joint variational density on the elements in $\boldsymbol{\Theta}$ is a distinguishing feature of our approach. The reason is twofold: first, it ensures that a complete inference a posteriori is available directly on the elements of $\boldsymbol{\Theta}$. Second, it enables to perform joint hypothesis testing procedures involving the entire vector $\boldsymbol{\vartheta} = \mathrm{vec}\,(\boldsymbol{\Theta}^\intercal)^\intercal$ or to obtain joint credibility intervals in a very straightforward way. In the main simulation and empirical application, we consider a further factorization that assumes independence a posteriori among the rows of $\boldsymbol{\Theta}$, i.e. $q(\boldsymbol{\vartheta}) = \prod_{j=1}^{d} q(\boldsymbol{\vartheta}_j)$. In this case the optimal variational density for each row $\boldsymbol{\vartheta}_j$ is again a Gaussian distribution $q^*(\boldsymbol{\vartheta}_j) \equiv \mathsf{N}_{d+p+1}(\boldsymbol{\mu}_{q(\vartheta_j)}, \boldsymbol{\Sigma}_{q(\vartheta_j)})$, for each $j = 1, \ldots, d$, which further reduces the computational burden of the estimation procedure.

Adaptive lasso

The Bayesian adaptive lasso (Leng et al., 2014) extends the lasso (Park and Casella, 2008) by imposing a different amount of shrinkage for each parameter. This is particularly suitable for financial returns forecasting given the endogenous and correlated nature of the covariates, which makes the usual practice of standardization a sub-optimal solution. For each entry of the matrix $\boldsymbol{\Theta}$, namely $\vartheta_{j,k}$, the Bayesian adaptive lasso prior assumes a laplace distribution with a different scaling parameter $\vartheta_{j,k}|\lambda_{j,k} \sim \mathsf{Lap}(\lambda_{j,k})$, for $j = 1, \ldots, d$ and $k = 1, \ldots, d + p + 1$. For the purpose of providing computationally feasible algorithms, we represent the Laplace distribution as a scale mixture of Gaussians with an exponential mixing density, $\vartheta_{j,k}|\upsilon_{j,k} \sim \mathsf{N}(0, \upsilon_{j,k})$, $\upsilon_{j,k}|\lambda^2_{j,k} \sim \mathsf{Exp}(\lambda^2_{j,k}/2)$. The choice of the scaling parameters $\lambda^2_{j,k}$ is crucial to recover the underlying signal and it is certainly non trivial in high-dimensional spaces. A common strategy consists to infer their values from the data by assuming a common hyper-prior distribution: $\lambda^2_{j,k} \sim \mathsf{Ga}(h_1, h_2)$, where $h_1, h_2 > 0$ are fixed hyper-parameters. Let $\boldsymbol{\xi}_{\mathrm{L}} = (\boldsymbol{\xi}^\intercal, \boldsymbol{\upsilon}^\intercal, (\boldsymbol{\lambda}^2)^\intercal))^\intercal$ be the vector of parameters $\boldsymbol{\xi}$ augmented with the latent parameters used in the hierarchical specification, we assume $q(\boldsymbol{\xi}_{\mathrm{L}})$ to be factorised as follows:

$$q(\boldsymbol{\xi}_{\mathrm{L}}) = q(\boldsymbol{\xi})q(\boldsymbol{\upsilon}, \boldsymbol{\lambda}^2), \qquad q(\boldsymbol{\upsilon}, \boldsymbol{\lambda}^2) = \prod_{j=1}^{d} \prod_{k=1}^{d+p+1} q(\upsilon_{j,k})q(\lambda^2_{j,k}), \tag{7}$$

where $q(\boldsymbol{\upsilon}, \boldsymbol{\lambda}^2)$ is factorised as the product of each component. Proposition 3.2 provides the optimal variational densities for the adaptive lasso prior specification. The proof and the analytical derivations of the parameters are available in Appendix B.2.

Proposition 3.2. *Under the adaptive Bayesian lasso prior, the optimal variational densities for* $(\boldsymbol{\nu}^\intercal, \boldsymbol{\beta}^\intercal, \boldsymbol{\vartheta}^\intercal, \boldsymbol{\upsilon}, \boldsymbol{\lambda}^2)^\intercal$ *in* (7) *are:* $q^*(\nu_j) \equiv \mathsf{Ga}(a_{q(\nu_j)}, b_{q(\nu_j)})$, $j = 1, \ldots, d$, *for the*

precision parameters, $q^*(\boldsymbol{\beta}_j) \equiv \mathsf{N}_{j-1}(\boldsymbol{\mu}_{q(\beta_j)}, \boldsymbol{\Sigma}_{q(\beta_j)})$*,* $j = 2, \ldots, d$*, for the rows of the Cholseky factor, and* $q^*(\boldsymbol{\vartheta}) \equiv \mathsf{N}_{d(d+p+1)}(\boldsymbol{\mu}_{q(\vartheta)}, \boldsymbol{\Sigma}_{q(\vartheta)})$ *for the regression parameters. Moreover,* $q^*(1/\upsilon_{j,k}) \equiv \mathsf{IG}(a_{q(\upsilon_{j,k})}, b_{q(\upsilon_{j,k})})$ *and* $q^*(\lambda^2_{j,k}) \equiv \mathsf{Ga}(a_{q(\lambda^2_{j,k})}, b_{q(\lambda^2_{j,k})})$*, for* $j = 1, \ldots, d$ *and* $k = 1, \ldots, d+p+1$ *and where* $\mathsf{IG}(\cdot, \cdot)$ *denotes the inverse-Gaussian distribution.*

Adaptive normal-gamma

A further extension of the adaptive lasso prior is the normal-gamma prior of Griffin and Brown (2010). We assume different shrinkage parameters in order to make this prior adaptive. The hierarchical specification of the adaptive version of the normal-gamma for the elements of $\boldsymbol{\Theta}$ requires that $\vartheta_{j,k}|\upsilon_{j,k} \sim \mathsf{N}(0, \upsilon_{j,k})$, and $\upsilon_{j,k}|\eta_j, \lambda_{j,k} \sim \mathsf{Ga}\left(\eta_j, \eta_j\lambda_{j,k}/2\right)$ for $j = 1, \ldots, d$ and $k = 1, \ldots, d+p+1$. Notice that by restricting $\eta_j = 1$ we obtain the adaptive lasso prior. Marginalization over the variance $\upsilon_{j,k}$ leads to $p(\vartheta_{j,k}|\eta_j, \lambda_{j,k})$ which corresponds to the density of a variance-gamma distribution. The hyper-parameters η_j and $\lambda_{j,k}$ determine the amount of shrinkage and should be carefully calibrated. To avoid the calibration of the hyper-parameters $(\lambda_{j,k}, \eta_j)$, which is non trivial in high-dimensional parameters spaces, we impose a further level of hierarchy by assuming $\lambda_{j,k} \sim \mathsf{Ga}(h_1, h_2)$ and $\eta_j \sim \mathsf{Exp}(h_3)$, where now (h_1, h_2, h_3) becomes a fixed vector of prior hyper-parameters with $h_l > 0$ for $l = 1, 2, 3$. Let $\boldsymbol{\xi}_{\mathrm{NG}} = (\boldsymbol{\xi}^\intercal, \boldsymbol{\upsilon}^\intercal, \boldsymbol{\lambda}^\intercal, \boldsymbol{\eta}^\intercal)^\intercal$ be the vector of parameters $\boldsymbol{\xi}$ augmented with the latent parameters used in the hierarchical specification of the adaptive normal-gamma prior, we assume that `MFVB` density $q(\boldsymbol{\xi}_{\mathrm{NG}})$ is factorised as follows:

$$q(\boldsymbol{\xi}_{\mathrm{NG}}) = q(\boldsymbol{\xi})q(\boldsymbol{\upsilon}, \boldsymbol{\lambda}, \boldsymbol{\eta}), \qquad q(\boldsymbol{\upsilon}, \boldsymbol{\lambda}, \boldsymbol{\eta}) = \prod_{j=1}^{d} q(\eta_j) \prod_{k=1}^{d+p+1} q(\upsilon_{j,k})q(\lambda_{j,k}). \tag{8}$$

where $q(\boldsymbol{\upsilon}, \boldsymbol{\lambda}, \boldsymbol{\eta})$ are factorised as the product of each component. Proposition 3.3 provides the optimal variational densities for the adaptive normal-gamma specification. The proof

and the analytical derivations of the parameters are available in Appendix B.3.

Proposition 3.3. *Under the adaptive normal-gamma prior, the optimal variational densities for* $(\boldsymbol{\nu}^\intercal, \boldsymbol{\beta}^\intercal, \boldsymbol{\vartheta}^\intercal, \boldsymbol{\upsilon}^\intercal, \boldsymbol{\lambda}^\intercal, \boldsymbol{\eta}^\intercal)^\intercal$ *in* (8) *are:* $q^*(\nu_j) \equiv \mathsf{Ga}(a_{q(\nu_j)}, b_{q(\nu_j)})$, $j = 1, \ldots, d$, *for the precision parameters,* $q^*(\boldsymbol{\beta}_j) \equiv \mathsf{N}_{j-1}(\boldsymbol{\mu}_{q(\beta_j)}, \boldsymbol{\Sigma}_{q(\beta_j)})$, $j = 2, \ldots, d$, *for the rows of the Cholesky factor and* $q^*(\boldsymbol{\vartheta}) \equiv \mathsf{N}_{d(d+p+1)}(\boldsymbol{\mu}_{q(\vartheta)}, \boldsymbol{\Sigma}_{q(\vartheta)})$ *for the regression parameters. Moreover,* $q^*(\upsilon_{j,k}) \equiv \mathsf{GIG}(\zeta_{q(\upsilon_{j,k})}, a_{q(\upsilon_{j,k})}, b_{q(\upsilon_{j,k})})$ *and* $q^*(\lambda_{j,k}) \equiv \mathsf{Ga}(a_{q(\lambda_{j,k})}, b_{q(\lambda_{j,k})})$, *for* $j = 1, \ldots, d$ *and* $k = 1, \ldots, d+p+1$, *where* $\mathsf{GIG}(\cdot,\cdot,\cdot)$ *denotes the generalized inverse-gaussian distribution. The optimal density for the parameter* η_j, *for* $j = 1, \ldots, d$, *is not a known distribution function. The analytical form of such distribution and its moments are calculated via numerical integration.*

Horseshoe prior

The last specification is the horseshoe prior originally proposed by Carvalho et al. (2009, 2010). Such continuous shrinkage prior assumes $\vartheta_{j,k}|\upsilon_{j,k}^2, \gamma^2 \sim \mathsf{N}(0, \gamma^2\upsilon_{j,k}^2)$, $\gamma \sim \mathsf{C}^+(0,1)$, $\upsilon_{j,k} \sim \mathsf{C}^+(0,1)$, where $\mathsf{C}^+(0,1)$ denotes the standard half-cauchy distribution having probability density function equal to $f(x) = 2/\{\pi(1+x^2)\}\mathbb{1}_{(0,\infty)}(x)$. The horseshoe is a global-local shrinkage procedure (Polson and Scott, 2011; Bhattacharya et al., 2016; Tang et al., 2018) that retrieves aggressive shrinkage of unimportant coefficients without affecting the largest ones. To simplify the computations under the horseshoe prior, Wand et al. (2011) suggests the following scale mixture representation of the half-Cauchy distribution:

$$\begin{aligned} \vartheta_{j,k}|\upsilon_{j,k}^2, \gamma^2 \sim \mathsf{N}(0, \gamma^2\upsilon_{j,k}^2), \quad \gamma^2|\eta \sim \mathsf{InvGa}(1/2, 1/\eta), \quad \upsilon_{j,k}^2|\lambda_{j,k} \sim \mathsf{InvGa}(1/2, 1/\lambda_{j,k}), \\ \eta \sim \mathsf{InvGa}(1/2, 1), \qquad \lambda_{j,k} \sim \mathsf{InvGa}(1/2, 1), \end{aligned} \tag{9}$$

where the local and overall shrinkage are determined by $\upsilon_{j,k}^2$ and γ^2 respectively, while $\mathsf{InvGa}(\cdot,\cdot)$ denotes the inverse gamma distribution. Let $\boldsymbol{\xi}_{\text{HS}} = (\boldsymbol{\xi}^\intercal, (\boldsymbol{\upsilon}^2)^\intercal, \gamma^2, \boldsymbol{\lambda}^\intercal, \eta)^\intercal$ be the vector of parameters $\boldsymbol{\xi}$ augmented with the latent parameters used in the hierarchical specification of the horseshoe prior, we assume the following factorization of the variational density for $\boldsymbol{\xi}_{\text{HS}}$:

$$q(\boldsymbol{\xi}_{\text{HS}}) = q(\boldsymbol{\xi})q(\boldsymbol{\upsilon}^2, \gamma^2, \boldsymbol{\lambda}, \eta), \qquad q(\boldsymbol{\upsilon}^2, \gamma^2, \boldsymbol{\lambda}, \eta) = q(\gamma^2)q(\eta)\prod_{j=1}^{d}\prod_{k=1}^{d+p+1} q(\upsilon_{j,k}^2)q(\lambda_{j,k}). \quad (10)$$

Proposition 3.4 provides the optimal variational densities for the horseshoe. The proof and the analytical derivations of the parameters are available in Appendix B.4.

Proposition 3.4. *Under the horseshoe prior, the optimal variational densities for the parameters* $(\boldsymbol{\nu}^\intercal, \boldsymbol{\beta}^\intercal, \boldsymbol{\vartheta}^\intercal, (\boldsymbol{\upsilon}^2)^\intercal, \gamma^2, \boldsymbol{\lambda}^\intercal, \eta)^\intercal$ *in* (10) *are:* $q^*(\nu_j) \equiv \mathsf{Ga}(a_{q(\nu_j)}, b_{q(\nu_j)})$, $j = 1, \ldots, d$, *for the precision parameters,* $q^*(\boldsymbol{\beta}_j) \equiv \mathsf{N}_{j-1}(\boldsymbol{\mu}_{q(\beta_j)}, \boldsymbol{\Sigma}_{q(\beta_j)})$, $j = 2, \ldots, d$, *for the rows of the Cholesky factor and* $q^*(\boldsymbol{\vartheta}) \equiv \mathsf{N}_{d(d+p+1)}(\boldsymbol{\mu}_{q(\vartheta)}, \boldsymbol{\Sigma}_{q(\vartheta)})$ *for the regression parameters. Moreover, we have* $q^*(\gamma^2) \equiv \mathsf{InvGa}(a_{q(\gamma^2)}, b_{q(\gamma^2)})$ *and* $q^*(\eta) \equiv \mathsf{InvGa}(1, b_{q(\eta)})$ *for the global shrinkage parameters, and* $q^*(\upsilon_{j,k}^2) \equiv \mathsf{InvGa}(1, b_{q(\upsilon_{j,k}^2)})$ *and* $q^*(\lambda_{j,k}) \equiv \mathsf{InvGa}(1, b_{q(\lambda_{j,k})})$ *for* $j = 1, \ldots, d$ *and* $k = 1, \ldots, d+p+1$ *for the local shrinkage parameters.*

3.2 Combining shrinkage and sparsity

The continuous shrinkage priors introduced in the previous section provide point estimates for the null parameters that are very close to zero, but with a small variance. Hence, we need a post-inference procedure that sets the unimportant parameters to be nulls according to a given threshold. The sparsification in the posterior estimates of the regression matrix $\boldsymbol{\Theta}$ based on shrinkage priors is implemented via the Signal Adaptive Variable Selector (SAVS)

algorithm of Ray and Bhattacharya (2018). The SAVS is a post-process approach to divide the estimates into signals and nulls on the basis of the magnitude of the posterior point estimates (see, e.g., Hauzenberger et al., 2021; Huber et al., 2021). In particular, consider a regression parameter ϑ_j and the associated vector of covariates $\mathbf{z}_j$, then if $|\widehat{\vartheta}_j|\,||\mathbf{z}_j||^2 \leq |\widehat{\vartheta}_j|^{-2}$ we set $\widehat{\vartheta}_j = 0$, where $||\cdot||$ denotes the euclidean norm. The amount of sparsity imposed by the SAVS directly depends on the posterior estimates from the algorithmic procedure. That is, a more precise identification of the weak signals in the vector of covariates $\mathbf{z}_j$ implies a more accurate identification of the sparse structure of the regression parameters. In addition, the SAVS can be implemented regardless the type of shrinkage prior specification. This makes it particularly suitable for our purpose.

3.3 Approximation of the distribution of the precision matrix

The variational inferential procedure explained above enables us to get an approximating density for the regression parameters in $\mathbf{\Theta}$, for the Cholesky factor $\mathbf{B}$ (and therefore for $\mathbf{L}$), and for the elements on the diagonal of $\mathbf{V}$. However, to obtain a complete inference on the parameters of the original model in (1) we stil miss an approximation for the density of the precision matrix $\mathbf{\Omega} = \mathbf{L}^\intercal\mathbf{V}\mathbf{L}$. Proposition 3.5 shows that, conditional on $\mathbf{L}$ and $\mathbf{V}$, the distribution of $\mathbf{\Omega}$ can be approximated by a d-dimensional whishart distribution $\mathsf{Wishart}_d(\delta, \mathbf{H})$, where δ and $\mathbf{H}$ are the degrees of freedom and the scaling matrix, respectively. The proof is based on the Expectation Propagation (EP) variational procedure of Minka (2001), which has the goal of minimizing the *KL* divergence between $p(\mathbf{\Omega})$, the true and unknown distribution of $\mathbf{\Omega}$, and the approximating distribution $q(\mathbf{\Omega})$.

Proposition 3.5. *The approximate distribution q of $\mathbf{\Omega}$ is $\mathsf{Wishart}_d(\widehat{\delta}, \widehat{\mathbf{H}})$, where the scaling matrix is given by $\widehat{\mathbf{H}} = \widehat{\delta}^{-1}\mathbb{E}_p\left[\mathbf{\Omega}\right]$ and $\widehat{\delta}$ can be obtained numerically as the solution of a*

convex optimization problem.

Proof. The Kullback-Leibler divergence between $p(\mathbf{\Omega})$ and the approximating distribution $q(\mathbf{\Omega})$ is $\mathcal{D}_{KL}(p(\mathbf{\Omega})\|q(\mathbf{\Omega})) \propto -\mathbb{E}_p(\log q(\mathbf{\Omega}))$, where the expectation is taken with respect to the distribution $p(\mathbf{\Omega})$. Therefore the optimal parameters are $(\widehat{\delta}, \widehat{\mathbf{H}}) = \arg\min_{\delta,\mathbf{H}} \psi(\delta, \mathbf{H})$, where $\psi(\delta, \mathbf{H}) = -\mathbb{E}_p(\log q(\mathbf{\Omega}))$:

$$\psi(\delta, \mathbf{H}) \propto \frac{d\delta}{2}\log 2 + \frac{\delta}{2}\log|\mathbf{H}| + \log\Gamma_d(\delta/2) - \frac{\delta}{2}\mathbb{E}_p\left[\log|\mathbf{\Omega}|\right] + \frac{1}{2}\mathrm{tr}\left\{\mathbf{H}^{-1}\mathbb{E}_p\left[\mathbf{\Omega}\right]\right\}. \quad (11)$$

Notice that $\mathbb{E}_p\left[\log|\mathbf{\Omega}|\right] = \mathbb{E}_{q(V)}\left[\log|\mathbf{V}|\right] = \sum_{j=1}^{d}\mu_{q(\log\nu_j)}$ and $\mathbb{E}_p\left[\mathbf{\Omega}\right] = \mathbb{E}_{q(L),q(V)}\left[\mathbf{L}^\intercal\mathbf{V}\mathbf{L}\right]$ are available as byproduct of the mean-field Variational Bayes algorithm. Differentiating (11) with respect to the scaling matrix $\mathbf{H}$, solving $\partial\psi(\delta, \mathbf{H})/\partial\mathbf{H} = 0$ provides $\widehat{\mathbf{H}}_\delta = \delta^{-1}\mathbb{E}_p\left[\mathbf{\Omega}\right]$ that depends on the degrees of freedom δ. Plugging-in the latter in the objective function $\psi(\delta, \widehat{\mathbf{H}}_\delta)$ and proceeding with the minimization of the resulting functional with respect to δ provides $\widehat{\delta}$, which completes the proof. □

To assess the goodness of the proposed approximation, we sample from $q(\mathbf{L})$ and $q(\mathbf{V})$ and then obtain values from $q(\mathbf{\Omega})$ exploiting the modified Cholesky decomposition. The left panel of Figure 2 compares the sampled distributions with the marginals of the Wishart with parameters $(\widehat{\delta}, \widehat{\mathbf{H}})$ in terms of approximation accuracy $\mathcal{ACC} = 100\left\{1 - 0.5\int|q(\omega) - p(\omega)|\,d\omega\right\}\%$, where ω is a generic element of $\mathbf{\Omega}$. The simulation results confirm the intuition in Proposition 3.5 that our method provides a good approximation of the posterior distribution of $\mathbf{\Omega}$ even when the dimension d increases.

3.4 Prediction

Let assume that the optimal variational density $q^*(\boldsymbol{\xi})$ that approximates the posterior distribution of $p(\boldsymbol{\xi}|\mathbf{z}_{1:t})$ given the information set up to time t, $\mathbf{z}_{1:t} = \{\mathbf{y}_{1:t}, \mathbf{x}_{1:t}\}$ is available, and denote by $p(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\xi})$ the likelihood for the new observation $\mathbf{y}_{t+1}$. The posterior predictive distribution takes the form

$$p(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = \int p(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\xi}) p(\boldsymbol{\xi}|\mathbf{z}_{1:t}) d\boldsymbol{\xi}, \tag{12}$$

A strategy to obtain the variational predictive distribution that approximates (12) is suggested by Gunawan et al. (2020) and consists to replace the posterior density $p(\boldsymbol{\xi}|\mathbf{z}_{1:t})$ in (12) by the optimal variational distribution $q^*(\boldsymbol{\xi})$, as follows

$$q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = \int p(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\xi}) q^*(\boldsymbol{\xi}) d\boldsymbol{\xi} = \int \int p(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta}, \boldsymbol{\Omega}) q(\boldsymbol{\vartheta}) q(\boldsymbol{\Omega}) d\boldsymbol{\vartheta}\, d\boldsymbol{\Omega}. \tag{13}$$

An analytical expression for the above integral is not available, and, as a first solution, it can be approximated through Monte Carlo integration exploiting the fact that the optimal variational density $q^*(\boldsymbol{\xi})$ is known and can be efficiently sampled. A simulation-based approximated estimator for the variational predictive distribution $q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t})$ is therefore obtained by averaging the likelihood of the new observations $p(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\xi}^{(i)})$ over the draws $\boldsymbol{\xi}^{(i)} \sim q^*(\boldsymbol{\xi})$, for $i = 1, \ldots, N$ from the optimal variational density, leading to $\widehat{q}(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = N^{-1} \sum_{i=1}^{N} p(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\xi}^{(i)})$. Notice that we can further simplify Eq.(13) by integrating $\boldsymbol{\Omega}$ such that:

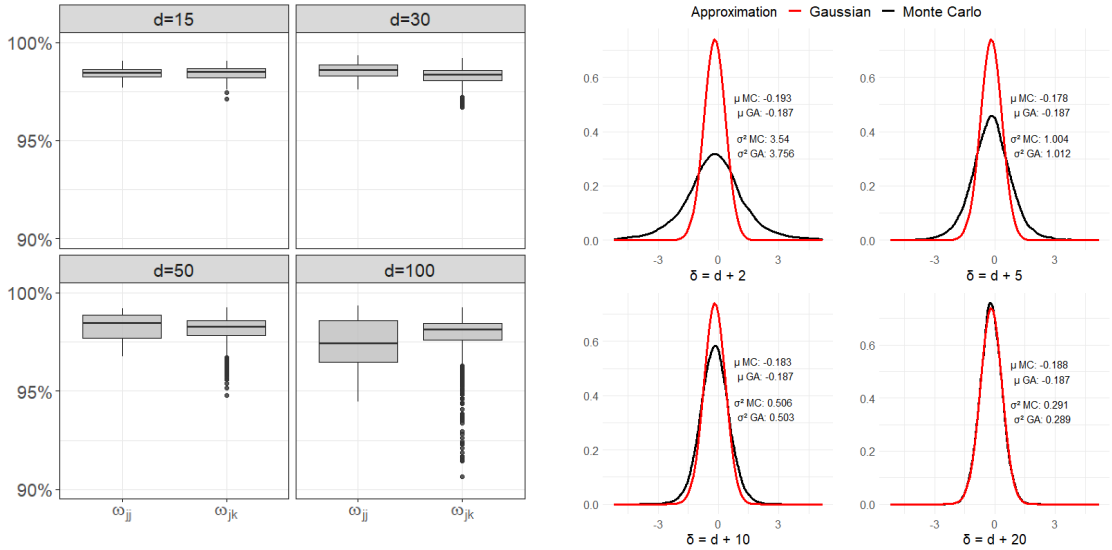
$$q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = \int h(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta}) q(\boldsymbol{\vartheta}) d\boldsymbol{\vartheta}, \tag{14}$$

where $h(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta})$ denotes the density function of a multivariate Student-t distribution $\mathrm{t}_v(\mathbf{m}, \mathbf{S})$ with $v = \widehat{\delta} - d + 1$ degrees of freedom, mean $\mathbf{m} = \boldsymbol{\Theta}\mathbf{z}_t$ and d-dimensional scaling matrix $\mathbf{S} = (v\widehat{\mathbf{H}})^{-1}$. As a result, an approximated estimator for the predictive distribution can be obtained through simulations by averaging the density of the multivariate Student-t $h(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta}^{(i)})$ over the draws $\boldsymbol{\vartheta}^{(i)} \sim q^*(\boldsymbol{\vartheta})$, for $i = 1, \ldots, N$, providing the approximation $\widehat{q}(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = N^{-1}\sum_{i=1}^{N} h(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta}^{(i)})$. This allows for a more efficient sampling from the predictive since we only need to sample values of $\boldsymbol{\vartheta}$ from a Gaussian distribution. However, to obtain a sufficiently good Monte Carlo approximation, especially in high dimensions, we require $N \gg 0$ which increase the computational burden. To increase the sampling efficiency, we approximate the multivariate Student-t in (14) with the closest multivariate normal distribution in terms of KL divergence, and obtain the following integral:

$$q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = \int \phi(\mathbf{y}_{t+1}|\boldsymbol{\Theta}\mathbf{z}_t, \mathbf{R}^{-1}) q(\boldsymbol{\vartheta}) d\boldsymbol{\vartheta}, \tag{15}$$

where now $\phi(\mathbf{y}_{t+1}|\boldsymbol{\Theta}\mathbf{z}_t, \mathbf{R}^{-1})$ denotes the density of the multivariate normal distribution that is closest in a KL sense to the multivariate Student-t $h(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta})$.

The advantage of this procedure is that the integral in (15) can be solved analytically leading to a variational predictive density $q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t})$ which is a multivariate gaussian distribution with variance matrix $\boldsymbol{\Sigma}_{pred}$ and mean vector $\boldsymbol{\mu}_{pred}$. The corresponding analytical expressions are provided in Appendix C. The right panel of Figure 2 shows the approximation of variational predictive posterior with Monte Carlo methods (MC) and via Gaussian approximation (GA) varying the degrees of freedom $\widehat{\delta}$ for the distribution of $\boldsymbol{\Omega}$. We can see that if $\widehat{\delta} \gg d$ the approximation is rather accurate, while the accuracy decreases as $\widehat{\delta}$ approaches d. However, even for the case $\widehat{\delta} \approx d$, we can still obtain rather precise estimates of the first and second moments of the variational predictive posterior.



(a) Accuracy for the approximation of Ω

(b) Approximation of the predictive density

Figure 2: Approximation of $\mathbf{\Omega}$ and variational predictive posterior. (a) plots the accuracy of the whishart approximation $q(\mathbf{\Omega})$ for dimensions $d = 15, 30, 50, 100$ separately for the diagonal (ω_{jj}) and out-of-diagonal (ω_{jk}) elements of $\mathbf{\Omega}$. (b) plots an example of the approximation of variational predictive posterior with Monte Carlo methods (MC) and via Gaussian approximation (GA) varying the degrees of freedom $\widehat{\delta}$ for the distribution of $\mathbf{\Omega}$.

4 Simulation study

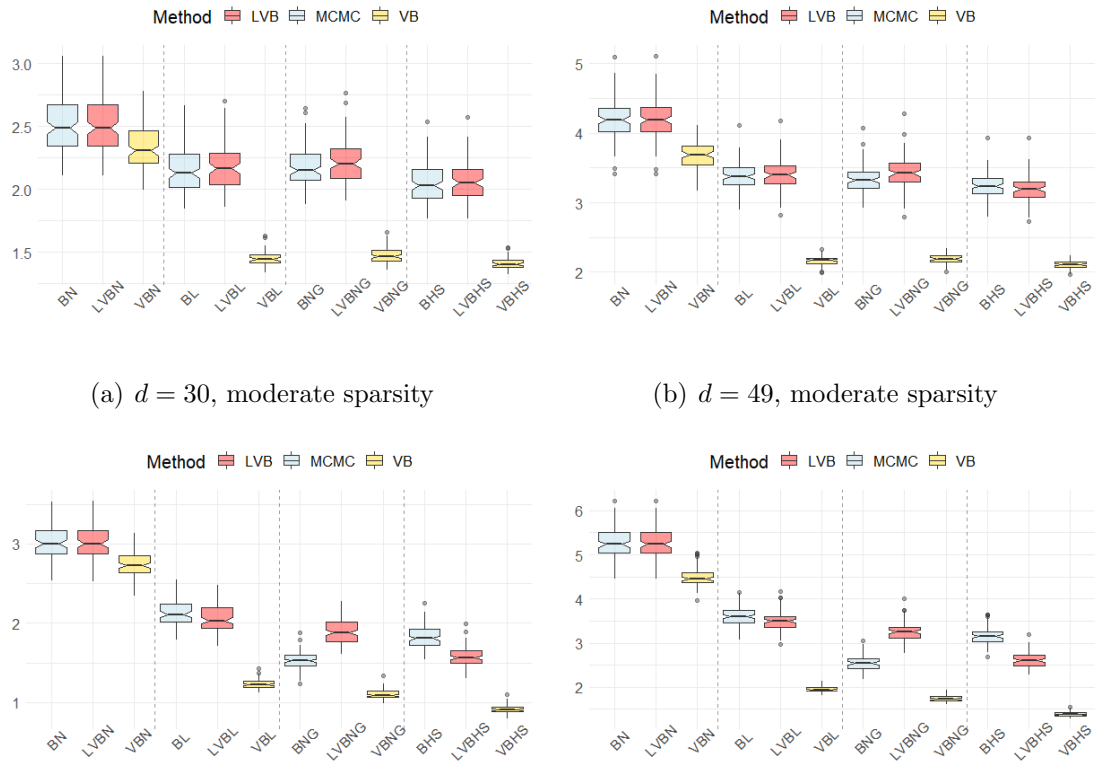
In order to compare the properties of the our proposed estimation method against standard MCMC and variational Bayes approaches in a controlled setting, we now perform an extensive simulation study. To this purpose, we consider different scenarios coherent with the setting of the main empirical application. In particular, we set the length of the time series $T = 360$, corresponding to 30 years of monthly data, the cross-sectional dimension of the data generating process $d = 15, 30, 49$ and we further assume either moderate (50% of true zeros) and high level of sparsity (90% of true zeros). Without loss of generality, the true matrix $\mathbf{\Theta}$ is generated assuming there are only endogenous variables (in particular we regress $\mathbf{y}_{t-1}$ on $\mathbf{y}_t$) in the following way: we fix to zero sd^2 entries at random, where $s = 0.5, 0.9$, while the remaining non zero coefficients are sampled from a mixture of two Gaussian distributions with means equal to ± 0.08, and standard deviation 0.1. The

normals are truncated at ± 0.05 respectively, to avoid very small values for the non zero parameters. The results of the simulation study for $N = 100$ replications are shown in Figure 3 and 4 for dimensions $d = 30, 49$, while the results for $d = 15$ are reported in Appendix D.

We compare the performance of our `VB` method against two main alternatives which are representative of state-of-the-art Bayesian estimation procedures within the context of multivariate time series models. First, we consider a linearization of the multivariate system of equations as proposed by Gefang et al. (2019). We assume different prior specifications for the elements in $\mathbf{A}$ such as *i).* Non-sparse normal prior, *ii).* Bayesian adaptive lasso, *iii).* Adaptive normal-gamma and *iv)* horseshoe. The latter are firstly estimated via MCMC (`MCMC`) with 5000 draws from the posterior, after 1000 of burn-in, (labelled as BN, BL, BNG and BHS). As a second approach, we consider the linearization of the model of Gefang et al. (2019), where parameters are estimated using a variational Bayes (`LVB`) method as in Gefang et al. (2019) and Chan and Yu (2020) (denoted with LVBN, LVBL, LVBNG and LVBHS). Our method instead considers the same prior specifications but the estimation is performed within the mean-field variational Bayes approach outlined in Section 3 (labelled VBN, VBL, VBNG and VBHS), which elicit shrinkage priors directly on the original parameters matrix $\mathbf{\Theta}$.

As a measure of point estimation accuracy of the regression matrix $\mathbf{\Theta}$, we first look at the Frobenius norm, denoted by $\|\cdot\|_F$. This measure the difference between the true matrix $\mathbf{\Theta}$, which is observed at each simulation, and the corresponding estimated $\widehat{\mathbf{\Theta}}$. Figure 3 shows the box charts for the $N = 100$ replications. Focusing first on the moderate sparsity case (i.e. 50% of zeros in the true matrix), the different priors performs equally given the estimation method. However, when the sparsity is higher (i.e. 90% of zeros in the true

matrix), the horseshoe tends to perform better that the adaptive lasso and normal-gamma. Perhaps more relevant for our estimation framework, by eliciting shrinkage priors directly on the regression matrix $\boldsymbol{\Theta}$ significantly improves the accuracy of the point estimates independently on the shrinkage prior utilised. This suggests that our procedure might lead to a more accurate identification of the cross-sectional lead-lag effects across predictors.



(a) $d = 30$, moderate sparsity

(b) $d = 49$, moderate sparsity

(c) $d = 30$, high sparsity

(d) $d = 49$, high sparsity

Figure 3: Frobenius norm of $\boldsymbol{\Theta} - \widehat{\boldsymbol{\Theta}}$ for different hierarchical shrinkage priors and different inference approaches.

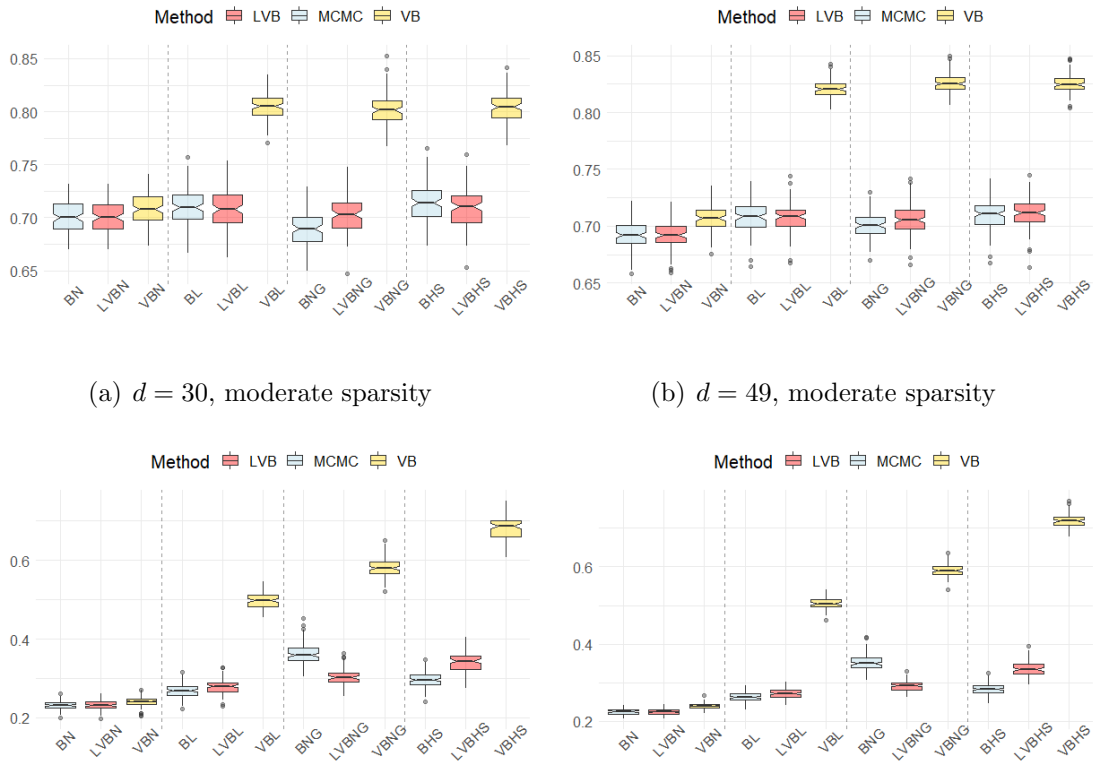
Figure 4 reports the F1-score across different estimation methods and for different simulation scenarios. The F1-score is a measure of model's accuracy used to evaluate binary classification systems, which classify examples into "positive" or "negative". Within the context of sparse regression matrix this boils down to quantifying the type I and type II errors in the identification of the significant predictors. As highlighted in Section 3, the

sparsification of the regression matrix $\widehat{\boldsymbol{\Theta}}$ is implemented by using the SAVS algorithm proposed by Ray and Bhattacharya (2018). The simulation show an interesting result: when the sparsity level is fixed at 50%, both the `MCMC` and the `LVB`, which are based on the Cholesky reparametrization of the regression matrix, provide results similar to the non-sparse priors. That is, the difference with our approach is quite astonishing. In fact, since we have a lot of non-zero elements, the linear combination $\boldsymbol{\Theta} = \mathbf{L}^{-1}\mathbf{A}$ has more terms to take into account and leads to a dense estimate of $\boldsymbol{\Theta}$ and larger variability in the estimates. More importantly, higher identification accuracy of our approach is not limited to the case of moderate sparsity. The bottom panels of Figure 4 show that, for both $d = 30$ and $d = 49$ dimension, our `VB` approach significantly outperform both the `MCMC` and `LVB` approaches independently on the shrinkage prior used.

To summarise, Figures 3 and 4 highlighted the clear advantage, from a pure inference perspective, of eliciting shrinkage priors on the regression matrix $\boldsymbol{\Theta}$ vs its linear transformation $\mathbf{A}$. More results are available in Appendix D, showing that the higher accuracy of our framework is preserved in smaller-dimensional settings (i.e. $d = 15$) and highlighting that `VB` algortihms are faster than the `MCMC` counterpart in terms of computational speed.

5 Cross-industry returns predictability

The simulation results in Section 4 suggest that by eliciting shrinkage priors on the regression matrix $\boldsymbol{\Theta}$ instead of its Cholesky factorization $\mathbf{A} = \mathbf{L}\boldsymbol{\Theta}$, one can possibly obtain a more accurate identification of the model structure. We now investigate the implications for forecasting stock returns within the context of a relatively large set of industry portfolios. Industry aggregation for portfolio returns is based on the four-digit SIC codes of the existing firm at each time t. More specifically, each NYSE, AMEX, and NASDAQ stock



(a) $d = 30$, moderate sparsity

(b) $d = 49$, moderate sparsity

(c) $d = 30$, high sparsity

(d) $d = 49$, high sparsity

Figure 4: F1 score computed looking at the true non-null parameters in $\boldsymbol{\Theta}$ and the non-null parameters in the estimated matrix $\widehat{\boldsymbol{\Theta}}$, for different hierarchical shrinkage priors and different inference approaches.

is assigned to an industry portfolio at the end of June of year t based on its four-digit SIC code at that time. The returns on a given portfolio are then computed from July of t to June of $t+1$. We consider both a longer time series of $d = 30$ industry portfolios from July 1926 to May 2020, and a larger cross section of $d = 49$ industry portfolios from July 1969 to May 2020. The difference sample length is due to data availability. The sample periods cover major macroeconomic events, from the great depression to the Covid-19 outbreak.

In addition to the lagged cross-industry portfolio returns, we consider as aggregate predictors a variety of common risk factors and macroeconomic variables. As far as risk factors are concerned, we consider the return on the value-weighted market portfolio in excess of the risk-free rate (`mkt`), and four alternative long-short investment strategies based on

market capitalization (`smb`), book-to-market ratios (`hml`), operating profitability (`rmw`) and investment (`cma`), as proposed by Fama and French (2015). The set of additional macroeconomic predictors is from Goyal and Welch (2008); this includes the log of the aggregate price-dividend ratio (`pd`), the term spread (`term`) (difference between the long term yield on government bonds and the T-bill), the credit spread (`credit`) (the BAA-AAA bond yields difference), the monthly change in inflation (`infl`) measured as the log change in the CPI, the aggregate market book-to-market ratio (`bm`), the net-equity issuing activity (`ntis`) and the long-term corporate bond returns (`corpr`).

5.1 Forecasting performance

For each industry, we calculate the out-of-sample predictive R^2, as suggested by Campbell and Thompson (2007). The R^2_{oos} is akin to the in-sample R^2 and is calculated as

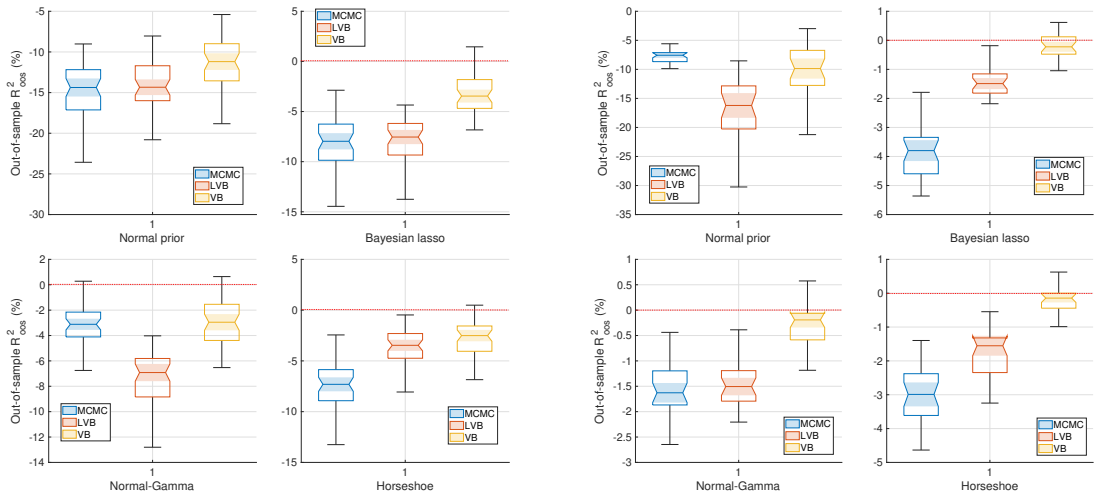
$$R^2_{i,oos} = 1 - \frac{\sum_{t_0=2}^{T} \left(y_{it} - \widehat{y}_{it}\left(\mathcal{M}_s\right)\right)^2}{\sum_{t_0=2}^{T} \left(y_{it} - \overline{y}_{it}\right)^2},$$

where $\overline{y}_{it}$ is the naive forecast from the recursive mean and $\widehat{y}_{it}\left(\mathcal{M}_s\right)$ is the forecast from a given model $\mathcal{M}_s$, for a given industry $i = 1, \ldots, d$, and t_0 is the date of the first prediction. We consider a 360 months rolling window period for the recursive mean and the model estimation, so that for instance for the 30 industry portfolio forecast errors are obtained by comparing each month the excess industry portfolio returns starting from 1957:07 to 2020:05. We compare the performance of our variational Bayes method `VB`, which allows to elicit a shrinkage prior directly on the matrix $\boldsymbol{\Theta}$, against both the `MCMC` and `LVB` algorithms outlined in the simulation analysis to estimate $\mathbf{A} = \mathbf{L}\boldsymbol{\Theta}$. Across estimation strategies we consider the same four prior specifications used in the simulation analysis, that is *i)* a non-sparse gaussian prior, *ii)* a Bayesian adaptive lasso, *iii)* an adaptive normal-gamma

and *iv)* the horseshoe (see, e.g., Gefang et al., 2019 and Chan and Yu, 2020).

5.1.1 Out-of-sample R^2_{oos}

Figure 5 reports the R^2_{oos} for both the 30 industry classification (right panels) and the 49 industry classification (left panels). Each box chart represents the cross-sectional distribution, that is the distribution of R^2_{oos} for $i = 1, \ldots, d$ industries.



(a) R^2_{oos} across 49 industries

(b) R^2_{oos} across 30 industries

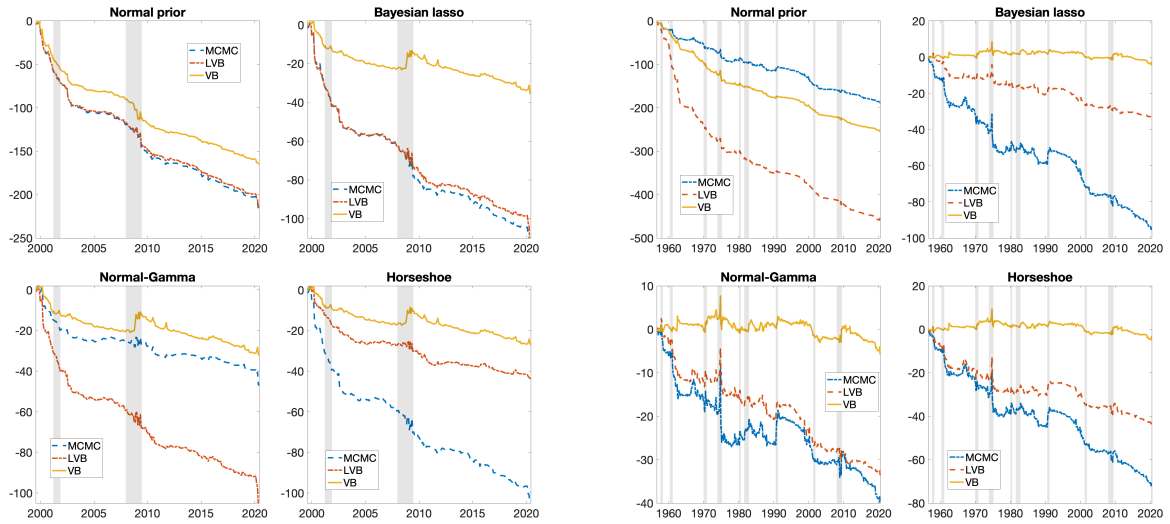
Figure 5: Cross-industry returns predictability. The left (right) four panels show the cross-sectional distribution of the R^2_{oos} for the 49 (30) industry portfolios. The red horizontal line separates the $R^2_{oos} > 0$ from $R^2_{oos} < 0$.

Consistent with the results in Campbell and Thompson (2007); Goyal and Welch (2008); Pettenuzzo et al. (2014), the simple naive forecast based on the rolling mean is a challenging benchmark to beat. Standard dense estimates, either through `MCMC` or a `LVB` approach, clearly suffer with the curse of dimensionality, that is, they perform poorly out-of-sample. This translates in largely negative out-of-sample R^2_{oos}. On the other hand, the forecasting performance of our `VB` approach outperform both competing approaches. For instance, for the cross section of 30 industry portfolios, the *median* R^2_{oos} is substantially higher, although slightly negative, than both the forecast from the `MCMC` or `LVB`. In fact, for about of a third

of the 30 industries, the out-of-sample R^2_{oos} from the `VB` is positive and is as high as 0.7% monthly, which means it outperforms the naive rolling mean forecast. This result holds across different prior specifications. As far as the cross section of 49 industry portfolios is concerned, the out-of-sample R^2_{oos} deteriorates across the board. Nevertheless, our `VB` estimation approach attains the highest *median* and average performance compared to `MCMC` and `LVB`, and in some cases generate a positive R^2_{oos} as high as 2% (cfr. Bayesian lasso and horseshoe). These results are key since they lie on the fact that unlike the `MCMC` and `LVB` approaches, our estimation procedure directly shrinkage the coefficients on the regression matrix $\mathbf{\Theta}$ rather than $\mathbf{A} = \mathbf{L\Theta}$. These results provide solid out-of-sample evidence to the in-sample perspective of Cohen and Frazzini (2008) and Menzly and Ozbas (2010), who find that economic links among certain individual firms and industries could possibly contribute to cross-industry return predictability.

Upon reporting the complete out-of-sample forecasting results, we follow Goyal and Welch (2008); Rapach et al. (2010) and present time series plots of the differences between the cumulative square prediction errors of the benchmark recursive mean forecast and each of the competing estimation strategies with different shrinkage priors. For the easy of exposition, individual industry forecasts are aggregated at each time t based on the average market capitalization (size) of a firm within a given industry. More precisely, the cumulative sum of squared errors (SSE) for the historical average versus the *ith* model is generally computed as: $\Delta\text{CumSSE}_{t,s} = \sum_{\tau=\underline{t}}^{t}(\overline{\boldsymbol{e}}_\tau)^2 - \sum_{\tau=\underline{t}}^{t}(\widehat{\boldsymbol{e}}_{\tau,s})^2$, where $\overline{\boldsymbol{e}}_\tau = \sum_{i=1}^{n} \mathrm{w}_{i\tau}(y_{i\tau} - \overline{y}_{i\tau})$ and $\widehat{\boldsymbol{e}}_{\tau,s} = \sum_{i=1}^{n} \mathrm{w}_{i\tau}(y_{i\tau} - \widehat{y}_{i\tau}(\mathcal{M}_s))$ are the weighted prediction error from the naive rolling mean benchmark and a given competing model $\mathcal{M}_s$ at time τ, respectively. Here, w_{it} represents the inverse of the average market capitalization (size) of a firm within a given industry. The role of the weight is to standardise the distance in a multivariate space,

and weight the assets' forecast errors differently depending on their relevance within a diversified market portfolio. Other things equal, there will be less (more) penalty for the forecast errors of a small (large) asset, which carries less (more) relevance in the dynamics of the market portfolio, but also less reward when accurate. Figure 6 reports the results.



(a) $\Delta\text{CumSSE}_{t,s}$ for 49 industry portfolios

(b) $\Delta\text{CumSSE}_{t,s}$ for the 30 industry portfolios

Figure 6: Cumulative squared error differentials. This figure plots the cumulative out-of-sample squared forecasting error differentials ($\Delta\text{CumSSE}_{t,s}$) for the weighted forecasting error. The left (right) panels shows the results for the cross section of 49 (30) industry portfolios.

When the curve Figure 6 is positive, the corresponding predictive model outperforms the historical mean benchmark, while the opposite holds when the curve is negative. The cumulative squared error differentials somewhat confirm the intuition from the individual R^2_{oos}. That is, while the rolling window mean forecast is a challenging benchmark, our `VB` estimation strategy for the $\boldsymbol{\Theta}$ regression matrix produces a positive out-of-sample performance for the majority of the out-of-sample period, at least for the cross section of 30 industry portfolios. Perhaps more important;y, our `VB` estimation strategy largely outperforms both the `MCMC` and the `LVB` approaches, irrespective of the prior specification. In fact, although the `VB` does not outperform the rolling window mean forecast, it cumulative

squared error differentials is substantially smaller than the one obtained from both competing approaches. In Appendix E we report the corresponding weighted $R^2_{W,oos}$ in addition to the results from an alternative weighting scheme where w_{it} represents the inverse of the sample volatility of the asset. The additional results largely confirm the superior performance of our `VB` estimation approach compared to `MCMC` and the `LVB`, irrespective of the shrinkage prior utilised.

Notably, the performance gap with respect to the naive rolling window forecast seems to decrease during recessions. For instance, for the 49 industry classification the cumulative squared error differential decreases during the great financial crisis of 2008/2009. In other words, the sparsity imposed on the regression matrix $\boldsymbol{\Theta}$ through our `VB` approach seems to pay off in terms of forecasting accuracy more during recessions vs expansions. This is consistent with recent studies, such as Rapach et al. (2010), Henkel et al. (2011), Dangl and Halling (2012), and Farmer et al. (2019) which show that the predictability of aggregate stock market returns is primarily concentrated in economic recessions, while it is largely absent during economic expansions. In Appendix E we provide additional results for the predictability of industry portfolio returns over the business cycle. The predictive ability of all prior specifications, including the non-sparse normal prior, substantially increases across industries during recessions. However, perhaps with the only exception of the normal-gamma for the 30 industry portfolios, our `VB` still outperforms both the `MCMC` and the `LVB` competing estimation strategies.

5.2 Economic significance

So far, our analysis has been concentrated on statistical measures of predictive accuracy. However, it is of paramount importance to evaluate the extent to which apparent gains

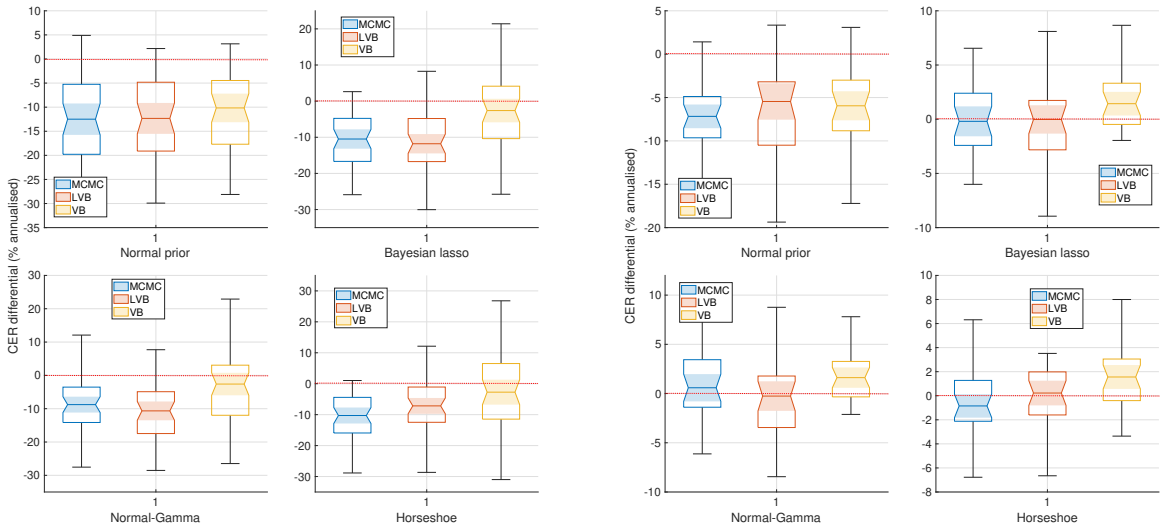
in predictive accuracy translates into better investment performances. Following existing literature (see, e.g., Goyal and Welch, 2008; Rapach et al., 2010; Dangl and Halling, 2012 and Pettenuzzo et al., 2014), we calculate the realised utility gains for a power utility investor within both a single- and multiple-asset setting. For the latter case, at each time t, the decision-maker selects the weights on risky assets to maximize the power utility (CRRA preferences) of the form, $\widehat{U}_{t,s} = \widehat{W}_t^{1-\gamma}(\mathcal{M}_s)/(1-\gamma)$, and $\widehat{W}_t(\mathcal{M}_s)$, the wealth generated by the competing model, s, at time, t. Campbell and Viceira (2004) show that the optimal portfolio allocation based on the conditional forecast can be expressed as $\mathrm{w}_t = \gamma^{-1}\Sigma_{t|t-1}^{-1}[\widehat{\boldsymbol{y}}_t + \boldsymbol{\sigma}_{t|t-1}^2/2]$, with $\widehat{\boldsymbol{y}}_t$ the vector of returns forecast at time t, $\boldsymbol{\sigma}_{t|t-1}^2$ a vector containing the diagonal elements of conditional covariance matrix $\Sigma_{t|t-1}$. Given the optimal weights, we compute the realised returns. We follow DeMiguel et al. (2014) and assume a flat 10 basis point transaction cost for each trade. Then, following Fleming et al. (2001), we obtain the certainty equivalent gains (annualized and in percentages) from the realised returns by equating the average utility of the historical average forecast, $u_{t,HA}$, with the average utility of the alternative models, $u_{t,s}$.

One comment is in order. This stylized optimal portfolio allocation exercise requires the investor to use some sort of time-varying volatility model to estimate $\Sigma_{t|t-1}$. However, different predictive strategies have different estimates of the conditional variance of the returns. In fact, the multivariate predictive regression framework we build upon assumes a constant variance for the residuals (see Section 2). Although modeling time-varying volatility is beyond the scope of the paper, in order to mitigate the effect of alternative volatility estimates on the resulting optimal weights, we consider, for each predictive strategy, a simple rolling window estimator of the conditional variance for the single-asset case and the conditional covariance for the multiple-asset case (see, e.g., Thornton and Valente, 2012).

Consistent with the recursive nature of the forecasts, the rolling-window estimates for the variance and the covariance are based on the past 360 months of realised excess returns. We also winsorize the weights for each of the industry to $-2 \leq w_t \leq 3$ to prevent extreme investments. Finally, to make our results directly comparable to other studies we assume a risk aversion $\gamma = 5$ (see, e.g., Pettenuzzo et al., 2014; Johannes et al., 2014).

5.2.1 Certainty equivalent returns

Figure 7 shows the certainty equivalent (CER) differentials across models and estimation strategies for both the cross section of 49 (left panels) and 30 (right panels) industry portfolios. As far as the 49 industry portfolios are concerned, the economic significance confirms the evidence for the R^2_{oos}; from a pure economic standpoint, the forecast from a recursive mean are quite challenging to beat by using sparse multivariate regression models irrespective of the shrinkage prior utilised.



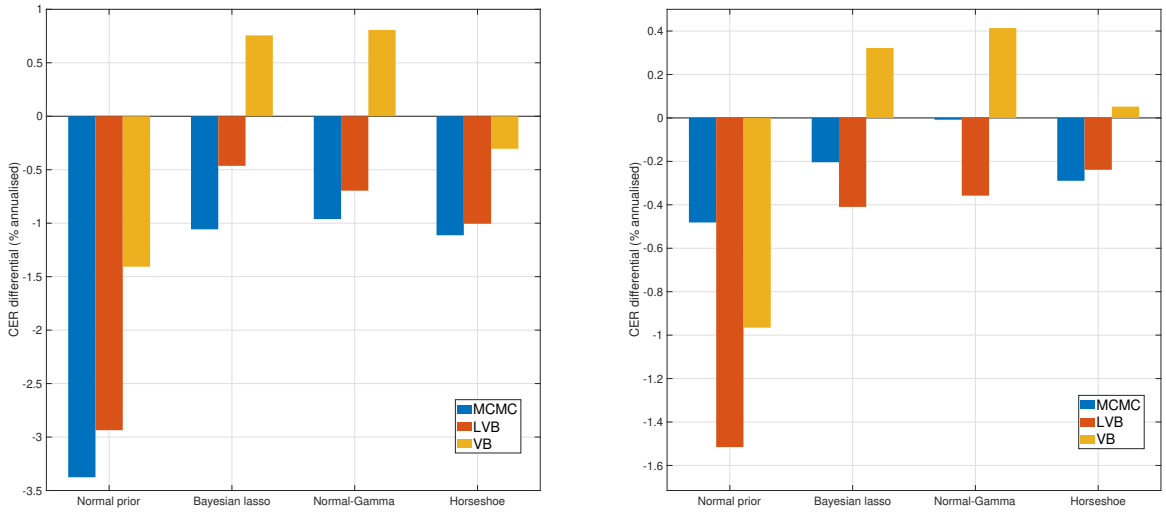
(a) CER differential across 49 industries

(b) CER differential across 30 industries

Figure 7: Cross-industry certainty equivalent differentials. The left (right) four panels show the results for the cross section 49 (30) industry portfolios.

Nevertheless, around a third of the investment signals generated from the `VB` approach

produce some positive economic value compared to the naive recursive mean benchmark. Perhaps more importantly, for both the adaptive Bayesian lasso, the adaptive normal-gamma and the horseshoe, our VB estimation approach produces a *median* CER differential which albeit slightly negative, is significantly higher than both the MCMC and the VB estimation strategies. The results for the cross section of 30 industry portfolios provide some clearer evidence. As far as our VB estimation strategy is concerned, the *median* CER differential is as high as 2% on an annualised basis, with values as high as 9% (annualized) for few industries. This compares to a value close to zero for the LVB and a negative CER differential for the MCMC across different shrinkage priors.



(a) Multiasset CER differential for 49 industries (b) Multiasset CER differential for 30 industries

Figure 8: Certainty equivalent differentials for the multivariate allocation. This figure plots the certainty equivalent returns in excess of the naive forecast, i.e., recursive mean, for each model specification.

Figure 8 reports the results for the multivariate allocation where all industry portfolios are jointly considered. The left panel shows the results for the cross section of 49 industry portfolios. Our VB estimation strategy outperforms both MCMC and LVB: it generates a CER differential of around 0.7% annualised for both the adaptive Bayesian lasso and the

normal-gamma priors. Economically, this means that a representative investor with power utility is willing to pay almost 1% annually to access the strategy based on a mean-field approximation shrinkage multivariate regression, while on the opposite the same investor would need to be paid as much as 1% annually to use the forecasting signal produced by either `LVB` or the `MCMC` approaches. None of the estimation procedures produce a sizable and positive return under the horseshoe prior specification.

The right panel of Figure 8 shows the results for the cross section of 30 industry portfolios. While the performance improves across estimation strategies and prior specifications, our `VB` approach still outperforms both competing methods by producing positive CER differentials irrespective of the type of shrinkage prior utilised. On the other hand, both the `LVB` and `MCMC` approaches deliver negative certainty equivalent returns of up to -40 basis points, annualised. That is, an investor would be willing to pay almost 1% annual fee to access the forecasts from `VB` against a negative fee to access the signals from `LVB` and `MCMC`.

6 Conclusion

We are interested in building a sparse multivariate predictive regression framework to model cross-industry stock returns predictability. Methodologically, we propose a novel variational Bayes estimation algorithm which allows for fast and accurate identification of the model parameters without leveraging on standard order-dependent Cholesky-type reparameterizations. Empirically, we show that our estimation approach substantially outperforms, both statistically and economically, forecasts from a variety of competing strategies, such as MCMC and variational Bayes methods. Delving further into the key computational features, we show in simulation that our variational Bayes inference is more accurate and less computationally expensive than Cholesky-based MCMC and variational Bayes approaches.

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Online appendix for:

Sparse multivariate modeling for stock returns predictability

This online appendix provide the derivation of the optimal densities used in the mean-field variational Bayes algorithms. The derivation concerns the optimal densities for both the normal prior as well as the adaptive Bayesian lasso, the adaptive normal-gamma and the horseshoe. In addition, in this appendix we provide additional simulation and empirical results.

A Theoretical results

This section provides major results that will be repeatedly used in the proofs of the derivation of the optimal densities used in the mean-field variational Bayes algorithms presented in Appendix B.

Result 1. *Assume that* $\mathbf{y}$ *is a* n*-dimensional vector,* $\mathbf{X}$ *a* $p \times n$ *matrix and* $\boldsymbol{\vartheta}$ *a* p*-dimensional vector of parameters whose distribution is denoted by* $q(\boldsymbol{\vartheta})$*.*
Define $\|\mathbf{y} - \boldsymbol{\vartheta}\mathbf{X}\|_2^2 = (\mathbf{y} - \boldsymbol{\vartheta}\mathbf{X})(\mathbf{y} - \boldsymbol{\vartheta}\mathbf{X})^\intercal$*, then it holds:*

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{\vartheta}}\left[\|\mathbf{y} - \boldsymbol{\vartheta}\mathbf{X}\|_2^2\right] &= \mathbf{y}\mathbf{y}^\intercal + \mathbb{E}_{\boldsymbol{\vartheta}}\left[\boldsymbol{\vartheta}\mathbf{X}\mathbf{X}^\intercal\boldsymbol{\vartheta}^\intercal\right] - 2\boldsymbol{\mu}_{q(\vartheta)}\mathbf{X}\mathbf{y}^\intercal \\
&= \mathbf{y}\mathbf{y}^\intercal + tr\left\{\mathbb{E}_{\boldsymbol{\vartheta}}\left[\boldsymbol{\vartheta}^\intercal\boldsymbol{\vartheta}\right]\mathbf{X}\mathbf{X}^\intercal\right\} - 2\boldsymbol{\mu}_{q(\vartheta)}\mathbf{X}\mathbf{y}^\intercal \\
&= \mathbf{y}\mathbf{y}^\intercal + \boldsymbol{\mu}_{q(\vartheta)}\mathbf{X}\mathbf{X}^\intercal\boldsymbol{\mu}_{q(\vartheta)}^\intercal + tr\left\{\boldsymbol{\Sigma}_{q(\vartheta)}\mathbf{X}\mathbf{X}^\intercal\right\} - 2\boldsymbol{\mu}_{q(\vartheta)}\mathbf{X}\mathbf{y}^\intercal \\
&= \|\mathbf{y} - \boldsymbol{\mu}_{q(\vartheta)}\mathbf{X}\|_2^2 + tr\left\{\boldsymbol{\Sigma}_{q(\vartheta)}\mathbf{X}\mathbf{X}^\intercal\right\},
\end{aligned}$$

where $\mathbb{E}_{\boldsymbol{\vartheta}}(f(\boldsymbol{\vartheta}))$ *denotes the expectation of the function* $f(\boldsymbol{\vartheta}) : \mathbb{R}^p \to \mathbb{R}^k$ *with respect to* $q(\boldsymbol{\vartheta})$*,* $tr(\cdot)$ *denotes the trace operator that returns the sum of the diagonal entries of a square matrix, and* $\boldsymbol{\mu}_{q(\vartheta)}$ *and* $\boldsymbol{\Sigma}_{q(\vartheta)}$ *denotes the mean and variance-covariance matrix of* $\boldsymbol{\vartheta}$*.*

Result 2. *Let* $\boldsymbol{\Theta}$ *be a* $d \times p$ *random matrix with elements* $\vartheta_{i,j}$*, for* $i = 1, \ldots, d$ *and* $j = 1, \ldots, p$*, and let* $\mathbf{A}$ *be a* $p \times p$ *matrix. Our interest relies on the computation of the expectation of* $\boldsymbol{\Theta}\mathbf{A}\boldsymbol{\Theta}^\intercal$ *with respect to the distribution of* $\boldsymbol{\Theta}$*, where the expectation is taken element-wise. The* (i, j)*-th entry of* $\boldsymbol{\Theta}\mathbf{A}\boldsymbol{\Theta}^\intercal$ *is equal to* $\boldsymbol{\vartheta}_i\mathbf{A}\boldsymbol{\vartheta}_j^\intercal$*, where* $\boldsymbol{\vartheta}_i$ *and* $\boldsymbol{\vartheta}_j$ *denote the* i*-th and* j*-th row of* $\boldsymbol{\Theta}$*, respectively.*
Therefore, the (i, j)*-th entry of* $\boldsymbol{\Theta}\mathbf{A}\boldsymbol{\Theta}^\intercal$ *is equal to:*

$$\mathbb{E}\big(\boldsymbol{\vartheta}_i\mathbf{A}\boldsymbol{\vartheta}_j^\intercal\big) = \mathbb{E}\big(tr\big\{\boldsymbol{\vartheta}_j^\intercal\boldsymbol{\vartheta}_i\mathbf{A}\big\}\big) = tr\big\{\mathbb{E}(\boldsymbol{\vartheta}_j^\intercal\boldsymbol{\vartheta}_i\mathbf{A})\big\} = tr\big\{\mathbb{E}(\boldsymbol{\vartheta}_j^\intercal\boldsymbol{\vartheta}_i)\mathbf{A}\big\}.$$

Let denote by $\boldsymbol{\mu}_{\vartheta_i} = \mathbb{E}(\boldsymbol{\vartheta}_i)$ *and* $\boldsymbol{\Sigma}_{\vartheta_i,\vartheta_j} = Cov(\boldsymbol{\vartheta}_i, \boldsymbol{\vartheta}_j)$*, then the previous expectation reduces to:*

$$\mathbb{E}(\boldsymbol{\vartheta}_i\mathbf{A}\boldsymbol{\vartheta}_j^\intercal) = tr\big\{\big(\boldsymbol{\mu}_{\vartheta_j}^\intercal\boldsymbol{\mu}_{\vartheta_i} + \boldsymbol{\Sigma}_{\vartheta_i,\vartheta_j}\big)\mathbf{A}\big\} = \boldsymbol{\mu}_{\vartheta_i}\mathbf{A}\boldsymbol{\mu}_{\vartheta_j}^\intercal + tr\left\{\boldsymbol{\Sigma}_{\vartheta_i,\vartheta_j}\mathbf{A}\right\}.$$

In matrix form, $\mathbb{E}(\boldsymbol{\Theta}\mathbf{A}\boldsymbol{\Theta}^{\intercal}) = \boldsymbol{\mu}_{\boldsymbol{\Theta}}\mathbf{A}\boldsymbol{\mu}_{\boldsymbol{\Theta}}^{\intercal} + \mathbf{K}_{\boldsymbol{\Theta}}$*, where* $\boldsymbol{\mu}_{\boldsymbol{\Theta}}$ *is a* $d \times p$ *matrix with elements* $\mu_{\vartheta_{i,j}}$*, while* $\mathbf{K}_{\boldsymbol{\Theta}}$ *is a* $d \times d$ *symmetric matrix with elements equal to* $tr\left\{\boldsymbol{\Sigma}_{\vartheta_i,\vartheta_j}\mathbf{A}\right\}$*.*

Result (2) can be further generalized to compute the expectation of $\boldsymbol{\Theta}_1\mathbf{A}\boldsymbol{\Theta}_2^{\intercal}$ *with respect to the joint distribution of* $(\boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2)$ *where* $\boldsymbol{\Theta}_1$ *is* $d_1 \times p$ *and* $\boldsymbol{\Theta}_2$ *is* $d_2 \times p$*.*

Result 3. *Let* $\boldsymbol{\vartheta}$ *be a* d*-dimesnional gaussian random vector with mean vector* $\boldsymbol{\mu}_{\vartheta}$ *and variance-covariance matrix* $\boldsymbol{\Sigma}_{\vartheta}$*. Then the expectation of the quadratic form* $(\boldsymbol{\vartheta}-\boldsymbol{\mu}_{\vartheta})^{\intercal}\boldsymbol{\Sigma}_{\vartheta}^{-1}(\boldsymbol{\vartheta}-\boldsymbol{\mu}_{\vartheta})$ *with respect to* $\boldsymbol{\vartheta}$ *is equal to* d*. Indeed:*

$$\mathbb{E}_{\vartheta}\left[(\boldsymbol{\vartheta}-\boldsymbol{\mu}_{\vartheta})^{\intercal}\boldsymbol{\Sigma}_{\vartheta}^{-1}(\boldsymbol{\vartheta}-\boldsymbol{\mu}_{\vartheta})\right] = tr\left\{\mathbb{E}_{\vartheta}\left[(\boldsymbol{\vartheta}-\boldsymbol{\mu}_{\vartheta})(\boldsymbol{\vartheta}-\boldsymbol{\mu}_{\vartheta})^{\intercal}\right]\boldsymbol{\Sigma}_{\vartheta}^{-1}\right\} = tr\left\{\boldsymbol{\Sigma}_{\vartheta}\boldsymbol{\Sigma}_{\vartheta}^{-1}\right\} = tr\left\{\mathbf{I}_d\right\} = d.$$

B Derivation of the variational Bayes algorithms

This appendix explains how to obtain the relevant quantities of the mean-field variational Bayes algorithms described in Section 3 for the prior distributions described in Section 3.1. We begin by discussing the non-informative prior, then turn to the adaptive Bayesian lasso, the adaptive normal-gamma and conclude with the horseshoe prior.

B.1 Non-informative prior specification

Proposition B.1.1. *The optimal variational density for the precision parameter* ν_j *is equal to* $q^*(\nu_j) \equiv \mathsf{Ga}(a_{q(\nu_j)}, b_{q(\nu_j)})$*, where, for* $j = 1, \ldots, d$*:*

$$a_{q(\nu_j)} = a_{\nu} + T/2, \quad b_{q(\nu_j)} = b_{\nu} + \frac{1}{2}\sum_{t=1}^{T}\mathbb{E}_{-\nu_j}\left[\varepsilon_{j,t}^2\right], \tag{B.1}$$

where

$$\begin{aligned}\mathbb{E}_{-\nu_j}\left[\varepsilon_{j,t}^2\right] = &\left(y_{j,t} - \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})} - \boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1}\right)^2 \\ &+ tr\left\{\boldsymbol{\Sigma}_{q(\vartheta_j)}\mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal}\right\} \\ &+ tr\left\{\left(\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)} + \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^{\intercal}\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\right)\mathbf{K}_{\vartheta,t}\right\} \\ &+ tr\left\{\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^{\intercal}\right\} \\ &- 2\mathbf{k}_{\boldsymbol{\vartheta},t}\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^{\intercal},\end{aligned}$$

where $\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})} = \mathbf{y}_t^j - \boldsymbol{\mu}_{q(\boldsymbol{\Theta}^j)}\mathbf{z}_{t-1}$*, and, for* $i = 1, \ldots, j-1$ *and* $k = 1, \ldots, j-1$*, the elements in the matrix* $\mathbf{K}_{\boldsymbol{\vartheta},t}$ *and in the row vector* $\mathbf{k}_{\boldsymbol{\vartheta},t}$ *are* $[\mathbf{K}_{\vartheta,t}]_{i,k} = tr\left\{Cov(\boldsymbol{\vartheta}_i, \boldsymbol{\vartheta}_k)\mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal}\right\}$ *and* $[\mathbf{k}_{\vartheta,t}]_i = tr\left\{Cov(\boldsymbol{\vartheta}_i, \boldsymbol{\vartheta}_j)\mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal}\right\}$ *respectively.*

Proof. Consider the model written for the j-th variable:

$$y_{j,t} = \boldsymbol{\beta}_j \mathbf{r}_{j,t} + \boldsymbol{\vartheta}_j \mathbf{z}_{t-1} + \varepsilon_{j,t}, \quad \varepsilon_{j,t} \sim \mathsf{N}(0, 1/\nu_j),$$

and notice that $\varepsilon_{j,t} = y_{j,t} - \boldsymbol{\beta}_j \mathbf{r}_{j,t} - \boldsymbol{\vartheta}_j \mathbf{z}_{t-1}$. Recall that a priori $\nu_j \sim \mathsf{Ga}(a_\nu, b_\nu)$ and compute $\log q^*(\nu_j) \propto \mathbb{E}_{-\nu_j}\left[\ell_j(\boldsymbol{\xi}; \mathbf{y}, \mathbf{x}) + \log p(\nu_j)\right]$:

$$\begin{aligned}
\log q^*(\nu_j) &\propto \mathbb{E}_{-\nu_j}\left[\frac{T}{2}\log \nu_j - \frac{\nu_j}{2}\sum_{t=1}^{T}\varepsilon_{j,t}^2 + (a_\nu - 1)\log \nu_j - b_\nu \nu_j\right] \\
&\propto \left(\frac{T}{2} + a_\nu - 1\right)\log \nu_j - \nu_j\left(b_\nu + \frac{1}{2}\sum_{t=1}^{T}\mathbb{E}_{-\nu_j}\left[\varepsilon_{j,t}^2\right]\right),
\end{aligned}$$

where

$$\begin{aligned}
\mathbb{E}_{-\nu_j}\left[\varepsilon_{j,t}^2\right] &= \mathbb{E}_{-\nu_j}\left[\left(y_{j,t} - \boldsymbol{\beta}_j \mathbf{r}_{j,t} - \boldsymbol{\vartheta}_j \mathbf{z}_{t-1}\right)^2\right] \\
&= y_{j,t}^2 + \mathbb{E}_{\vartheta}\left[\boldsymbol{\vartheta}_j \mathbf{z}_{t-1}\mathbf{z}_{t-1}^\intercal \boldsymbol{\vartheta}_j\right] + \overbrace{\mathbb{E}_{\vartheta,\boldsymbol{\beta}_j}\left[\boldsymbol{\beta}_j \mathbf{r}_{j,t}\mathbf{r}_{j,t}^\intercal \boldsymbol{\beta}_j^\intercal\right]}^{\text{A}} \\
&\quad - 2y_{j,t}\mathbb{E}_{\vartheta}\left[\boldsymbol{\vartheta}_j\right]\mathbf{z}_{t-1} - 2y_{j,t}\mathbb{E}_{\boldsymbol{\beta}_j}\left[\boldsymbol{\beta}_j\right]\mathbb{E}_{\vartheta}\left[\mathbf{r}_{j,t}\right] \\
&\quad + 2\underbrace{\mathbb{E}_{\vartheta}\left[\boldsymbol{\vartheta}_j \mathbf{z}_{t-1}\mathbf{r}_{j,t}^\intercal\right]\mathbb{E}_{\boldsymbol{\beta}_j}\left[\boldsymbol{\beta}_j^\intercal\right]}_{\text{B}} \\
&= y_{j,t}^2 + \boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1}\mathbf{z}_{t-1}^\intercal \boldsymbol{\mu}_{q(\vartheta_j)} + \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^\intercal \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^\intercal \\
&\quad - 2y_{j,t}\boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1} - 2y_{j,t}\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})} \\
&\quad + 2\boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^\intercal \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^\intercal \\
&\quad + \operatorname{tr}\left\{\boldsymbol{\Sigma}_{q(\vartheta_j)}\mathbf{z}_{t-1}\mathbf{z}_{t-1}^\intercal\right\} + \operatorname{tr}\left\{\left(\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)} + \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^\intercal \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\right)\mathbf{K}_{\vartheta,t}\right\} \\
&\quad + \operatorname{tr}\left\{\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^\intercal\right\} - 2\mathbf{k}_{\vartheta,t}\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^\intercal \\
&= \left(y_{j,t} - \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})} - \boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1}\right)^2 \\
&\quad + \operatorname{tr}\left\{\boldsymbol{\Sigma}_{q(\vartheta_j)}\mathbf{z}_{t-1}\mathbf{z}_{t-1}^\intercal\right\} + \operatorname{tr}\left\{\left(\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)} + \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^\intercal \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\right)\mathbf{K}_{\vartheta,t}\right\} \\
&\quad + \operatorname{tr}\left\{\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^\intercal\right\} - 2\mathbf{k}_{\vartheta,t}\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^\intercal,
\end{aligned}$$

where $\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})} = \mathbf{y}_t^j - \boldsymbol{\mu}_{q(\boldsymbol{\Theta}^j)}\mathbf{z}_{t-1}$. The computations involving terms A and B are presented in the following equations. Firs of all, define $\boldsymbol{\beta}_j \mathbf{r}_{j,t}\mathbf{r}_{j,t}^\intercal \boldsymbol{\beta}_j^\intercal = \|\boldsymbol{\beta}_j \mathbf{r}_{j,t}\|_2^2$, then the term A above

is equal to:

$$
\begin{aligned}
\mathbb{E}_{\vartheta,\boldsymbol{\beta}_j}\left[\|\boldsymbol{\beta}_j\mathbf{r}_{j,t}\|_2^2\right] &= \mathbb{E}_{\boldsymbol{\beta}_j}\Big[\boldsymbol{\beta}_j \overbrace{\mathbb{E}_{\vartheta}\left[\mathbf{r}_{j,t}\mathbf{r}_{j,t}^{\intercal}\right]}^{\text{See Results 1 and 2}} \boldsymbol{\beta}_j^{\intercal}\Big] \\
&= \mathbb{E}_{\boldsymbol{\beta}_j}\left[\boldsymbol{\beta}_j\left\{\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^{\intercal}+\mathbf{K}_{\vartheta,t}\right\}\boldsymbol{\beta}_j^{\intercal}\right] \\
&= \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\left\{\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^{\intercal}+\mathbf{K}_{\vartheta,t}\right\}\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^{\intercal}+\operatorname{tr}\left\{\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}\left[\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^{\intercal}+\mathbf{K}_{\vartheta,t}\right]\right\} \\
&= \|\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\|_2^2+\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}+\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^{\intercal}\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}\right)\mathbf{K}_{\vartheta,t}\right\} \\
&\quad+\operatorname{tr}\left\{\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^{\intercal}\right\},
\end{aligned}
$$

while the term B is:

$$
\begin{aligned}
\mathbb{E}_{\vartheta}\left[\boldsymbol{\vartheta}_j\mathbf{z}_{t-1}\mathbf{r}_{j,t}^{\intercal}\right]\mathbb{E}_{\boldsymbol{\beta}_j}\left[\boldsymbol{\beta}_j^{\intercal}\right] &= \mathbb{E}_{\vartheta}\Big[\boldsymbol{\vartheta}_j\mathbf{z}_{t-1}\mathbf{y}_t^{j\intercal}-\overbrace{\boldsymbol{\vartheta}_j\mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal}\boldsymbol{\Theta}^{j\intercal}}^{\text{See Result 2}}\Big]\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^{\intercal} \\
&= \left(\boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1}\mathbf{y}_t^{j\intercal}-\boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal}\boldsymbol{\mu}_{q(\boldsymbol{\Theta}^j)}^{\intercal}-\mathbf{k}_{\vartheta,t}\right)\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^{\intercal} \\
&= \boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^{\intercal}\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^{\intercal}-\mathbf{k}_{\vartheta,t}\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}^{\intercal}.
\end{aligned}
$$

Notice that for the latter derivation we use Results 1 and 2. To conclude, we obtain:

$$
\log q^*(\nu_j)\propto\left(\frac{T}{2}+a_\nu-1\right)\log\nu_j-\nu_j\left(b_\nu+\frac{1}{2}\sum_{t=1}^{T}\mathbb{E}_{-\nu_j}\left[\varepsilon_{j,t}^2\right]\right),
$$

then take the exponential and notice that the latter is the kernel of a gamma random variable $\mathsf{Ga}(a_{q(\nu_j)},b_{q(\nu_j)})$ as defined in Proposition B.1.1. □

Proposition B.1.2. *The optimal variational density for the parameter $\boldsymbol{\beta}_j$ for $j=2,\dots,d$ is equal to $q^*(\boldsymbol{\beta}_j)\equiv\mathsf{N}_{j-1}(\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)},\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)})$, where:*

$$
\begin{aligned}
\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)} &= \left(\mu_{q(\nu_j)}\sum_{t=1}^{T}\left(\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^{\intercal}+\mathbf{K}_{\vartheta,t}\right)+1/\tau\mathbf{I}_{j-1}\right)^{-1}, \\
\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)} &= \boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}\mu_{q(\nu_j)}\sum_{t=1}^{T}\left(\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}(y_{j,t}-\boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1})^{\intercal}+\mathbf{k}_{\vartheta,t}\right).
\end{aligned}
\tag{B.2}
$$

Proof. Consider the model written for the j-th variable:

$$
y_{j,t}=\boldsymbol{\beta}_j\mathbf{r}_{j,t}+\boldsymbol{\vartheta}_j\mathbf{z}_{t-1}+\varepsilon_{j,t},\quad \varepsilon_{j,t}\sim\mathsf{N}(0,1/\nu_j).
$$

Recall that a priori $\boldsymbol{\beta}_j\sim\mathsf{N}_{j-1}(\mathbf{0},\tau\mathbf{I}_{j-1})$ and compute the optimal variational density as

$\log q^*(\boldsymbol{\beta}_j) \propto \mathbb{E}_{-\boldsymbol{\beta}_j}\left[\ell_j(\boldsymbol{\xi};\mathbf{y},\mathbf{x}) + \log p(\boldsymbol{\beta}_j)\right]$:

$$\begin{aligned}\log q^*(\boldsymbol{\beta}_j) &\propto \mathbb{E}_{-\boldsymbol{\beta}_j}\left[-\frac{\nu_j}{2}\sum_{t=1}^{T}\left(y_{j,t} - \boldsymbol{\vartheta}_j\mathbf{z}_{t-1} - \boldsymbol{\beta}_j\mathbf{r}_{j,t}\right)^2 - \frac{1}{2\tau}\boldsymbol{\beta}_j\boldsymbol{\beta}_j^\intercal\right] \\ &\propto \mathbb{E}_{-\boldsymbol{\beta}_j}\left[-\frac{1}{2}\left\{\boldsymbol{\beta}_j\left(\nu_j\sum_{t=1}^{T}\mathbf{r}_{j,t}\mathbf{r}_{j,t}^\intercal + 1/\tau\mathbf{I}_{j-1}\right)\boldsymbol{\beta}_j^\intercal - 2\boldsymbol{\beta}_j\nu_j\sum_{t=1}^{T}\mathbf{r}_{j,t}(y_{j,t} - \boldsymbol{\vartheta}_j\mathbf{z}_{t-1})^\intercal\right\}\right],\end{aligned}$$

and, applying some results defined is Appendix A, we get:

$$\begin{aligned}\log q^*(\boldsymbol{\beta}_j) &\propto -\frac{1}{2}\left\{\boldsymbol{\beta}_j\left(\mu_{q(\nu_j)}\sum_{t=1}^{T}\mathbb{E}_{\vartheta}\overbrace{\left[\mathbf{r}_{j,t}\mathbf{r}_{j,t}^\intercal\right]}^{\text{Result 2}} + 1/\tau\mathbf{I}_{j-1}\right)\boldsymbol{\beta}_j^\intercal - 2\boldsymbol{\beta}_j\mu_{q(\nu_j)}\sum_{t=1}^{T}\mathbb{E}_{\vartheta}\overbrace{\left[\mathbf{r}_{j,t}(y_{j,t} - \boldsymbol{\vartheta}_j\mathbf{z}_{t-1})^\intercal\right]}^{\text{Result 2}}\right\} \\ &\propto -\frac{1}{2}\left\{\boldsymbol{\beta}_j\left(\mu_{q(\nu_j)}\sum_{t=1}^{T}\left(\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}^\intercal + \mathbf{K}_{\vartheta,t}\right) + 1/\tau\mathbf{I}_{j-1}\right)\boldsymbol{\beta}_j^\intercal \right. \\ &\qquad \left. - 2\boldsymbol{\beta}_j\mu_{q(\nu_j)}\sum_{t=1}^{T}\left(\boldsymbol{\mu}_{q(\mathbf{r}_{j,t})}(y_{j,t} - \boldsymbol{\mu}_{q(\vartheta_j)}\mathbf{z}_{t-1})^\intercal + \mathbf{k}_{\vartheta,t}\right)\right\}.\end{aligned}$$

Take the exponential and notice that the latter is the kernel of a gaussian random variable $\mathsf{N}_{j-1}(\boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}, \boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)})$, as defined in Proposition B.1.2. □

Proposition B.1.3. *The optimal variational density for the parameter $\boldsymbol{\vartheta}$ is equal to a multivariate gaussian $q^*(\boldsymbol{\vartheta}) \equiv \mathsf{N}_{d(d+p+1)}(\boldsymbol{\mu}_{q(\vartheta)}, \boldsymbol{\Sigma}_{q(\vartheta)})$, where:*

$$\boldsymbol{\Sigma}_{q(\vartheta)} = \left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} \otimes \sum_{t=1}^{T}\mathbf{z}_{t-1}\mathbf{z}_{t-1}^\intercal + 1/\upsilon\mathbf{I}_{d(d+p+1)}\right)^{-1}, \qquad \boldsymbol{\mu}_{q(\vartheta)} = \boldsymbol{\Sigma}_{q(\vartheta)}\sum_{t=1}^{T}\left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} \otimes \mathbf{z}_{t-1}\right)\mathbf{y}_t, \tag{B.3}$$

where $\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} = \mathbb{E}_q\left[\boldsymbol{\Omega}\right] = \mathbb{E}_q\left[\mathbf{L}^\intercal\mathbf{V}\mathbf{L}\right] = (\mathbf{I}_d - \boldsymbol{\mu}_{q(\mathbf{B})})^\intercal\boldsymbol{\mu}_{q(\mathbf{V})}(\mathbf{I}_d - \boldsymbol{\mu}_{q(\mathbf{B})}) + \mathbf{C}_\vartheta$ and $\mathbf{C}_\vartheta$ is a $d \times d$ symmetric matrix whose generic element is given by:

$$[\mathbf{C}_\vartheta]_{i,j} = \sum_{k=j+1}^{d} Cov(\beta_{k,i}, \beta_{k,j})\mu_{q(\nu_k)}.$$

Proof. Consider the model written as $\mathbf{L}\mathbf{y}_t = \mathbf{L}\boldsymbol{\Theta}\mathbf{z}_{t-1} + \boldsymbol{\varepsilon}_t$ with $\boldsymbol{\varepsilon}_t \sim \mathsf{N}_d(0, \mathbf{V}^{-1})$ and then apply the vectorisation operation on the transposed and get:

$$\mathbf{L}\mathbf{y}_t = (\mathbf{L} \otimes \mathbf{z}_{t-1}^\intercal)\boldsymbol{\vartheta} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathsf{N}_d(0, \mathbf{V}^{-1}).$$

Recall that a priori $\boldsymbol{\vartheta} \sim \mathsf{N}_{d(d+p+1)}(\mathbf{0}, \upsilon\mathbf{I}_{d(d+p+1)})$. Compute the optimal variational density

for the parameter $\boldsymbol{\vartheta}$ as $\log q^*(\boldsymbol{\vartheta}) \propto \mathbb{E}_{-\vartheta}\left[\ell(\boldsymbol{\xi}; \mathbf{y}, \mathbf{x}) + \log p(\boldsymbol{\vartheta})\right]$:

$$\begin{aligned}
\log q^*(\boldsymbol{\vartheta}) &\propto -\frac{1}{2}\mathbb{E}_{-\vartheta}\left[\sum_{t=1}^{T}\left(\mathbf{L}\mathbf{y}_t - (\mathbf{L}\otimes\mathbf{z}_{t-1}^{\intercal})\boldsymbol{\vartheta}\right)^{\intercal}\mathbf{V}\left(\mathbf{L}\mathbf{y}_t - (\mathbf{L}\otimes\mathbf{z}_{t-1}^{\intercal})\boldsymbol{\vartheta}\right)\right] - \frac{1}{2\upsilon}\mathbb{E}_{-\vartheta}\left[\boldsymbol{\vartheta}^{\intercal}\boldsymbol{\vartheta}\right] \\
&\propto -\frac{1}{2}\mathbb{E}_{-\vartheta}\left[\sum_{t=1}^{T}\left(\boldsymbol{\vartheta}^{\intercal}(\mathbf{L}^{\intercal}\mathbf{V}\mathbf{L}\otimes\mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal})\boldsymbol{\vartheta}\right) - 2\sum_{t=1}^{T}\boldsymbol{\vartheta}^{\intercal}\Big((\mathbf{L}^{\intercal}\mathbf{V}\mathbf{L}\otimes\mathbf{z}_{t-1})\mathbf{y}_t\Big)\right] - \frac{1}{2\upsilon}\boldsymbol{\vartheta}^{\intercal}\boldsymbol{\vartheta} \\
&\propto -\frac{1}{2}\left\{\boldsymbol{\vartheta}^{\intercal}\left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})}\otimes\sum_{t=1}^{T}\mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal} + 1/\upsilon\mathbf{I}_{d(d+p+1)}\right)\boldsymbol{\vartheta} - 2\boldsymbol{\vartheta}^{\intercal}\sum_{t=1}^{T}\left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})}\otimes\mathbf{z}_{t-1}\right)\mathbf{y}_t\right\}.
\end{aligned}$$

To compute the expectation $\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} = \mathbb{E}_{-\vartheta}\left[(\mathbf{I}_d - \mathbf{B})^{\intercal}\mathbf{V}(\mathbf{I}_d - \mathbf{B})\right]$ we use the following:

$$\begin{aligned}
\mathbb{E}_{\mathbf{B},\mathbf{V}}\left[(\mathbf{I}_d - \mathbf{B})^{\intercal}\mathbf{V}(\mathbf{I}_d - \mathbf{B})\right] &= \mathbb{E}_{\mathbf{B},\mathbf{V}}\left[\mathbf{V} - 2\mathbf{B}^{\intercal}\mathbf{V} - \mathbf{B}^{\intercal}\mathbf{V}\mathbf{B}\right] \\
&= \boldsymbol{\mu}_{q(\mathbf{V})} - 2\boldsymbol{\mu}_{q(\mathbf{B})}^{\intercal}\boldsymbol{\mu}_{q(\mathbf{V})} - \mathbb{E}_{\mathbf{B},\mathbf{V}}\left[\mathbf{B}^{\intercal}\mathbf{V}\mathbf{B}\right] \\
&= \boldsymbol{\mu}_{q(\mathbf{V})} - 2\boldsymbol{\mu}_{q(\mathbf{B})}^{\intercal}\boldsymbol{\mu}_{q(\mathbf{V})} + \boldsymbol{\mu}_{q(\mathbf{B})}^{\intercal}\boldsymbol{\mu}_{q(\mathbf{V})}\boldsymbol{\mu}_{q(\mathbf{B})} + \mathbf{C}_{\vartheta} \\
&= (\mathbf{I}_d - \boldsymbol{\mu}_{q(\mathbf{B})})^{\intercal}\boldsymbol{\mu}_{q(\mathbf{V})}(\mathbf{I}_d - \boldsymbol{\mu}_{q(\mathbf{B})}) + \mathbf{C}_{\vartheta},
\end{aligned}$$

where we exploit the fact that the (i,j)-th element of $\mathbf{B}^{\intercal}\mathbf{V}\mathbf{B}$ is given by:

$$\left[\mathbf{B}^{\intercal}\mathbf{V}\mathbf{B}\right]_{i,j} = \sum_{k=j+1}^{d}\beta_{k,i}\beta_{k,j}\nu_k, \quad i \leq j \quad \text{and} \quad \left[\mathbf{B}^{\intercal}\mathbf{V}\mathbf{B}\right]_{i,j} = \left[\mathbf{B}^{\intercal}\mathbf{V}\mathbf{B}\right]_{j,i}$$

hence

$$\begin{aligned}
\mathbb{E}_{\mathbf{B},\mathbf{V}}\left[\mathbf{B}^{\intercal}\mathbf{V}\mathbf{B}\right]_{i,j} &= \mathbb{E}_{\mathbf{B},\mathbf{V}}\left[\sum_{k=j+1}^{d}\beta_{k,i}\beta_{k,j}\nu_k\right] \\
&= \sum_{k=j+1}^{d}\left(\mu_{q(\beta_{k,i})}\mu_{q(\beta_{k,j})} + \mathrm{Cov}(\beta_{k,i},\beta_{k,j})\right)\mu_{q(\nu_k)} \\
&= \sum_{k=j+1}^{d}\mu_{q(\beta_{k,i})}\mu_{q(\beta_{k,j})}\mu_{q(\nu_k)} + \sum_{k=j+1}^{d}\mathrm{Cov}\left(\beta_{k,i},\beta_{k,j}\right)\mu_{q(\nu_k)} \\
&= \left[\boldsymbol{\mu}_{q(\mathbf{B}^{\intercal})}\boldsymbol{\mu}_{q(\mathbf{V})}\boldsymbol{\mu}_{q(\mathbf{B})}\right]_{i,j} + \sum_{k=j+1}^{d}\mathrm{Cov}\left(\beta_{k,i},\beta_{k,j}\right))\mu_{q(\nu_k)}.
\end{aligned}$$

Thus, each element of $\mathbf{C}_{\vartheta}$ is given by

$$\left[\mathbf{C}_{\vartheta}\right]_{i,j} = \sum_{k=j+1}^{d}\mathrm{Cov}(\beta_{k,i},\beta_{k,j})\mu_{q(\nu_k)} = \left[\mathbf{C}_{\vartheta}\right]_{j,i}.$$

Take the exponential of the $\log q^*(\boldsymbol{\vartheta})$ derived above and notice that it coincides with the kernel of a gaussian random variable $\mathsf{N}_{d(d+p+1)}(\boldsymbol{\mu}_{q(\vartheta)}, \boldsymbol{\Sigma}_{q(\vartheta)})$, as defined in Proposition B.1.3. □

Proposition B.1.4. *The optimal variational density for the parameter $\boldsymbol{\vartheta}_j$ is equal to a multivariate gaussian $q^*(\boldsymbol{\vartheta}_j) \equiv \mathsf{N}_{d+p+1}(\boldsymbol{\mu}_{q(\vartheta_j)}, \boldsymbol{\Sigma}_{q(\vartheta_j)})$, where, for each row $j = 1, \ldots, d$ of $\boldsymbol{\Theta}$:*

$$\begin{aligned}
\boldsymbol{\Sigma}_{q(\vartheta_j)} &= \left(\boldsymbol{\mu}_{q(\omega_{j,j})} \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal} + 1/\upsilon \mathbf{I}_{d+p+1} \right)^{-1}, \\
\boldsymbol{\mu}_{q(\vartheta_j)} &= \boldsymbol{\Sigma}_{q(\vartheta_j)} \left(\sum_{t=1}^{T} \left(\boldsymbol{\mu}_{q(\omega_j)} \otimes \mathbf{z}_{t-1} \right) \mathbf{y}_t - \left(\boldsymbol{\mu}_{q(\omega_{j,-j})} \otimes \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal} \right) \boldsymbol{\mu}_{q(\vartheta_{-j})} \right).
\end{aligned} \tag{B.4}$$

Under this setting the vector $\mathbf{k}_{\vartheta,t}$ computed for $q^*(\nu_j)$ and $q^*(\boldsymbol{\beta}_j)$ is a null vector since the independence among rows of $\boldsymbol{\Theta}$ is assumed.

Proof. Consider the setting as in Proposition B.1.3, define $\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} = \mathbb{E}_{-\vartheta}\left[(\mathbf{I}_d - \mathbf{B})^{\intercal}\mathbf{V}(\mathbf{I}_d - \mathbf{B})\right]$ the expectation of the precision matrix and compute the optimal variational density for the parameter $\boldsymbol{\vartheta}_j$ as $\log q^*(\boldsymbol{\vartheta}_j) \propto \mathbb{E}_{-\vartheta_j}\left[\ell(\boldsymbol{\xi}; \mathbf{y}, \mathbf{x}) + \log p(\boldsymbol{\vartheta}_j)\right]$:

$$\begin{aligned}
\log q^*(\boldsymbol{\vartheta}_j) \propto & -\frac{1}{2}\mathbb{E}_{-\boldsymbol{\vartheta}_j}\left[\boldsymbol{\vartheta}\right]^{\intercal} \left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} \otimes \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal} \right) \mathbb{E}_{-\boldsymbol{\vartheta}_j}\left[\boldsymbol{\vartheta}\right] - \frac{1}{2\upsilon}\boldsymbol{\vartheta}_j^{\intercal}\boldsymbol{\vartheta}_j \\
& + \mathbb{E}_{-\boldsymbol{\vartheta}_j}\left[\boldsymbol{\vartheta}\right]^{\intercal} \sum_{t=1}^{T} \left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} \otimes \mathbf{z}_{t-1} \right) \mathbf{y}_t \\
\propto & -\frac{1}{2}\boldsymbol{\vartheta}_j^{\intercal} \left(\boldsymbol{\mu}_{q(\omega_{j,j})} \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal} \right) \boldsymbol{\vartheta}_j - \frac{1}{2\upsilon}\boldsymbol{\vartheta}_j^{\intercal}\boldsymbol{\vartheta}_j \\
& + \boldsymbol{\vartheta}_j^{\intercal} \sum_{t=1}^{T} \left(\boldsymbol{\mu}_{q(\omega_j)} \otimes \mathbf{z}_{t-1} \right) \mathbf{y}_t - \boldsymbol{\vartheta}_j^{\intercal} \left(\boldsymbol{\mu}_{q(\omega_{j,-j})} \otimes \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal} \right) \boldsymbol{\mu}_{q(\vartheta_{-j})}.
\end{aligned}$$

Where we used the following partitions:

$$\boldsymbol{\vartheta} = \begin{pmatrix} \boldsymbol{\vartheta}_j \\ \boldsymbol{\vartheta}_{-j} \end{pmatrix}, \qquad \boldsymbol{\Omega} = \begin{pmatrix} \omega_{j,j} & \boldsymbol{\omega}_{j,-j} \\ \boldsymbol{\omega}_{-j,j} & \boldsymbol{\Omega}_{-j,-j} \end{pmatrix},$$

and we denote with $\boldsymbol{\omega}_j$ the j-th row of $\boldsymbol{\Omega}$. Re-arrange the terms, take the exponential of the $\log q^*(\boldsymbol{\vartheta}_j)$ derived above and notice that it coincides with the kernel of a gaussian random variable $\mathsf{N}_{d+p+1}(\boldsymbol{\mu}_{q(\vartheta_j)}, \boldsymbol{\Sigma}_{q(\vartheta_j)})$, as defined in Proposition B.1.4. □

Proposition B.1.5. *The variational lower bound for the non-sparse multivariate regression*

model can be derived analytically and it is equal to:

$$\log \underline{p}(\mathbf{y}; q) = d\left(-\frac{T}{2}\log 2\pi + a_\nu \log b_\nu - \log\Gamma(a_\nu)\right) - \sum_{j=1}^{d}\left(a_{q(\nu_j)}\log b_{q(\nu_j)} - \log\Gamma(a_{q(\nu_j)})\right)$$
$$- \frac{1}{2}\sum_{j=2}^{d}\sum_{k=1}^{j-1}\left(\log\tau + 1/\tau\mu_{q(\beta_{j,k}^2)}\right) + \frac{1}{2}\sum_{j=2}^{d}\left(\log|\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}| + (j-1)\right)$$
$$- \frac{1}{2}\sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\left(\log\upsilon + 1/\upsilon\mu_{q(\vartheta_{j,k}^2)}\right) + \frac{1}{2}\left(\log|\boldsymbol{\Sigma}_{q(\boldsymbol{\vartheta})}| + d(d+p+1)\right). \tag{B.5}$$

Proof. First of all, notice that the lower bound can be written in terms of expected values with respect to the density q as:

$$\log \underline{p}(\mathbf{y}; q) = \int q(\boldsymbol{\xi})\log\frac{p(\boldsymbol{\xi}, \mathbf{y})}{q(\boldsymbol{\xi})}\, d\boldsymbol{\xi} = \mathbb{E}_q\left[\log p(\boldsymbol{\xi}, \mathbf{y})\right] - \mathbb{E}_q\left[\log q(\boldsymbol{\xi})\right],$$

where $\log p(\boldsymbol{\xi}, \mathbf{y}) = \ell(\boldsymbol{\xi}; \mathbf{y}) + \log p(\boldsymbol{\xi})$. Following our model specification, we have that

$$\log p(\boldsymbol{\xi}, \mathbf{y}) = \sum_{j=1}^{d}\left(\ell_j(\boldsymbol{\xi}; \mathbf{y}, \mathbf{x}) + \log p(\nu_j)\right) + \sum_{j=2}^{d}\log p(\boldsymbol{\beta}_j) + \log p(\boldsymbol{\vartheta}),$$

where $\ell_j(\boldsymbol{\vartheta}; \mathbf{y}, \mathbf{x})$ denotes the log-likelihood for the j-th variable:

$$\ell_j(\boldsymbol{\xi}; \mathbf{y}, \mathbf{x}) = -\frac{T}{2}\log 2\pi + \frac{T}{2}\log\nu_j - \frac{\nu_j}{2}\sum_{t=1}^{T}\left(y_{j,t} - \boldsymbol{\beta}_j\mathbf{r}_{j,t} - \boldsymbol{\vartheta}_j\mathbf{z}_{t-1}\right)^2.$$

Similarly for the variational density we have:

$$\log q(\boldsymbol{\xi}) = \sum_{j=1}^{d}\log q(\nu_j) + \sum_{j=2}^{d}\log q(\boldsymbol{\beta}_j) + \log q(\boldsymbol{\vartheta}),$$

and the lower bound can be divided into terms referring to each parameter:

$$
\begin{aligned}
\log \underline{p}(\mathbf{y}; q) &= \sum_{j=1}^{d} \mathbb{E}_q \left[\ell_j(\boldsymbol{\xi}; \mathbf{y}, \mathbf{x}) + \log p(\nu_j) - \log q(\nu_j)\right] \\
&\quad + \sum_{j=2}^{d} \mathbb{E}_q \left[\log p(\boldsymbol{\beta}_j) - \log q(\boldsymbol{\beta}_j)\right] + \mathbb{E}_q \left[\log p(\boldsymbol{\vartheta}) - \log q(\boldsymbol{\vartheta})\right] \\
&= \sum_{j=1}^{d} \Big(\underbrace{\mathbb{E}_q \left[\ell_j(\boldsymbol{\xi}; \mathbf{y}, \mathbf{x}) + \log \underline{p}(\mathbf{y}; \nu_j)\right]}_{A} + \sum_{j=2}^{d} \underbrace{\mathbb{E}_q \left[\log \underline{p}(\mathbf{y}; \boldsymbol{\beta}_j)\right]}_{B} + \underbrace{\mathbb{E}_q \left[\log \underline{p}(\mathbf{y}; \boldsymbol{\vartheta})\right]}_{C},
\end{aligned}
\tag{B.6}
$$

thus our strategy will be to evaluate each piece in the latter separately and then put the results together. The first part of the lower bound we compute is $A = \ell_j(\boldsymbol{\xi}; \mathbf{y}, \mathbf{x}) + \log \underline{p}(\mathbf{y}; \nu_j)$:

$$
\begin{aligned}
A &= \mathbb{E}_q \left[-\frac{T}{2} \log 2\pi + \frac{T}{2} \log \nu_j - \frac{\nu_j}{2} \sum_{t=1}^{T} \left(y_{j,t} - \boldsymbol{\beta}_j \mathbf{r}_{j,t} - \boldsymbol{\vartheta}_j \mathbf{z}_{t-1}\right)^2 \right] \\
&\quad + \mathbb{E}_q \left[a_\nu \log b_\nu - \log \Gamma(a_\nu) + (a_\nu - 1) \log \nu_j - \nu_j b_\nu\right] \\
&\quad - \mathbb{E}_q \left[a_{q(\nu_j)} \log b_{q(\nu_j)} - \log \Gamma(a_{q(\nu_j)}) + (a_{q(\nu_j)} - 1) \log \nu_j - \nu_j b_{q(\nu_j)}\right] \\
&= -\frac{T}{2} \log 2\pi + \frac{T}{2} \mu_{q(\log \nu_j)} - \frac{\mu_{q(\nu_j)}}{2} \sum_{t=1}^{T} \mathbb{E}_q \left[\varepsilon_{j,t}^2\right] \\
&\quad + a_\nu \log b_\nu - \log \Gamma(a_\nu) + (a_\nu - 1) \mu_{q(\log \nu_j)} - \mu_{q(\nu_j)} b_\nu \\
&\quad - a_{q(\nu_j)} \log b_{q(\nu_j)} + \log \Gamma(a_{q(\nu_j)}) - (a_{q(\nu_j)} - 1) \mu_{q(\log \nu_j)} + \mu_{q(\nu_j)} b_{q(\nu_j)} \\
&= -\frac{T}{2} \log 2\pi + a_\nu \log b_\nu - \log \Gamma(a_\nu) - a_{q(\nu_j)} \log b_{q(\nu_j)} + \log \Gamma(a_{q(\nu_j)}),
\end{aligned}
$$

where we exploit the definitions of $\mathbb{E}_q \left[\varepsilon_{j,t}^2\right], a_{q(\nu_j)}, b_{q(\nu_j)}$ given in Proposition B.1.1. The second term to compute is equal to:

$$
\begin{aligned}
B &= \mathbb{E}_q \left[-\frac{j-1}{2} \log 2\pi - \frac{1}{2} \sum_{k=1}^{j-1} \log \tau - \frac{1}{2\tau} \sum_{k=1}^{j-1} \beta_{j,k}^2 \right] \\
&\quad - \mathbb{E}_q \Bigg[-\frac{j-1}{2} \log 2\pi - \frac{1}{2} \log |\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}| - \frac{1}{2} \overbrace{(\boldsymbol{\beta}_j - \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)}) \boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}^{-1} (\boldsymbol{\beta}_j - \boldsymbol{\mu}_{q(\boldsymbol{\beta}_j)})^\intercal}^{\text{See Result 3}} \Bigg] \\
&= -\frac{1}{2} \sum_{k=1}^{j-1} \log \tau - \frac{1}{2\tau} \sum_{k=1}^{j-1} \mu_{q(\beta_{j,k}^2)} + \frac{1}{2} \log |\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}| + \frac{j-1}{2},
\end{aligned}
$$

where $\mu_{q(\beta_{j,k}^2)} = \mu_{q(\beta_{j,k})}^2 + \sigma_{q(\beta_{j,k})}^2$ and $\sigma_{q(\beta_{j,k})}^2$ denotes the k-th element on the diagonal of

$\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}$. To conclude, we compute the last term:

$$
\begin{aligned}
C = \mathbb{E}_q &\left[-\frac{d(d+p+1)}{2}\log 2\pi - \frac{1}{2}\sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\log \upsilon - \frac{1}{2\upsilon}\sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\vartheta_{j,k}^2 \right] \\
&- \mathbb{E}_q\left[-\frac{d(d+p+1)}{2}\log 2\pi - \frac{1}{2}\log|\boldsymbol{\Sigma}_{q(\vartheta)}| - \frac{1}{2}\overbrace{(\boldsymbol{\vartheta}-\boldsymbol{\mu}_{q(\vartheta)})^{\intercal}\boldsymbol{\Sigma}_{q(\vartheta)}^{-1}(\boldsymbol{\vartheta}-\boldsymbol{\mu}_{q(\vartheta)})}^{\text{See Result 3}}\right] \\
&= -\frac{1}{2}\sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\log \upsilon - \frac{1}{2\upsilon}\sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\mu_{q(\vartheta_{j,k}^2)} + \frac{1}{2}\log|\boldsymbol{\Sigma}_{q(\vartheta)}| + \frac{d(d+p+1)}{2}.
\end{aligned}
$$

Put together the terms A, B, C as in (B.6) and notice that the variational lower bound here computed coincides with the one presented in Proposition B.1.5. □

The moments of the optimal variational densities are updated at each iteration of the Algorithm 1 and the convergence is assessed by checking the variation both in the lower bound and the parameters.

Algorithm 1: MFVB with non-informative prior.

Initialize: $q^*(\boldsymbol{\xi})$, Δ_ξ, Δ_{ELBO}
while $\left(\widehat{\Delta}_{ELBO} > \Delta_{ELBO}\right) \vee \left(\widehat{\Delta}_\xi > \Delta_\xi\right)$ **do**
  Update $q^*(\nu_1)$ as in (B.1);
  for $j = 2, \ldots, d$ **do**
    Update $q^*(\nu_j)$ and $q^*(\boldsymbol{\beta}_j)$ as in (B.1)-(B.2);
  end
  Update $q^*(\boldsymbol{\vartheta})$ as in (B.3) or (B.4);
  Compute $\log \underline{p}\,(\mathbf{y}; q)$ as in (B.5);
  Compute $\widehat{\Delta}_{\text{ELBO}} = \log \underline{p}\,(\mathbf{y}; q)^{(\mathsf{iter})} - \log \underline{p}\,(\mathbf{y}; q)^{(\mathsf{iter}-1)}$;
  Compute $\widehat{\Delta}_\xi = q^*(\boldsymbol{\xi})^{(\mathsf{iter})} - q^*(\boldsymbol{\xi})^{(\mathsf{iter}-1)}$;
end

B.2 Adaptive Bayesian lasso

In order to induce shrinkage towards zero in the estimates of the coefficients $\boldsymbol{\vartheta}$, we assume an adaptive lasso prior. Notice that the optimal densities for the variances ν_j and for the cholesky factor rows $\boldsymbol{\beta}_j$ remain exactly the same computed in Section B.1. The changes in the optimal densities $q^*(\boldsymbol{\vartheta})$ consist in the fact that now the prior variances are no more fixed, but random variables themselves.

Proposition B.2.1. *The joint optimal variational density for the parameter $\boldsymbol{\vartheta}$ is equal to*

$q^*(\boldsymbol{\vartheta}) \equiv \mathsf{N}_{d(d+p+1)}(\boldsymbol{\mu}_{q(\vartheta)}, \boldsymbol{\Sigma}_{q(\vartheta)})$, *where:*

$$\boldsymbol{\Sigma}_{q(\vartheta)} = \left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} \otimes \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal} + \mathsf{diag}(\boldsymbol{\mu}_{q(1/\upsilon)})\right)^{-1}, \qquad \boldsymbol{\mu}_{q(\vartheta)} = \boldsymbol{\Sigma}_{q(\vartheta)} \sum_{t=1}^{T} \left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} \otimes \mathbf{z}_{t-1}\right) \mathbf{y}_t, \tag{B.7}$$

where $diag(\boldsymbol{\mu}_{q(1/\upsilon)})$ *is a diagonal matrix where* $\boldsymbol{\mu}_{q(1/\upsilon)} = (\mu_{q(1/\upsilon_{1,1})}, \mu_{q(1/\upsilon_{1,2})}, \ldots, \mu_{q(1/\upsilon_{d,d+p+1})})$.

Under the row-independence assumption, the optimal variational density for the parameter $\boldsymbol{\vartheta}_j$ *is equal to* $q^*(\boldsymbol{\vartheta}_j) \equiv \mathsf{N}_{d+p+1}(\boldsymbol{\mu}_{q(\vartheta_j)}, \boldsymbol{\Sigma}_{q(\vartheta_j)})$, *where:*

$$\begin{aligned} \boldsymbol{\Sigma}_{q(\vartheta_j)} &= \left(\boldsymbol{\mu}_{q(\omega_{j,j})} \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal} + diag(\boldsymbol{\mu}_{q(1/\upsilon_j)})\right)^{-1}, \\ \boldsymbol{\mu}_{q(\vartheta_j)} &= \boldsymbol{\Sigma}_{q(\vartheta_j)} \left(\sum_{t=1}^{T} \left(\boldsymbol{\mu}_{q(\omega_j)} \otimes \mathbf{z}_{t-1}\right) \mathbf{y}_t - \left(\boldsymbol{\mu}_{q(\omega_{j,-j})} \otimes \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal}\right) \boldsymbol{\mu}_{q(\vartheta_{-j})}\right), \end{aligned} \tag{B.8}$$

where $diag(\boldsymbol{\mu}_{q(1/\upsilon_j)})$ *is a diagonal matrix where* $\boldsymbol{\mu}_{q(1/\upsilon_j)} = (\mu_{q(1/\upsilon_{j,1})}, \mu_{q(1/\upsilon_{j,2})}, \ldots, \mu_{q(1/\upsilon_{j,d+p+1})})$. Hereafter we describe the optimal densities for the parameters used in hierarchical specification of the prior here assumed.

Proposition B.2.2. *The optimal density for the prior variance* $1/\upsilon_{j,k}$ *is equal to an inverse gaussian distribution* $q^*(1/\upsilon_{j,k}) \equiv \mathsf{IG}(a_{q(1/\upsilon_{j,k})}, b_{q(1/\upsilon_{j,k})})$, *where, for each* $j = 1, \ldots, d$ *and* $k = 1, \ldots, d+p+1$*:*

$$a_{q(1/\upsilon_{j,k})} = \mu_{q(\vartheta_{j,k}^2)}, \quad b_{q(1/\upsilon_{j,k})} = \mu_{q(\lambda_{j,k}^2)}. \tag{B.9}$$

Moreover, it is useful to know that

$$\mu_{q(1/\upsilon_{j,k})} = \sqrt{b_{q(1/\upsilon_{j,k})}/a_{q(1/\upsilon_{j,k})}}, \quad \mu_{q(\upsilon_{j,k})} = \sqrt{a_{q(1/\upsilon_{j,k})}/b_{q(1/\upsilon_{j,k})}} + 1/b_{q(1/\upsilon_{j,k})}.$$

Proof. Consider the prior specification which involves the parameter $\upsilon_{j,k}$:

$$\vartheta_{j,k}|\upsilon_{j,k} \sim \mathsf{N}(0, \upsilon_{j,k}), \qquad \upsilon_{j,k}|\lambda_{j,k}^2 \sim \mathsf{Exp}\left(\lambda_{j,k}^2/2\right).$$

Compute the optimal variational density $\log q^*(\upsilon_{j,k}) \propto \mathbb{E}_{-\upsilon_{\mathbf{j},\mathbf{k}}}\left[\log p(\vartheta_{j,k}) + \log p(\upsilon_{j,k})\right]$:

$$\begin{aligned} \log q^*(\upsilon_{j,k}) &\propto \mathbb{E}_{-\upsilon_{j,k}}\left[-\frac{1}{2}\log \upsilon_{j,k} - \frac{1}{2\upsilon_{j,k}}\vartheta_{j,k}^2 - \upsilon_{j,k}\frac{\lambda_{j,k}^2}{2}\right] \\ &\propto -1/2 \log \upsilon_{j,k} - \frac{1}{2\upsilon_{j,k}}\mu_{q(\vartheta_{j,k}^2)} - \upsilon_{j,k}\frac{\mu_{q(\lambda_{j,k}^2)}}{2}, \end{aligned}$$

and, as a consequence, we obtain:

$$\log q^*(1/\upsilon_{j,k}) \propto -3/2\log(1/\upsilon_{j,k}) - \frac{1}{2}(1/\upsilon_{j,k})\mu_{q(\vartheta^2_{j,k})} - \frac{\mu_{q(\lambda^2_{j,k})}}{2(1/\upsilon_{j,k})}.$$

Take the exponential and notice that the latter is the kernel of an inverse gaussian random variable $\mathsf{IG}(a_{q(1/\upsilon_{j,k})}, b_{q(1/\upsilon_{j,k})})$, as defined in Proposition B.2.2. □

Proposition B.2.3. *The optimal density for the latent parameter $\lambda^2_{j,k}$ for $j = 1, \ldots, d$ and $k = 1, \ldots, d+p+1$ is equal to a $q^*(\lambda^2_{j,k}) \equiv \mathsf{Ga}(a_{q(\lambda^2_{j,k})}, b_{q(\lambda^2_{j,k})})$, where:*

$$a_{q(\lambda^2_{j,k})} = h_1 + 1, \quad b_{q(\lambda^2_{j,k})} = \mu_{q(\upsilon_{j,k})}/2 + h_2. \tag{B.10}$$

Proof. Consider the prior specification which involves the parameter $\lambda^2_{j,k}$:

$$\upsilon_{j,k}|\lambda^2_{j,k} \sim \mathsf{Exp}\left(\lambda^2_{j,k}/2\right), \quad \lambda^2_{j,k} \sim \mathsf{Ga}(h_1, h_2).$$

Compute the optimal variational density as $\log q^*(\lambda^2_{j,k}) \propto \mathbb{E}_{-\lambda^2_{\mathbf{j},\mathbf{k}}}\left[\log p(\upsilon_{j,k}) + \log p(\lambda^2_{j,k})\right]$:

$$\begin{aligned}\log q^*(\lambda^2_{j,k}) &\propto \mathbb{E}_{-\lambda^2_{j,k}}\left[h_1 \log \lambda^2_{j,k} - \lambda^2_{j,k}\left(\upsilon_{j,k}/2 + h_2\right)\right] \\ &\propto h_1 \log \lambda^2_{j,k} - \lambda^2_{j,k}\left(\mu_{q(\upsilon_{j,k})}/2 + h_2\right),\end{aligned}$$

then take the exponential and notice that the latter is the kernel of a gamma random variable $\mathsf{Ga}(a_{q(\lambda^2_{j,k})}, b_{q(\lambda^2_{j,k})})$, as defined in Proposition B.2.3. □

Proposition B.2.4. *The variational lower bound for the multivariate regression model with adaptive Bayesian lasso prior can be derived analytically and it is equal to:*

$$\begin{aligned}\log \underline{p}(\mathbf{y}; q) = d\left(-\frac{T}{2}\log 2\pi + a_\nu \log b_\nu - \log\Gamma(a_\nu)\right) - \sum_{j=1}^{d}\left(a_{q(\nu_j)}\log b_{q(\nu_j)} - \log\Gamma(a_{q(\nu_j)})\right) \\ - \frac{1}{2}\sum_{j=2}^{d}\sum_{k=1}^{j-1}\left(\log\tau + 1/\tau\mu_{q(\beta^2_{j,k})}\right) + \frac{1}{2}\sum_{j=2}^{d}\left(\log|\mathbf{\Sigma}_{q(\boldsymbol{\beta}_j)}| + (j-1)\right) \\ + \frac{1}{2}\left(\log|\mathbf{\Sigma}_{q(\boldsymbol{\vartheta})}| + d(d+p+1)\right) + \sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\frac{1}{2}\mu_{q(\lambda^2_{j,k})}\mu_{q(\upsilon_{j,k})} \\ - \sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\left(1/4\log(b_{q(1/\upsilon_{j,k})}/a_{q(1/\upsilon_{j,k})}) - \log K_{1/2}(\sqrt{b_{q(1/\upsilon_{j,k})}a_{q(1/\upsilon_{j,k})}})\right) \\ + d(d+p+1)\left(h_1\log h_2 - \log\Gamma(h_1)\right) - \sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\left(a_{q(\lambda^2_{j,k})}\log b_{q(\lambda^2_{j,k})} - \log\Gamma(a_{q(\lambda^2_{j,k})})\right).\end{aligned} \tag{B.11}$$

Proof. As we did in (B.6) for Proposition B.1.5, the lower bound can be divided into terms referring to each parameter:

$$\log \underline{p}(\mathbf{y}; q) = A + \sum_{j=1}^{d} \sum_{k=1}^{d+p+1} \Bigg(\underbrace{\mathbb{E}_q \left[\log \underline{p}(\mathbf{y}; \upsilon_{j,k}) \right]}_{B} + \underbrace{\mathbb{E}_q \left[\log \underline{p}(\mathbf{y}; \lambda_{j,k}^2) \right]}_{C} \Bigg),$$

where A is equal to (B.6) in the previous non-informative model specification. Our strategy will be to evaluate each piece in the latter separately and then put the results together. Notice that the computations for the piece A are already available from Proposition B.1.5 and they are equal to the lower bound for the model with the non-informative prior where we still have to take the expectations with respect to the latent parameters $\upsilon_{j,k}$. Thus, we have that:

$$\begin{aligned} A = d \left(-\frac{T}{2} \log 2\pi + a_\nu \log b_\nu - \log \Gamma(a_\nu) \right) - \sum_{j=1}^{d} \left(a_{q(\nu_j)} \log b_{q(\nu_j)} - \log \Gamma(a_{q(\nu_j)}) \right) \\ - \frac{1}{2} \sum_{j=2}^{d} \sum_{k=1}^{j-1} \left(\log \tau + 1/\tau \mu_{q(\beta_{j,k}^2)} \right) + \frac{1}{2} \sum_{j=2}^{d} \left(\log |\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}| + (j-1) \right) \\ - \frac{1}{2} \sum_{j=1}^{d} \sum_{k=1}^{d+p+1} \left(\mu_{q(\log \upsilon_{j,k})} + \mu_{q(1/\upsilon_{j,k})} \mu_{q(\vartheta_{j,k}^2)} \right) + \frac{1}{2} \left(\log |\boldsymbol{\Sigma}_{q(\boldsymbol{\vartheta})}| + d(d+p+1) \right). \end{aligned}$$

Consider now the piece B and recall that, since $q^*(1/\upsilon_{j,k}) \equiv \mathsf{IG}(a_{q(\upsilon_{j,k})}, b_{q(\upsilon_{j,k})})$, then its inverse follows $q^*(\upsilon_{j,k}) \equiv \mathsf{GIG}(1/2, b_{q(1/\upsilon_{j,k})}, a_{q(1/\upsilon_{j,k})})$. We have that

$$\begin{aligned} B &= \mathbb{E}_q \left[\log \lambda_{j,k}^2 - \log 2 - \upsilon_{j,k} \frac{\lambda_{j,k}^2}{2} \right] \\ &\quad - \mathbb{E}_q \left[h(1/2, b_{q(1/\upsilon_{j,k})}, a_{q(1/\upsilon_{j,k})}) - 1/2 \log \upsilon_{j,k} - \frac{1}{2} \left(b_{q(1/\upsilon_{j,k})} \upsilon_{j,k} + \frac{a_{q(1/\upsilon_{j,k})}}{\upsilon_{j,k}} \right) \right] \\ &= \mu_{q(\log \lambda_{j,k}^2)} - \log 2 - h(1/2, b_{q(1/\upsilon_{j,k})}, b_{q(1/\upsilon_{j,k})}) + 1/2 \mu_{q(\log \upsilon_{j,k})} \\ &\quad - \frac{1}{2} \left(\mu_{q(\upsilon_{j,k})} \mu_{q(\lambda_{j,k}^2)} - b_{q(1/\upsilon_{j,k})} \mu_{q(\upsilon_{j,k})} - a_{q(1/\upsilon_{j,k})} \mu_{q(1/\upsilon_{j,k})} \right), \end{aligned}$$

where $h(\zeta, a, b)$ denotes the logarithm of the normalizing constant of a GIG distribution, i.e.

$$h(\zeta, a, b) = \zeta/2 \log(a/b) - \log 2 - \log K_\zeta(\sqrt{ab}).$$

The term involving $\lambda^2_{j,k}$, for $j = 1, \dots, d$ and $k = 1, \dots, d+p+1$, is equal to:

$$
\begin{aligned}
C &= \mathbb{E}_q \left[h_1 \log h_2 - \log \Gamma(h_1) + (h_1 - 1) \log \lambda^2_{j,k} - \lambda^2_{j,k} h_2 \right] \\
&\quad - \mathbb{E}_q \left[a_{q(\lambda^2_{j,k})} \log b_{q(\lambda^2_{j,k})} - \log \Gamma(a_{q(\lambda^2_{j,k})}) + (a_{q(\lambda^2_{j,k})} - 1) \log \lambda^2_{j,k} - \lambda^2_{j,k} b_{q(\lambda^2_{j,k})} \right] \\
&= h_1 \log h_2 - \log \Gamma(h_1) + (h_1 - 1) \mu_{q(\log \lambda^2_{j,k})} - \mu_{q(\lambda^2_{j,k})} h_2 \\
&\quad - a_{q(\lambda^2_{j,k})} \log b_{q(\lambda^2_{j,k})} + \log \Gamma(a_{q(\lambda^2_{j,k})}) - (a_{q(\lambda^2_{j,k})} - 1) \mu_{q(\log \lambda^2_{j,k})} + \mu_{q(\lambda^2_{j,k})} b_{q(\lambda^2_{j,k})}.
\end{aligned}
$$

Group together the terms and exploit the analytical form of the optimal parameters to perform some simplifications. The remaining terms form the lower bound for a multivariate regression model with adaptive lasso prior. □

The moments of the optimal variational densities are updated at each iteration of the Algorithm 2 and the convergence is assessed by checking the variation both in the lower bound and the parameters.

Algorithm 2: MFVB with Bayesian adaptive lasso prior.

Initialize: $q^*(\boldsymbol{\xi})$, Δ_ξ, Δ_{ELBO}
while $\left(\widehat{\Delta}_{ELBO} > \Delta_{ELBO}\right) \vee \left(\widehat{\Delta}_\xi > \Delta_\xi\right)$ **do**
  Update $q^*(\nu_1)$ as in (B.1);
  for $j = 2, \dots, d$ **do**
    Update $q^*(\nu_j)$ and $q^*(\boldsymbol{\beta}_j)$ as in (B.1)-(B.2);
  end
  Update $q^*(\boldsymbol{\vartheta})$ as in (B.7) or (B.8);
  for $j = 1, \dots, d$ **do**
    for $k = 1, \dots, d+p+1$ **do**
      Update $q^*(\upsilon_{j,k})$, $q^*(\lambda^2_{j,k})$ as in (B.9)-(B.10);
    end
  end
  Compute $\log \underline{p}(\mathbf{y}; q)$ as in (B.11);
  Compute $\widehat{\Delta}_{\text{ELBO}} = \log \underline{p}(\mathbf{y}; q)^{(\mathsf{iter})} - \log \underline{p}(\mathbf{y}; q)^{(\mathsf{iter}-1)}$;
  Compute $\widehat{\Delta}_\xi = q^*(\boldsymbol{\xi})^{(\mathsf{iter})} - q^*(\boldsymbol{\xi})^{(\mathsf{iter}-1)}$;
end

B.3 Adaptive normal-gamma

In order to induce shrinkage towards zero in the estimates of the coefficients, we assume an adaptive normal-gamma prior on $\boldsymbol{\vartheta}$. Notice that the optimal densities for the variances ν_j and for $\boldsymbol{\beta}_j$ remain exactly the same computed in Section B.1. The optimal density $q^*(\boldsymbol{\vartheta})$ has the same structure as the one computed in Proposition (B.2.1) for the lasso prior.

Hereafter we describe the optimal densities for the parameters used in hierarchical specification of the normal-gamma prior.

Proposition B.3.1. *The optimal density for the prior variance $\upsilon_{j,k}$ is equal to a generalized inverse gaussian distribution $q^*(\upsilon_{j,k}) \equiv \mathsf{GIG}(\zeta_{q(\upsilon_{j,k})}, a_{q(\upsilon_{j,k})}, b_{q(\upsilon_{j,k})})$, where, for $j = 1, \ldots, d$ and $k = 1, \ldots, d+p+1$:*

$$\zeta_{q(\upsilon_{j,k})} = \mu_{q(\eta_j)} - 1/2, \quad a_{q(\upsilon_{j,k})} = \mu_{q(\eta_j)}\mu_{q(\lambda_{j,k})}, \quad b_{q(\upsilon_{j,k})} = \mu_{q(\vartheta^2_{j,k})}. \tag{B.12}$$

Moreover, it is useful to know that

$$\begin{aligned}
\mu_{q(\upsilon_{j,k})} &= \frac{\sqrt{b_{q(\upsilon_{j,k})}} K_{\zeta_{q(\upsilon_{j,k})}+1}\left(\sqrt{a_{q(\upsilon_{j,k})} b_{q(\upsilon_{j,k})}}\right)}{\sqrt{a_{q(\upsilon_{j,k})}} K_{\zeta_{q(\upsilon_{j,k})}}\left(\sqrt{a_{q(\upsilon_{j,k})} b_{q(\upsilon_{j,k})}}\right)}, \\
\mu_{q(1/\upsilon_{j,k})} &= \frac{\sqrt{a_{q(\upsilon_{j,k})}} K_{\zeta_{q(\upsilon_{j,k})}+1}\left(\sqrt{a_{q(\upsilon_{j,k})} b_{q(\upsilon_{j,k})}}\right)}{\sqrt{b_{q(\upsilon_{j,k})}} K_{\zeta_{q(\upsilon_{j,k})}}\left(\sqrt{a_{q(\upsilon_{j,k})} b_{q(\upsilon_{j,k})}}\right)} - \frac{2\zeta_{q(\upsilon_{j,k})}}{b_{q(\upsilon_{j,k})}}, \\
\mu_{q(\log \upsilon_{j,k})} &= \log \frac{\sqrt{b_{q(\upsilon_{j,k})}}}{\sqrt{a_{q(\upsilon_{j,k})}}} + \frac{\partial}{\partial \zeta_{q(\upsilon_{j,k})}} \log K_{\zeta_{q(\upsilon_{j,k})}}\left(\sqrt{a_{q(\upsilon_{j,k})} b_{q(\upsilon_{j,k})}}\right),
\end{aligned}$$

where $K_\zeta(\cdot)$ denotes the modified Bessel function of second kind.

Proof. Consider the prior specification which involves the parameter $\upsilon_{j,k}$:

$$\vartheta_{j,k}|\upsilon_{j,k} \sim \mathsf{N}(0, \upsilon_{j,k}), \qquad \upsilon_{j,k}|\eta_j, \lambda_{j,k} \sim \mathsf{Ga}\left(\eta_j, \frac{\eta_j \lambda_{j,k}}{2}\right).$$

Compute the optimal variational density as $\log q^*(\upsilon_{j,k}) \propto \mathbb{E}_{-\upsilon_{\mathbf{j,k}}}[\log p(\vartheta_{j,k}) + \log p(\upsilon_{j,k})]$:

$$\begin{aligned}
\log q^*(\upsilon_{j,k}) &\propto \mathbb{E}_{-\upsilon_{j,k}}\left[-\frac{1}{2}\log \upsilon_{j,k} - \frac{1}{2\upsilon_{j,k}}\beta^2_{j,k} + (\eta_j - 1)\log \upsilon_{j,k} - \upsilon_{j,k}\frac{\eta_j \lambda_{j,k}}{2}\right] \\
&\propto \left(\mu_{q(\eta_j)} - \frac{1}{2} - 1\right)\log \upsilon_{j,k} - \frac{1}{2\upsilon_{j,k}}\mu_{q(\vartheta^2_{j,k})} - \upsilon_{j,k}\frac{\mu_{q(\eta_j)}\mu_{q(\lambda_{j,k})}}{2},
\end{aligned}$$

where $\mu_{q(\vartheta^2_{j,k})} = \sigma^2_{q(\vartheta_{j,k})} + \mu^2_{q(\vartheta_{j,k})}$. Take the exponential and notice that the latter is the kernel of a generalized inverse gaussian random variable $\mathsf{GIG}(\zeta_{q(\upsilon_{j,k})}, a_{q(\upsilon_{j,k})}, b_{q(\upsilon_{j,k})})$, as defined in Proposition B.3.1. □

Proposition B.3.2. *The optimal density for the latent parameter $\lambda_{j,k}$ for $j = 1, \ldots, d$ and $k = 1, \ldots, d+p+1$ is equal to a $q^*(\lambda_{j,k}) \equiv \mathsf{Ga}(a_{q(\lambda_{j,k})}, b_{q(\lambda_{j,k})})$, where:*

$$a_{q(\lambda_{j,k})} = \mu_{q(\eta_j)} + h_1, \quad b_{q(\lambda_{j,k})} = \frac{\mu_{q(\eta_j)}\mu_{q(\upsilon_{j,k})}}{2} + h_2. \tag{B.13}$$

Moreover, it is useful to know that

$$\mu_{q(\lambda_{j,k})} = \frac{a_{q(\lambda_{j,k})}}{b_{q(\lambda_{j,k})}}, \quad \mu_{q(\log \lambda_{j,k})} = -\log b_{q(\lambda_{j,k})} + \frac{\Gamma'(a_{q(\lambda_{j,k})})}{\Gamma(a_{q(\lambda_{j,k})})}.$$

Proof. Consider the prior specification which involves the parameter $\lambda_{j,k}$:

$$\upsilon_{j,k}|\eta_j, \lambda_{j,k} \sim \mathsf{Ga}\left(\eta_j, \frac{\eta_j \lambda_{j,k}}{2}\right), \quad \lambda_{j,k} \sim \mathsf{Ga}(h_1, h_2).$$

Compute the optimal variational density as $\log q^*(\lambda_{j,k}) \propto \mathbb{E}_{-\lambda_{\mathbf{j},\mathbf{k}}}\left[\log p(\upsilon_{j,k}) + \log p(\lambda_{j,k})\right]$:

$$\begin{aligned} \log q^*(\lambda_{j,k}) &\propto \mathbb{E}_{-\lambda_{j,k}}\left[(\eta_j + h_1 - 1)\log \lambda_{j,k} - \lambda_{j,k}\left(\frac{\eta_j \upsilon_{j,k}}{2} + h_2\right)\right] \\ &\propto \left(\mu_{q(\eta_j)} + h_1 - 1\right)\log \lambda_{j,k} - \lambda_{j,k}\left(\frac{\mu_{q(\eta_j)}\mu_{q(\upsilon_{j,k})}}{2} + h_2\right), \end{aligned} \tag{B.14}$$

then take the exponential and notice that the latter is the kernel of a gamma random variable $\mathsf{Ga}(a_{q(\lambda_{j,k})}, b_{q(\lambda_{j,k})})$, as defined in Proposition B.3.2. □

Proposition B.3.3. *The optimal density for the latent parameter η_j for $j = 1, \ldots, d$ is equal to:*

$$q^*(\eta_j) = \frac{h(\eta_j)}{c_{\eta_j}} \exp\left\{-\eta_j \sum_{k=1}^{d+p+1}\left(\frac{\mu_{q(\lambda_{j,k})}\mu_{q(\upsilon_{j,k})}}{2} - \mu_{q(\log \lambda_{j,k})} - \mu_{q(\log \upsilon_{j,k})} + \log 2 + h_3\right)\right\}, \tag{B.15}$$

where $\log h(\eta_j) = (d+p+1)(\eta_j \log \eta_j - \log \Gamma(\eta_j))$ *and*

$$c_{\eta_j} = \int_{\mathbb{R}^+} h(\eta_j) \exp\left\{-\eta_j \sum_{k=1}^{d+p+1}\left(\frac{\mu_{q(\lambda_{j,k})}\mu_{q(\upsilon_{j,k})}}{2} - \mu_{q(\log \lambda_{j,k})} - \mu_{q(\log \upsilon_{j,k})} + (d+p+1)\log 2 + h_3\right)\right\} d\eta_j.$$

Then, we have that $\mu_{q(\eta_j)} = \int_{\mathbb{R}^+} \eta_j q^*(\eta_j)\, d\eta_j$.

Proof. Consider the prior specification which involves the parameter η_j:

$$\upsilon_{j,k}|\eta_j, \lambda_{j,k} \sim \mathsf{Ga}\left(\eta_j, \frac{\eta_j \lambda_{j,k}}{2}\right), \quad \eta_j \sim \mathsf{Exp}(h_3).$$

Compute the optimal variational density as $\log q^*(\eta_j) \propto \mathbb{E}_{-\eta_j}\left[\sum_{k=1}^{d+p+1} \log p(\upsilon_{j,k}) + \log p(\eta_j)\right]$:

$$\begin{aligned}
\log q^*(\eta_j) &\propto \mathbb{E}_{-\eta_j}\left[(d+p+1)\left(\eta_j \log \eta_j - \log \Gamma(\eta_j)\right) - \eta_j \sum_{k=1}^{d+p+1}\left(\left(\frac{\lambda_{j,k}\upsilon_{j,k}}{2} - \log \frac{\lambda_{j,k}\upsilon_{j,k}}{2}\right) + h_3\right)\right] \\
&= (d+p+1)\left(\eta_j \log \eta_j - \log \Gamma(\eta_j)\right) \\
&\quad - \eta_j \sum_{k=1}^{d+p+1}\left(\frac{\mu_{q(\lambda_{j,k})}\mu_{q(\upsilon_{j,k})}}{2} - \mathbb{E}_{\upsilon_{j,k}\lambda_{j,k}}\left[\log \frac{\lambda_{j,k}\upsilon_{j,k}}{2}\right] + h_3\right),
\end{aligned} \tag{B.16}$$

which is not the kernel of a know distribution, but since $\mathbb{E}[\log x] \leq \log \mathbb{E}[x] < \mathbb{E}[x]$, it holds that

$$\frac{\mu_{q(\lambda_{j,k})}\mu_{q(\upsilon_{j,k})}}{2} > \mathbb{E}_{\upsilon_{j,k}\lambda_{j,k}}\left[\log \frac{\lambda_{j,k}\upsilon_{j,k}}{2}\right] = \mu_{q(\log \lambda_{j,k})} + \mu_{q(\log \upsilon_{j,k})} - \log 2,$$

hence the exponential of term in (B.16) is integrable and thus we can compute the normalizing constant and its expectation. □

Proposition B.3.4. *The variational lower bound for the multivariate regression model with adaptive normal-gamma prior can be derived analytically and it is equal to:*

$$\begin{aligned}
\log \underline{p}(\mathbf{y}; q) = d\left(-\frac{T}{2}\log 2\pi + a_\nu \log b_\nu - \log \Gamma(a_\nu)\right) - \sum_{j=1}^{d}\left(a_{q(\nu_j)} \log b_{q(\nu_j)} - \log \Gamma(a_{q(\nu_j)})\right) \\
- \frac{1}{2}\sum_{j=2}^{d}\sum_{k=1}^{j-1}\left(\log \tau + 1/\tau \mu_{q(\beta_{j,k}^2)}\right) + \frac{1}{2}\sum_{j=2}^{d}\left(\log |\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}| + (j-1)\right) \\
+ \frac{1}{2}\left(\log |\boldsymbol{\Sigma}_{q(\boldsymbol{\vartheta})}| + d(d+p+1)\right) - \sum_{j=1}^{d}\sum_{k=1}^{d+p+1} h(\zeta_{q(\upsilon_{j,k})}, a_{q(\upsilon_{j,k})}, b_{q(\upsilon_{j,k})}) \\
+ d(d+p+1)\left(h_1 \log h_2 - \log \Gamma(h_1)\right) - \sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\left(a_{q(\lambda_{j,k})} \log b_{q(\lambda_{j,k})} - \log \Gamma(a_{q(\lambda_{j,k})})\right) \\
+ d \log h_3 + \sum_{j=1}^{d} \log c_{\eta_j} + \sum_{j=1}^{d} \mu_{q(\eta_j)} \sum_{k=1}^{d+p+1}\left(\mu_{q(\lambda_{j,k})}\mu_{q(\upsilon_{j,k})} - \mu_{q(\log \lambda_{j,k})} - \mu_{q(\log \upsilon_{j,k})}\right).
\end{aligned} \tag{B.17}$$

Proof. As we did in (B.6) for Proposition B.1.5, the lower bound can be divided into terms

referring to each parameter:

$$\log \underline{p}(\mathbf{y}; q) = A + \sum_{j=1}^{d} \sum_{k=1}^{d+p+1} \Bigg(\underbrace{\mathbb{E}_q \left[\log \underline{p}(\mathbf{y}; \upsilon_{j,k}) \right]}_{B} + \underbrace{\mathbb{E}_q \left[\log \underline{p}(\mathbf{y}; \lambda_{j,k}) \right]}_{C} + \underbrace{\mathbb{E}_q \left[\log \underline{p}(\mathbf{y}; \eta_j) \right]}_{D} \Bigg), \tag{B.18}$$

where A is equal to (B.2). Our strategy will be to evaluate each piece in the latter separately and then put the results together. Consider the piece B:

$$\begin{aligned}
B &= \mathbb{E}_q \left[\eta_j \log \eta_j + \eta_j \left(\log \lambda_{j,k} - \log 2 \right) - \log \Gamma(\eta_j) + (\eta_j - 1) \log \upsilon_{j,k} - \upsilon_{j,k} \frac{\eta_j \lambda_{j,k}}{2} \right] \\
&\quad - \mathbb{E}_q \left[h(\zeta_{q(\upsilon_{j,k})}, a_{q(\upsilon_{j,k})}, b_{q(\upsilon_{j,k})}) + (\zeta_{q(\upsilon_{j,k})} - 1) \log \upsilon_{j,k} - \frac{a_{q(\upsilon_{j,k})} \upsilon_{j,k}}{2} - \frac{b_{q(\upsilon_{j,k})}}{2 \upsilon_{j,k}} \right] \\
&= \mu_{q(\eta_j \log \eta_j)} + \mu_{q(\eta_j)} \left(\mu_{q(\log \lambda_{j,k})} - \log 2 \right) - \mu_{q(\log \Gamma(\eta_j))} - h(\zeta_{q(\upsilon_{j,k})}, a_{q(\upsilon_{j,k})}, b_{q(\upsilon_{j,k})}) \\
&\quad + (\mu_{q(\eta_j)} - 1) \mu_{q(\log \upsilon_{j,k})} - (\zeta_{q(\upsilon_{j,k})} - 1) \mu_{q(\log \upsilon_{j,k})} \\
&\quad - \frac{1}{2} \left(\mu_{q(\upsilon_{j,k})} \mu_{q(\eta_j)} \mu_{q(\lambda_{j,k})} - a_{q(\upsilon_{j,k})} \mu_{q(\upsilon_{j,k})} - b_{q(\upsilon_{j,k})} \mu_{q(1/\upsilon_{j,k})} \right),
\end{aligned}$$

where $h(\zeta, a, b)$ denotes the logarithm of the normalizing constant of a GIG distribution, i.e.

$$h(\zeta, a, b) = \zeta/2 \log(a/b) - \log 2 - \log K_\zeta(\sqrt{ab}).$$

The term involving $\lambda_{j,k}$, for $j = 1, \ldots, d$ and $k = 1, \ldots, d+p+1$, is equal to:

$$\begin{aligned}
C &= \mathbb{E}_q \left[h_1 \log h_2 - \log \Gamma(h_1) + (h_1 - 1) \log \lambda_{j,k} - \lambda_{j,k} h_2 \right] \\
&\quad - \mathbb{E}_q \left[a_{q(\lambda_{j,k})} \log b_{q(\lambda_{j,k})} - \log \Gamma(a_{q(\lambda_{j,k})}) + (a_{q(\lambda_{j,k})} - 1) \log \lambda_{j,k} - \lambda_{j,k} b_{q(\lambda_{j,k})} \right] \\
&= h_1 \log h_2 - \log \Gamma(h_1) + (h_1 - 1) \mu_{q(\log \lambda_{j,k})} - \mu_{q(\lambda_{j,k})} h_2 \\
&\quad - a_{q(\lambda_{j,k})} \log b_{q(\lambda_{j,k})} + \log \Gamma(a_{q(\lambda_{j,k})}) - (a_{q(\lambda_{j,k})} - 1) \mu_{q(\log \lambda_{j,k})} + \mu_{q(\lambda_{j,k})} b_{q(\lambda_{j,k})},
\end{aligned}$$

and, to conclude, compute the term D:

$$\begin{aligned}
D &= \mathbb{E}_q \left[\log h_3 - \eta_j h_3 \right] \\
&\quad - \mathbb{E}_q \left[\log h(\eta_j) - \log c_{\eta_j} - \eta_j \sum_{k=1}^{d+p+1} \left(\frac{\mu_{q(\lambda_{j,k})} \mu_{q(\upsilon_{j,k})}}{2} - \mu_{q(\log \lambda_{j,k})} - \mu_{q(\log \upsilon_{j,k})} + \log 2 + h_3 \right) \right] \\
&= \log h_3 - \mu_{q(\eta_j)} h_3 \\
&\quad - \mu_{q(\log h(\eta_j))} + \log c_{\eta_j} + \mu_{q(\eta_j)} \sum_{k=1}^{d+p+1} \left(\frac{\mu_{q(\lambda_{j,k})} \mu_{q(\upsilon_{j,k})}}{2} - \mu_{q(\log \lambda_{j,k})} - \mu_{q(\log \upsilon_{j,k})} + \log 2 + h_3 \right).
\end{aligned}$$

Group together the terms and exploit the analytical form of the optimal parameters to

perform some simplifications. The remaining terms form the lower bound for a multivariate regression model with adaptive normal-gamma prior. □

The moments of the optimal variational densities are updated at each iteration of the Algorithm 3 and the convergence is assessed by checking the variation both in the lower bound and the parameters.

Algorithm 3: MFVB with adaptive normal-gamma prior.

Initialize: $q^*(\boldsymbol{\xi})$, Δ_ξ, Δ_{ELBO}
while $\left(\widehat{\Delta}_{ELBO} > \Delta_{ELBO}\right) \vee \left(\widehat{\Delta}_\xi > \Delta_\xi\right)$ **do**
  Update $q^*(\nu_1)$ as in (B.1);
  for $j = 2, \ldots, d$ **do**
    Update $q^*(\nu_j)$ and $q^*(\boldsymbol{\beta}_j)$ as in (B.1)-(B.2);
  end
  Update $q^*(\boldsymbol{\vartheta})$ as in (B.7) or (B.8);
  for $j = 1, \ldots, d$ **do**
    for $k = 1, \ldots, d+p+1$ **do**
      Update $q^*(\upsilon_{j,k})$, $q^*(\lambda_{j,k})$ as in (B.12)-(B.13);
    end
    Update $q^*(\eta_j)$ as in (B.15);
  end
  Compute $\log \underline{p}\,(\mathbf{y}; q)$ as in (B.17);
  Compute $\widehat{\Delta}_{\text{ELBO}} = \log \underline{p}\,(\mathbf{y}; q)^{(\mathsf{iter})} - \log \underline{p}\,(\mathbf{y}; q)^{(\mathsf{iter}-1)}$;
  Compute $\widehat{\Delta}_\xi = q^*(\boldsymbol{\xi})^{(\mathsf{iter})} - q^*(\boldsymbol{\xi})^{(\mathsf{iter}-1)}$;
end

B.4 Horseshoe prior

First of all, notice that the optimal densities for the variances ν_j and for the coefficients $\boldsymbol{\beta}_j$ remain the same computed in Section B.1. The changes in the optimal densities $q^*(\boldsymbol{\vartheta})$ are stated in the next proposition.

Proposition B.4.1. *The joint optimal variational density for the parameter $\boldsymbol{\vartheta}$ is equal to $q^*(\boldsymbol{\vartheta}) \equiv \mathsf{N}_{d(d+p+1)}(\boldsymbol{\mu}_{q(\vartheta)}, \boldsymbol{\Sigma}_{q(\vartheta)})$, where:*

$$\begin{aligned}
\boldsymbol{\Sigma}_{q(\vartheta)} &= \left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} \otimes \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\mathsf{T}} + \boldsymbol{\mu}_{q(1/\gamma^2)}\mathsf{diag}(\boldsymbol{\mu}_{q(1/\upsilon^2)})\right)^{-1}, \\
\boldsymbol{\mu}_{q(\vartheta)} &= \boldsymbol{\Sigma}_{q(\vartheta)} \sum_{t=1}^{T} \left(\boldsymbol{\mu}_{q(\boldsymbol{\Omega})} \otimes \mathbf{z}_{t-1}\right) \mathbf{y}_t,
\end{aligned} \tag{B.19}$$

where $diag(\boldsymbol{\mu}_{q(1/\upsilon^2)})$ is a diagonal matrix and $\boldsymbol{\mu}_{q(1/\upsilon^2)} = (\mu_{q(1/\upsilon_{1,1}^2)}, \mu_{q(1/\upsilon_{1,2}^2)}, \ldots, \mu_{q(1/\upsilon_{d,d+p+1}^2)})$.

Under the row-independence assumption, the optimal variational density for the parameter $\boldsymbol{\vartheta}_j$ is equal to $q^*(\boldsymbol{\vartheta}_j) \equiv \mathsf{N}_{d+p+1}(\boldsymbol{\mu}_{q(\vartheta_j)}, \boldsymbol{\Sigma}_{q(\vartheta_j)})$, where:

$$\begin{aligned}
\boldsymbol{\Sigma}_{q(\vartheta_j)} &= \left(\boldsymbol{\mu}_{q(\omega_{j,j})} \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal} + \boldsymbol{\mu}_{q(1/\gamma^2)} \mathsf{diag}(\boldsymbol{\mu}_{q(1/\upsilon_j^2)})\right)^{-1}, \\
\boldsymbol{\mu}_{q(\vartheta_j)} &= \boldsymbol{\Sigma}_{q(\vartheta_j)} \left(\sum_{t=1}^{T} \left(\boldsymbol{\mu}_{q(\omega_j)} \otimes \mathbf{z}_{t-1}\right) \mathbf{y}_t - \left(\boldsymbol{\mu}_{q(\omega_{j,-j})} \otimes \sum_{t=1}^{T} \mathbf{z}_{t-1}\mathbf{z}_{t-1}^{\intercal}\right) \boldsymbol{\mu}_{q(\vartheta_{-j})}\right),
\end{aligned} \tag{B.20}$$

where diag$(\boldsymbol{\mu}_{q(1/\upsilon_j^2)})$ is a diagonal matrix and $\boldsymbol{\mu}_{q(1/\upsilon_j^2)} = (\mu_{q(1/\upsilon_{j,1}^2)}, \mu_{q(1/\upsilon_{j,2}^2)}, \ldots, \mu_{q(1/\upsilon_{j,d+p+1}^2)})$.

Hereafter we describe the optimal densities for the parameters used in hierarchical specification of the prior.

Proposition B.4.2. *The optimal density for the prior local variance $\upsilon_{j,k}^2$ is equal to an inverse gamma distribution $q^*(\upsilon_{j,k}^2) \equiv \mathsf{InvGa}(1, b_{q(\upsilon_{j,k}^2)})$, where, for $j = 1, \ldots, d$ and $k = 1, \ldots, d+p+1$:*

$$b_{q(\upsilon_{j,k}^2)} = \mu_{q(1/\lambda_{j,k})} + \frac{1}{2}\mu_{q(\vartheta_{j,k}^2)}\mu_{q(1/\gamma^2)}. \tag{B.21}$$

Proof. Consider the prior specification which involves the parameter $\upsilon_{j,k}^2$:

$$\vartheta_{j,k}|\gamma^2, \upsilon_{j,k}^2 \sim \mathsf{N}(0, \gamma^2\upsilon_{j,k}^2), \qquad \upsilon_{j,k}^2|\lambda_{j,k} \sim \mathsf{InvGa}\left(1/2, 1/\lambda_{j,k}\right).$$

Compute the optimal variational density $\log q^*(\upsilon_{j,k}^2) \propto \mathbb{E}_{-\upsilon_{\mathbf{j,k}}^2}\left[\log p(\vartheta_{j,k}) + \log p(\upsilon_{j,k}^2)\right]$:

$$\begin{aligned}
\log q^*(\upsilon_{j,k}^2) &\propto \mathbb{E}_{-\upsilon_{j,k}^2}\left[-\frac{1}{2}\log \upsilon_{j,k}^2 - \frac{1}{2\gamma^2\upsilon_{j,k}^2}\vartheta_{j,k}^2 - (1/2+1)\log \upsilon_{j,k}^2 - \frac{1}{\upsilon_{j,k}^2\lambda_{j,k}}\right] \\
&\propto -2\log \upsilon_{j,k}^2 - \frac{1}{\upsilon_{j,k}^2}\left(\mu_{q(1/\gamma^2)}\mu_{q(\vartheta_{j,k}^2)}/2 + \mu_{q(1/\lambda_{j,k})}\right).
\end{aligned}$$

Take the exponential and notice that the latter is the kernel of an inverse gamma random variable $\mathsf{InvGa}(1, b_{q(\upsilon_{j,k}^2)})$, as defined in Proposition B.4.2. □

Proposition B.4.3. *The optimal density for the prior global variance γ^2 is equal to an inverse gamma distribution $q^*(\gamma^2) \equiv \mathsf{InvGa}(a_{q(\gamma^2)}, b_{q(\gamma^2)})$, where:*

$$a_{q(\gamma^2)} = \frac{d(d+p+1)+1}{2}, \quad b_{q(\gamma^2)} = \mu_{q(1/\eta)} + \frac{1}{2}\sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\mu_{q(1/\upsilon_{j,k}^2)}\mu_{q(\vartheta_{j,k}^2)}. \tag{B.22}$$

Proof. Consider the prior specification which involves the parameter γ^2:

$$\vartheta_{j,k}|\gamma^2, \upsilon_{j,k}^2 \sim \mathsf{N}(0, \gamma^2 \upsilon_{j,k}^2), \qquad \gamma^2|\eta \sim \mathsf{InvGa}\left(1/2, 1/\eta\right).$$

Compute the optimal variational density $\log q^*(\gamma^2) \propto \mathbb{E}_{-\gamma^2}\left[\sum_{j=1}^{d}\sum_{k=1}^{d+p+1} \log p(\vartheta_{j,k}) + \log p(\gamma^2)\right]$:

$$\begin{aligned}\log q^*(\gamma^2) &\propto \mathbb{E}_{-\gamma^2}\left[-\frac{d(d+p+1)}{2}\log\gamma^2 - \frac{1}{2\gamma^2\upsilon_{j,k}^2}\vartheta_{j,k}^2 - (1/2+1)\log\gamma^2 - \frac{1}{\gamma^2\eta}\right] \\ &\propto -\left(\frac{d(d+p+1)+1}{2}+1\right)\log\gamma^2 - \frac{1}{\gamma^2}\left(\sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\mu_{q(1/\upsilon_{j,k}^2)}\mu_{q(\vartheta_{j,k}^2)}/2 + \mu_{q(1/\eta)}\right).\end{aligned}$$

Take the exponential and notice that the latter is the kernel of an inverse gamma random variable $\mathsf{InvGa}(a_{q(\gamma^2)}, b_{q(\gamma^2)})$, as defined in Proposition B.4.3. □

Proposition B.4.4. *The optimal density for the latent parameter $\lambda_{j,k}$ is equal to an inverse gamma distribution $q^*(\lambda_{j,k}) \equiv \mathsf{InvGa}(1, b_{q(\lambda_{j,k})})$, where, for $j = 1, \ldots, d$ and $k = 1, \ldots, d+p+1$:*

$$b_{q(\lambda_{j,k})} = 1 + \mu_{q(1/\upsilon_{j,k}^2)}. \tag{B.23}$$

Proof. Consider the prior specification which involves the parameter $\lambda_{j,k}$:

$$\upsilon_{j,k}^2|\lambda_{j,k} \sim \mathsf{InvGa}\left(1/2, 1/\lambda_{j,k}\right), \qquad \lambda_{j,k} \sim \mathsf{InvGa}\left(1/2, 1\right).$$

Compute the optimal variational density $\log q^*(\lambda_{j,k}) \propto \mathbb{E}_{-\lambda_{\mathbf{j},\mathbf{k}}}\left[\log p(\upsilon_{j,k}^2) + \log p(\lambda_{j,k})\right]$:

$$\begin{aligned}\log q^*(\lambda_{j,k}) &\propto \mathbb{E}_{-\lambda_{j,k}}\left[-\frac{1}{2}\log\lambda_{j,k} - \frac{1}{\upsilon_{j,k}^2\lambda_{j,k}} - (1/2+1)\log\lambda_{j,k} - \frac{1}{\lambda_{j,k}}\right] \\ &\propto -2\log\lambda_{j,k} - \frac{1}{\lambda_{j,k}}\left(1 + \mu_{q(1/\upsilon_{j,k}^2)}\right).\end{aligned}$$

Take the exponential and notice that the latter is the kernel of an inverse gamma random variable $\mathsf{InvGa}(1, b_{q(\lambda_{j,k})})$, as defined in Proposition B.4.4. □

Proposition B.4.5. *The optimal density for the latent parameter η is equal to an inverse gamma distribution $q^*(\eta) \equiv \mathsf{InvGa}(1, b_{q(\eta)})$, where:*

$$b_{q(\eta)} = 1 + \mu_{q(1/\gamma^2)}. \tag{B.24}$$

Proof. Consider the prior specification which involves the parameter η:

$$\gamma^2|\eta \sim \mathsf{InvGa}\left(1/2, 1/\eta\right), \qquad \eta \sim \mathsf{InvGa}\left(1/2, 1\right).$$

Compute the optimal variational density $\log q^*(\eta) \propto \mathbb{E}_{-\eta}\left[\log p(\gamma^2) + \log p(\eta)\right]$:

$$\begin{aligned}\log q^*(\eta) &\propto \mathbb{E}_{-\eta}\left[-\frac{1}{2}\log\eta - \frac{1}{\gamma^2\eta} - (1/2+1)\log\eta - \frac{1}{\eta}\right] \\ &\propto -2\log\eta - \frac{1}{\eta}\left(1 + \mu_{q(1/\gamma^2)}\right).\end{aligned}$$

Take the exponential and notice that the latter is the kernel of an inverse gamma random variable $\mathsf{InvGa}(1, b_{q(\eta)})$, as defined in Proposition B.4.5. □

Proposition B.4.6. *The variational lower bound for the multivariate regression model with Horseshoe prior can be derived analytically and it is equal to:*

$$\begin{aligned}\log \underline{p}(\mathbf{y}; q) = d\left(-\frac{T}{2}\log 2\pi + a_\nu \log b_\nu - \log\Gamma(a_\nu)\right) - \sum_{j=1}^{d}\left(a_{q(\nu_j)}\log b_{q(\nu_j)} - \log\Gamma(a_{q(\nu_j)})\right) \\ - \frac{1}{2}\sum_{j=2}^{d}\sum_{k=1}^{j-1}\left(\log\tau + 1/\tau\mu_{q(\beta_{j,k}^2)}\right) + \frac{1}{2}\sum_{j=2}^{d}\left(\log|\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}| + (j-1)\right) \\ + \frac{1}{2}\left(\log|\boldsymbol{\Sigma}_{q(\boldsymbol{\vartheta})}| + d(d+p+1)\right) + \mu_{q(1/\gamma^2)}\left(\mu_{q(1/\eta)} + \sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\mu_{q(\vartheta_{j,k}^2)}\mu_{q(1/\upsilon_{j,k}^2)}\right) \\ + \sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\left(\mu_{q(1/\upsilon_{j,k}^2)}\mu_{q(1/\lambda_{j,k})} - \log b_{q(\upsilon_{j,k}^2)} - \log b_{q(\lambda_{j,k})} - \log\pi\right) \\ - a_{q(\gamma^2)}\log b_{q(\gamma^2)} - \log b_{q(\eta)} - \log\pi.\end{aligned} \tag{B.25}$$

Proof. As we did in (B.6) for Proposition B.1.5, the lower bound can be divided into terms referring to each parameter:

$$\begin{aligned}\log \underline{p}(\mathbf{y}; q) = A + \underbrace{\mathbb{E}_q\left[\log \underline{p}(\mathbf{y}; \gamma^2)\right]}_{B} + \underbrace{\mathbb{E}_q\left[\log \underline{p}(\mathbf{y}; \eta)\right]}_{C} \\ + \sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\left(\underbrace{\mathbb{E}_q\left[\log \underline{p}(\mathbf{y}; \upsilon_{j,k}^2)\right]}_{D} + \underbrace{\mathbb{E}_q\left[\log \underline{p}(\mathbf{y}; \lambda_{j,k})\right]}_{E}\right),\end{aligned} \tag{B.26}$$

where A is similar to (B.6) in the previous non-informative model specification. Our strategy will be to evaluate each piece in the latter separately and then put the results together. Notice that the computations for the piece A are similar to Proposition B.1.5. Hence, we

have that:

$$
\begin{aligned}
A = d\left(-\frac{T}{2}\log 2\pi + a_\nu \log b_\nu - \log\Gamma(a_\nu)\right) - \sum_{j=1}^{d}\left(a_{q(\nu_j)}\log b_{q(\nu_j)} - \log\Gamma(a_{q(\nu_j)})\right) \\
- \frac{1}{2}\sum_{j=2}^{d}\sum_{k=1}^{j-1}\left(\log\tau + 1/\tau\mu_{q(\beta_{j,k}^2)}\right) + \frac{1}{2}\sum_{j=2}^{d}\left(\log|\boldsymbol{\Sigma}_{q(\boldsymbol{\beta}_j)}| + (j-1)\right) \\
- \frac{1}{2}\sum_{j=1}^{d}\sum_{k=1}^{d+p+1}\left(\mu_{q(\log\delta^2)} + \mu_{q(\log\upsilon_{j,k}^2)} + \mu_{q(1/\delta^2)}\mu_{q(1/\upsilon_{j,k}^2)}\mu_{q(\vartheta_{j,k}^2)}\right) + \frac{1}{2}\left(\log|\boldsymbol{\Sigma}_{q(\boldsymbol{\vartheta})}| + d(d+p+1)\right).
\end{aligned}
\tag{B.27}
$$

Consider now the piece B. We have that:

$$
\begin{aligned}
B &= \mathbb{E}_q\left[-\frac{1}{2}\log\eta - \frac{1}{2}\log\pi - (1/2+1)\log\gamma^2 - 1/(\gamma^2\eta)\right] \\
&\quad - \mathbb{E}_q\left[a_{q(\gamma^2)}\log b_{q(\gamma^2)} - \log\Gamma(a_{q(\gamma^2)}) - (a_{q(\gamma^2)}+1)\log\gamma^2 - b_{q(\gamma^2)}/\gamma^2\right] \\
&= -\frac{1}{2}\mu_{q(\log\eta)} - \frac{1}{2}\log\pi - (1/2+1)\mu_{q(\log\gamma^2)} - \mu_{q(1/\gamma^2)}\mu_{q(1/\eta)} \\
&\quad - a_{q(\gamma^2)}\log b_{q(\gamma^2)} + \log\Gamma(a_{q(\gamma^2)}) + (a_{q(\gamma^2)}+1)\mu_{q(\log\gamma^2)} + \mu_{q(1/\gamma^2)}b_{q(\gamma^2)},
\end{aligned}
$$

while, C reduces to:

$$
\begin{aligned}
C &= \mathbb{E}_q\left[-\frac{1}{2}\log\pi - (1/2+1)\log\eta - 1/\eta\right] - \mathbb{E}_q\left[\log b_{q(\eta)} - 2\log\eta - b_{q(\eta)}/\eta\right] \\
&= -\frac{1}{2}\log\pi - (1/2+1)\mu_{q(\log\eta)} - \mu_{q(1/\eta)} - \log b_{q(\eta)} + 2\mu_{q(\log\eta)} + \mu_{q(1/\eta)}b_{q(\eta)}.
\end{aligned}
$$

The remaining terms behave likely B and C. In particular, for $j = 1,\ldots,d$ and $k = 1,\ldots,d+p+1$:

$$
\begin{aligned}
D &= \mathbb{E}_q\left[-\frac{1}{2}\log\lambda_{j,k} - \frac{1}{2}\log\pi - (1/2+1)\log\upsilon_{j,k}^2 - 1/(\upsilon_{j,k}^2\lambda_{j,k})\right] \\
&\quad - \mathbb{E}_q\left[\log b_{q(\upsilon_{j,k}^2)} - 2\log\upsilon_{j,k}^2 - b_{q(\upsilon_{j,k}^2)}/\upsilon_{j,k}^2\right] \\
&= -\frac{1}{2}\mu_{q(\log\lambda_{j,k})} - \frac{1}{2}\log\pi - (1/2+1)\mu_{q(\log\upsilon_{j,k}^2)} - \mu_{q(1/\upsilon_{j,k}^2)}\mu_{q(1/\lambda_{j,k})} \\
&\quad - \log b_{q(\upsilon_{j,k}^2)} + 2\mu_{q(\log\upsilon_{j,k}^2)} + \mu_{q(1/\upsilon_{j,k}^2)}b_{q(\upsilon_{j,k}^2)},
\end{aligned}
$$

and

$$
\begin{aligned}
E &= \mathbb{E}_q\left[-\frac{1}{2}\log\pi - (1/2+1)\log\lambda_{j,k} - 1/\lambda_{j,k}\right] - \mathbb{E}_q\left[\log b_{q(\lambda_{j,k})} - 2\log\lambda_{j,k} - b_{q(\lambda_{j,k})}/\lambda_{j,k}\right] \\
&= -\frac{1}{2}\log\pi - (1/2+1)\mu_{q(\log\lambda_{j,k})} - \mu_{q(1/\lambda_{j,k})} - \log b_{q(\lambda_{j,k})} + 2\mu_{q(\log\lambda_{j,k})} + \mu_{q(1/\lambda_{j,k})}b_{q(\lambda_{j,k})}.
\end{aligned}
$$

Group together the terms and exploit the analytical form of the optimal parameters to perform some simplifications. The remaining terms form the lower bound for a multivariate regression model with Horseshoe prior. □

The moments of the optimal variational densities are updated at each iteration of the Algorithm 4 and the convergence is assessed by checking the variation both in the lower bound and the parameters.

Algorithm 4: MFVB with Horseshoe prior.

Initialize: $q^*(\boldsymbol{\xi})$, Δ_ξ, Δ_{ELBO}
while $\left(\widehat{\Delta}_{ELBO} > \Delta_{ELBO}\right) \vee \left(\widehat{\Delta}_\xi > \Delta_\xi\right)$ **do**
  Update $q^*(\nu_1)$ as in (B.1);
  for $j = 2, \ldots, d$ **do**
    Update $q^*(\nu_j)$ and $q^*(\boldsymbol{\beta}_j)$ as in (B.1)-(B.2);
  end
  Update $q^*(\boldsymbol{\vartheta})$ as in (B.19) or (B.20) ;
  for $j = 1, \ldots, d$ **do**
    for $k = 1, \ldots, d + p + 1$ **do**
      Update $q^*(\upsilon_{j,k}^2)$, $q^*(\lambda_{j,k})$ as in (B.21)-(B.23);
    end
  end
  Update $q^*(\gamma^2)$, $q^*(\eta)$ as in (B.22)-(B.24);
  Compute $\log \underline{p}\,(\mathbf{y}; q)$ as in (B.25);
  Compute $\widehat{\Delta}_{\text{ELBO}} = \log \underline{p}\,(\mathbf{y}; q)^{(\mathsf{iter})} - \log \underline{p}\,(\mathbf{y}; q)^{(\mathsf{iter}-1)}$;
  Compute $\widehat{\Delta}_\xi = q^*(\boldsymbol{\xi})^{(\mathsf{iter})} - q^*(\boldsymbol{\xi})^{(\mathsf{iter}-1)}$;
end

C Variational predictive posterior

Recall that the variational predictive posterior can be computed as:

$$q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = \int p(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\xi}) q^*(\boldsymbol{\xi}) d\boldsymbol{\xi} = \int \int p(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta}, \boldsymbol{\Omega}) q(\boldsymbol{\vartheta}) q(\boldsymbol{\Omega}) d\boldsymbol{\vartheta}\, d\boldsymbol{\Omega}, \tag{C.1}$$

which requires only a simulation step according to the first methodology presented in the main paper. If we wish to make the estimation simpler, we can integrate out the precision parameter $\boldsymbol{\Omega}$ in the following way:

$$q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = \int q(\boldsymbol{\vartheta}) \underbrace{\left[\int \mathsf{N}_d(\mathbf{y}_{t+1}; \boldsymbol{\Theta}\mathbf{z}_t, \boldsymbol{\Omega}^{-1}) \mathsf{Wishart}_d(\boldsymbol{\Omega}; \delta, \mathbf{H}) d\boldsymbol{\Omega}\right]}_{A} d\boldsymbol{\vartheta}, \tag{C.2}$$

where

$$A = \frac{2^{-d(\delta+1)/2}|\mathbf{H}|^{\delta/2}}{\pi^{d/2}\Gamma_d(\delta/2)} \int \underbrace{|\boldsymbol{\Omega}|^{(\delta-d)/2} \exp\left\{-\frac{1}{2}\mathrm{tr}\left\{\boldsymbol{\Omega}\left(\mathbf{H}^{-1} + (\mathbf{y}_{t+1} - \boldsymbol{\Theta}\mathbf{z}_t)(\mathbf{y}_{t+1} - \boldsymbol{\Theta}\mathbf{z}_t)^\intercal\right)\right\}\right\}}_{\text{Kernel of a } \mathsf{Wishart}_d(\delta+1,(\mathbf{H}^{-1}+(\mathbf{y}_{t+1}-\boldsymbol{\Theta}\mathbf{z}_t)(\mathbf{y}_{t+1}-\boldsymbol{\Theta}\mathbf{z}_t)^\intercal)^{-1})} d\boldsymbol{\Omega}$$

$$= \frac{|1 + \frac{1}{v}(\mathbf{y}_{t+1} - \boldsymbol{\Theta}\mathbf{z}_t)^\intercal v\mathbf{H}(\mathbf{y}_{t+1} - \boldsymbol{\Theta}\mathbf{z}_t)|^{-\frac{v+d}{2}}\Gamma(\frac{v+d}{2})}{\pi^{d/2}v^{d/2}|\mathbf{H}^{-1}|^{1/2}\Gamma(v/2)} = h(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta}), \tag{C.3}$$

is the density function of a multivariate Student-t distribution with dimension d, $v = \delta - d + 1$ degrees of freedom, mean vector $\boldsymbol{\Theta}\mathbf{z}_t$ and scaling matrix $\mathbf{S} = (v\mathbf{H})^{-1}$, i.e. $\mathsf{t}_v(\boldsymbol{\Theta}\mathbf{z}_t, \mathbf{S})$. Then, the integral in (C.1) becomes

$$q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = \int h(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta})q(\boldsymbol{\vartheta})d\boldsymbol{\vartheta}, \tag{C.4}$$

which requires to simulate from the optimal multivariate gaussian distribution of $\boldsymbol{\vartheta}$ according to the second methodology presented in the main paper.

The last step requires a further approximation. We consider a multivariate gaussian approximation to the multivariate Student-t which minimizes the Kullback-Leibler divergence:

$$\begin{aligned} \mathcal{D}_{KL}(h\|\phi) &\propto -\int \log\phi(\mathbf{y}_{t+1}|\mathbf{m}, \mathbf{R}^{-1})h(\mathbf{y}_{t+1}|\mathbf{z}_t, \boldsymbol{\vartheta})\, d\mathbf{y}_{t+1} \\ &= -\mathbb{E}_h(\log\phi(\mathbf{y}_{t+1}|\mathbf{m}, \mathbf{R}^{-1})) = \psi(\mathbf{m}, \mathbf{R}), \end{aligned} \tag{C.5}$$

where, in particular,

$$\begin{aligned} \psi(\mathbf{m}, \mathbf{R}) &\propto \mathbb{E}_h\left(-\frac{1}{2}\log\mathbf{R} + \frac{1}{2}(\mathbf{y}_{t+1} - \mathbf{m})^\intercal\mathbf{R}(\mathbf{y}_{t+1} - \mathbf{m})\right) \\ &= -\frac{1}{2}\log\mathbf{R} + \frac{1}{2}(\boldsymbol{\Theta}\mathbf{z}_t - \mathbf{m})^\intercal\mathbf{R}(\boldsymbol{\Theta}\mathbf{z}_t - \mathbf{m}) + \frac{v}{2(v-2)}\mathrm{tr}\left\{\mathbf{RS}\right\}, \end{aligned} \tag{C.6}$$

which turns out to be minimized when $\mathbf{m} = \boldsymbol{\Theta}\mathbf{z}_t$ and $\mathbf{R} = \frac{v-2}{v}\mathbf{S}^{-1}$. If we substitute the function $h(\cdot)$ with its gaussian approximation we get

$$q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) = \int \phi(\mathbf{y}_{t+1}|\mathbf{m}, \mathbf{R}^{-1})q(\boldsymbol{\vartheta})d\boldsymbol{\vartheta}. \tag{C.7}$$

Define $\mathbf{Z}_t = (\mathbf{I}_d \otimes \mathbf{z}_t^\intercal)$ and compute the integral above:

$$
\begin{aligned}
q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) &\propto \int \exp\left\{-\frac{1}{2}\left[(\mathbf{y}_{t+1} - \mathbf{Z}_t\boldsymbol{\vartheta})^\intercal \mathbf{R}(\mathbf{y}_{t+1} - \mathbf{Z}_t\boldsymbol{\vartheta}) + (\boldsymbol{\vartheta} - \boldsymbol{\mu}_{q(\vartheta)})^\intercal \boldsymbol{\Sigma}_{q(\vartheta)}^{-1}(\boldsymbol{\vartheta} - \boldsymbol{\mu}_{q(\vartheta)})\right]\right\} d\boldsymbol{\vartheta} \\
&\propto \exp\left\{-\frac{1}{2}\mathbf{y}_{t+1}^\intercal \mathbf{R}\mathbf{y}_{t+1}\right\} \\
&\qquad \times \int \exp\left\{-\frac{1}{2}\left[\boldsymbol{\vartheta}^\intercal(\boldsymbol{\Sigma}_{q(\vartheta)}^{-1} + \mathbf{Z}_t^\intercal \mathbf{R}\mathbf{Z}_t)\boldsymbol{\vartheta} - 2\boldsymbol{\vartheta}^\intercal(\boldsymbol{\Sigma}_{q(\vartheta)}^{-1}\boldsymbol{\mu}_{q(\vartheta)} + \mathbf{Z}_t\mathbf{R}\mathbf{y}_{t+1})\right]\right\} d\boldsymbol{\vartheta},
\end{aligned}
\tag{C.8}
$$

where the term in the integral is the kernel of a multivariate gaussian random variable with variance matrix $\tilde{\boldsymbol{\Sigma}} = (\boldsymbol{\Sigma}_{q(\vartheta)}^{-1} + \mathbf{Z}_t^\intercal \mathbf{R}\mathbf{Z}_t)^{-1}$ and mean $\tilde{\boldsymbol{\mu}} = \tilde{\boldsymbol{\Sigma}}(\boldsymbol{\Sigma}_{q(\vartheta)}^{-1}\boldsymbol{\mu}_{q(\vartheta)} + \mathbf{Z}_t\mathbf{R}\mathbf{y}_{t+1})$. Solve the integral and get:

$$
\begin{aligned}
q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) &\propto \exp\left\{-\frac{1}{2}(\mathbf{y}_{t+1}^\intercal \mathbf{R}\mathbf{y}_{t+1} - \tilde{\boldsymbol{\mu}}^\intercal \tilde{\boldsymbol{\Sigma}}\tilde{\boldsymbol{\mu}})\right\} \\
&\propto \exp\left\{-\frac{1}{2}(\mathbf{y}_{t+1}^\intercal \mathbf{R}\mathbf{y}_{t+1} - \mathbf{y}_{t+1}^\intercal \mathbf{R}\mathbf{Z}_t\tilde{\boldsymbol{\Sigma}}\mathbf{Z}_t^\intercal \mathbf{R}\mathbf{y}_{t+1} - 2\mathbf{y}_{t+1}\mathbf{R}\mathbf{Z}_t\tilde{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}_{q(\vartheta)}^{-1}\boldsymbol{\mu}_{q(\vartheta)})\right\} \\
&= \exp\left\{-\frac{1}{2}(\mathbf{y}_{t+1}^\intercal(\mathbf{R} - \mathbf{R}\mathbf{Z}_t\tilde{\boldsymbol{\Sigma}}\mathbf{Z}_t^\intercal \mathbf{R})\mathbf{y}_{t+1} - 2\mathbf{y}_{t+1}\mathbf{R}\mathbf{Z}_t\tilde{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}_{q(\vartheta)}^{-1}\boldsymbol{\mu}_{q(\vartheta)})\right\},
\end{aligned}
\tag{C.9}
$$

which is the kernel of a multivariate gaussian with variance matrix $\boldsymbol{\Sigma}_{pred} = (\mathbf{R} - \mathbf{R}\mathbf{Z}_t\tilde{\boldsymbol{\Sigma}}\mathbf{Z}_t^\intercal \mathbf{R})^{-1}$ and mean $\boldsymbol{\mu}_{pred} = \boldsymbol{\Sigma}_{pred}\mathbf{R}\mathbf{Z}_t\tilde{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}_{q(\vartheta)}^{-1}\boldsymbol{\mu}_{q(\vartheta)}$. To conclude, the gaussian approximation to the variational predictive posterior is such that $q(\mathbf{y}_{t+1}|\mathbf{z}_{1:t}) \equiv \mathsf{N}_d(\boldsymbol{\mu}_{pred}, \boldsymbol{\Sigma}_{pred})$.

D Additional simulation results

In this section we report additional details and results on the simulation study we highlighted in Section 4. We set the length of the time series equal to $T = 360$, corresponding to 30 years of monthly data, the dimension of the multivariate regression model equal to $d = 15, 30, 49$ and we further assume both moderate level of sparsity (50% of true zeros) and high level of sparsity (90% of true zeros).

The true matrix $\boldsymbol{\Theta}$ is generated as follows: we fix to zero sd^2 entries at random, where $s = 0.5, 0.9$, while the remaining non zero coefficients are sampled from a mixutre of two gaussian with means -0.08 and 0.08, and standard deviation 0.1. The normals are truncated at -0.05 and 0.05 respectively, to avoid very small values for the non zero parameters. Figure D.1 reports the distribution of the non-zero parameters in the true regression matrix used to simulate the data are shown in D.2.

In the main paper, we report the results of the simulation study for $N = 100$ replications for dimensions $d = 30, 49$. We now show some of the additional results on smaller dimensional

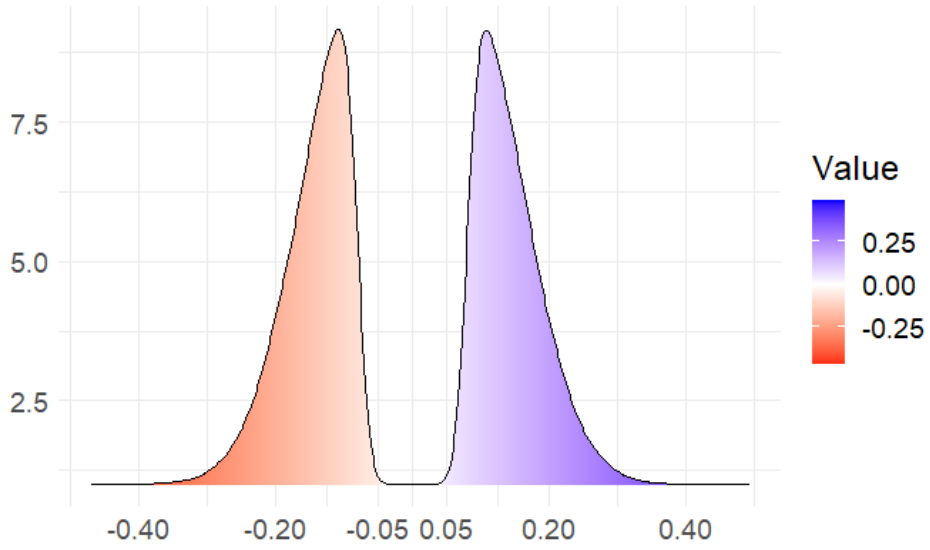
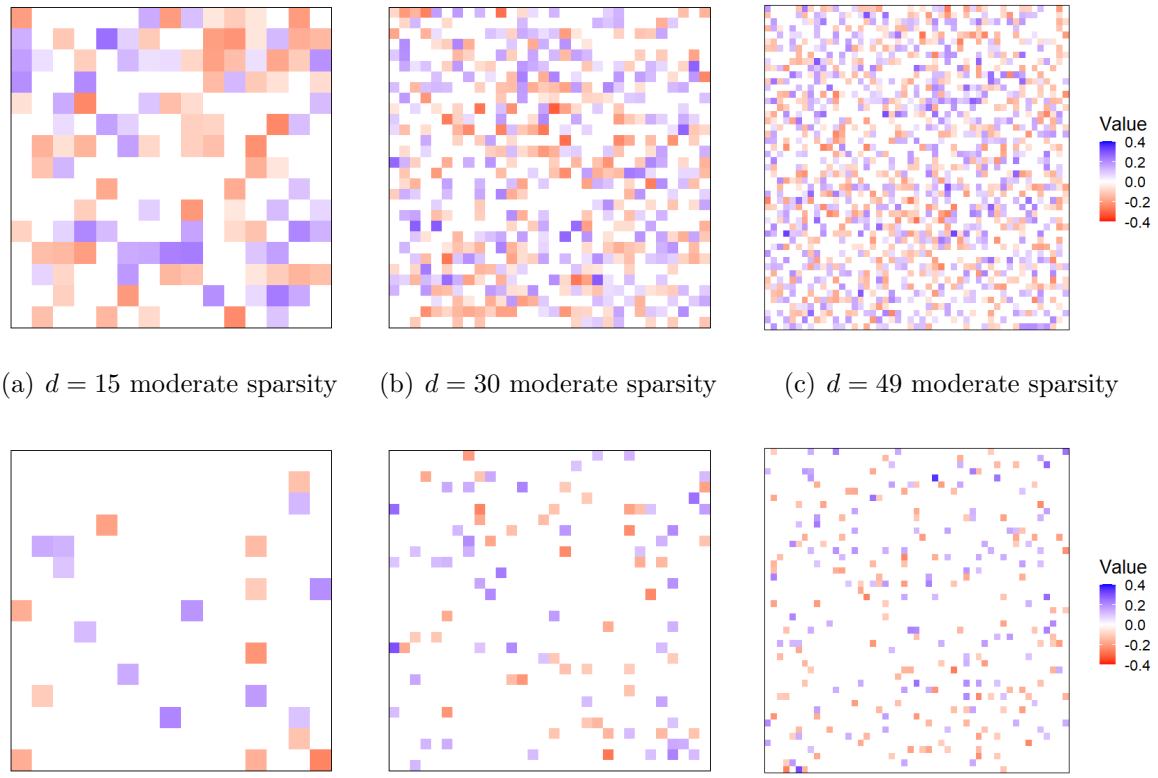


Figure D.1: Distribution of non-zero parameters in the true regression matrix. This figure plots the distribution from which we sample the non-zero entries of the regression matrices used to generate the data for the simulation study.

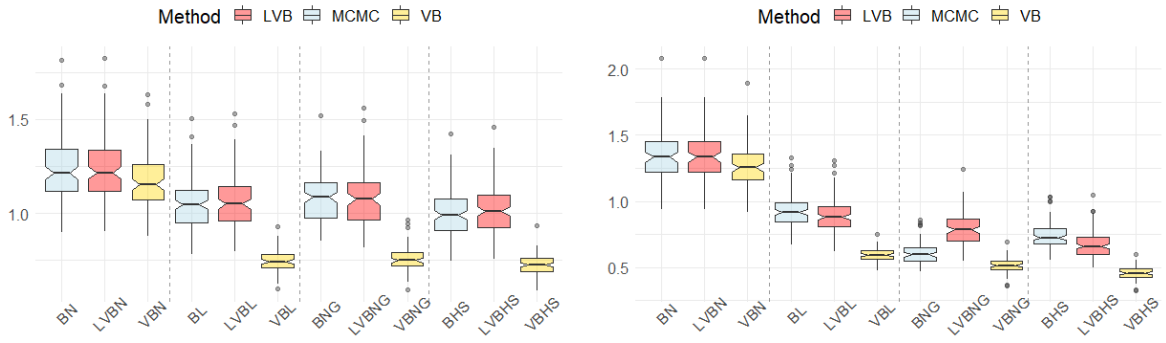
simulation cases. Figure D.2 shows the true regression matrix for a smaller dimensional $d = 15$ case. Similar to the main simulation reported in the paper, we look at the Frobenius norm, denoted by $\|\cdot\|_F$, of the difference between the true matrix $\boldsymbol{\Theta}$ and the estimated one $\widehat{\boldsymbol{\Theta}}$ as a measure of point estimation accuracy. Moreover, we implement the Signal Adaptive Variable Selector (SAVS) of Ray and Bhattacharya (2018) to induce sparse estimates in $\widehat{\boldsymbol{\Theta}}$ and then we compute the F1-score. Finally, we compare also the proposed methods in terms of computational speed. Our `VB` estimation procedure that makes inference directly on the regression matrix $\boldsymbol{\Theta}$ outperform those who consider $\mathbf{A} = \mathbf{L}\boldsymbol{\Theta}$ looking both at the point estimates accuracy (see frobenius norm results in Figure D.3) and at the identification of the sparse structure of the matrix (see F1-score results in Figure D.4). Focusing on the moderate sparsity case (i.e. 50% of zeros in the true matrix), the different priors performs equally given the estimation method, while when the sparsity is high horseshoe tends to perform better that lasso and normal-gamma. Another interesting result is that when the sparsity level is fixed at 50%, methods that work with the reparametrization of the regression matrix provide results similar to the non-informative priors and the difference with our approach is evident. Infact, since we have a lot of non-zero elements, the linear combination $\boldsymbol{\Theta} = \mathbf{L}^{-1}\mathbf{A}$ has more terms to take into account and leads to a dense estimate of $\boldsymbol{\Theta}$ and larger variability in the estimates.



(a) $d = 15$ moderate sparsity (b) $d = 30$ moderate sparsity (c) $d = 49$ moderate sparsity

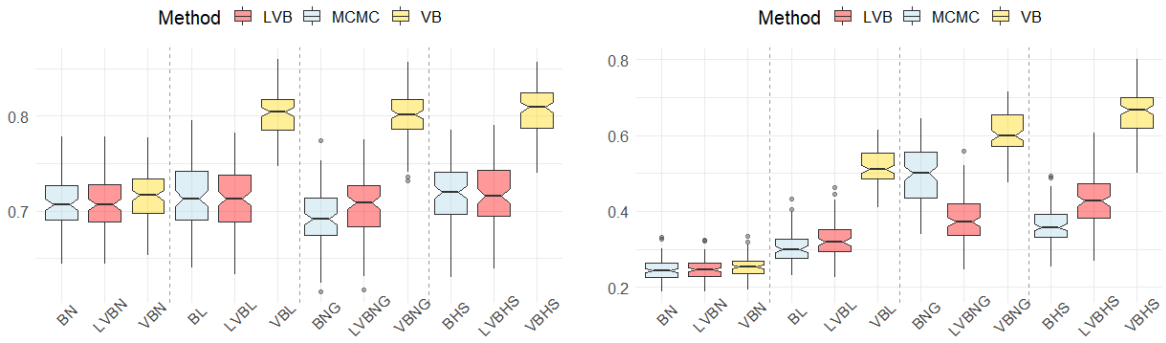
(d) $d = 15$ high sparsity (e) $d = 30$ high sparsity (f) $d = 49$ high sparsity

Figure D.2: True regression matrices for the simulation study. This figure plots the regression matrices used in the simulation study. We assume both moderate level of sparsity (top panels, 50% of true zeros) and high level of sparsity (bottom panels, 90% of true zeros).



(a) Moderate sparsity (b) High sparsity

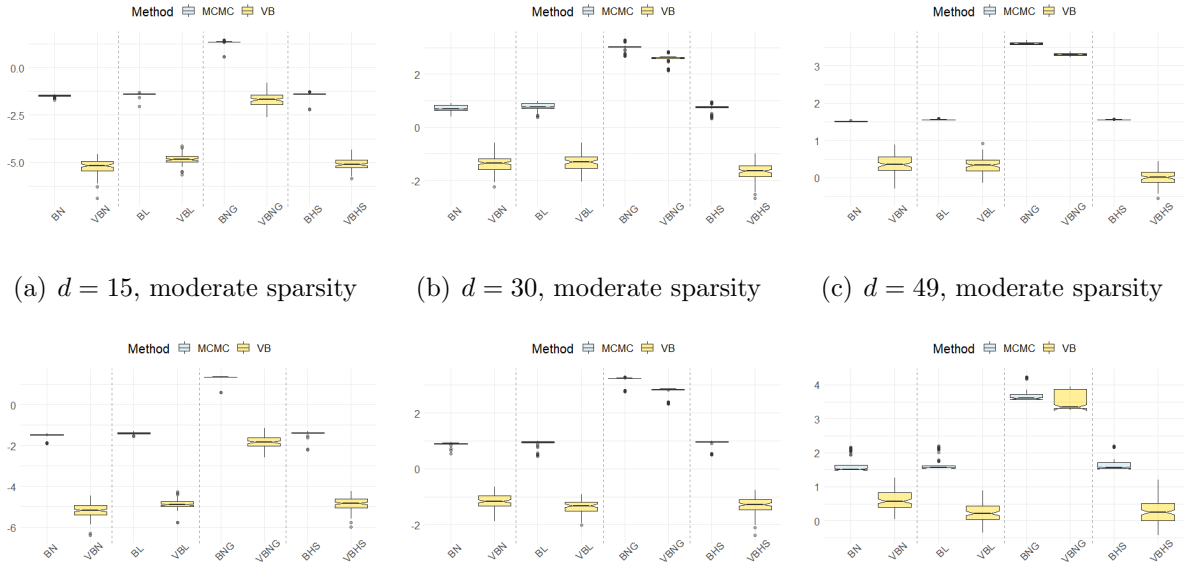
Figure D.3: Frobenius norm on the regression matrix. Boxplots of the frobenius norm of the difference among the true $\boldsymbol{\Theta}$ and the estimated one $\widehat{\boldsymbol{\Theta}}$ for different hierarchical shrinkage priors and different inference paradigms when the dimension is $d = 15$.



(a) Moderate sparsity

(b) High sparsity

Figure D.4: F1 score computed on the regression matrix. Boxplots of the F1 score computed looking at the true non-null parameters in $\boldsymbol{\Theta}$ and the non-null parameters in the estimated matrix $\widehat{\boldsymbol{\Theta}}$, for different hierarchical shrinkage priors and different inference paradigms when the dimension is $d = 15$.



(a) $d = 15$, moderate sparsity

(b) $d = 30$, moderate sparsity

(c) $d = 49$, moderate sparsity

(d) $d = 15$, high sparsity

(e) $d = 30$, high sparsity

(f) $d = 49$, high sparsity

Figure D.5: Boxplots of the computational time required by the algorithms `MCMC` against the variational methods (`VB`) for different hierarchical shrinkage priors. The time is expressed on the logarithmic scale.

E Additional empirical results

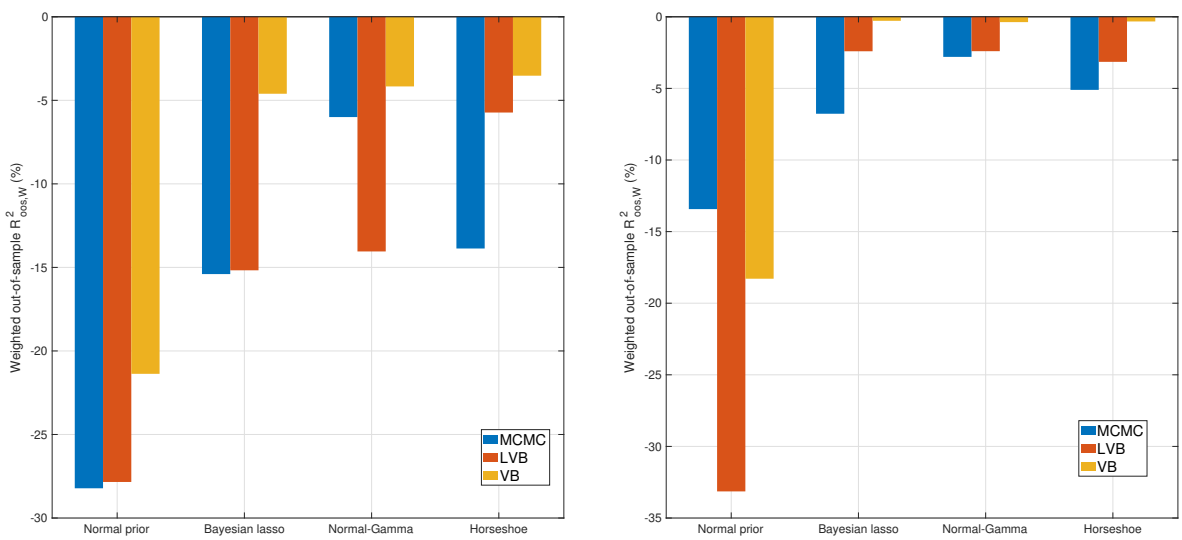
In this section we report some additional result concerning an alternative aggregation scheme for the individual forecasting errors along the lines of Christoffersen and Diebold (1998) and Fisher et al. (2020). In addition, we report the forecasting performance across different estimation strategies and prior specifications focusing on recession periods only.

E.1 Aggregate out-of-sample $R^2_{W,oos}$

We aggregate the individual forecasting errors for each industry portfolios as follows,

$$R^2_{W,oos} = 1 - \frac{\sum_{t_0=2}^{T} \sum_{i=1}^{n} \mathrm{w}_{it} \widehat{e}^2_{it}}{\sum_{t_0=2}^{T} \sum_{i=1}^{n} \mathrm{w}_{it} \overline{e}^2_{it}},$$

where $\widehat{e}_{it} = y_{it} - \widehat{y}_{it}(\mathcal{M}_s)$ the forecasting error from the $\mathcal{M}_s$ model and $\overline{e}_{it} = y_{it} - \overline{y}_{it}$ the forecasting error associated to the naive forecast $\overline{y}_{it}$. Figure E.6 represents the results for an aggregation where the weights w_{it} represent the inverse of the average market capitalization (size) of a firm within a given industry. All things equal, there will be less (more) penalty for the forecast errors of a small (large) asset, which carries less (more) relevance in the dynamics of the market portfolio, but also less reward when accurate.



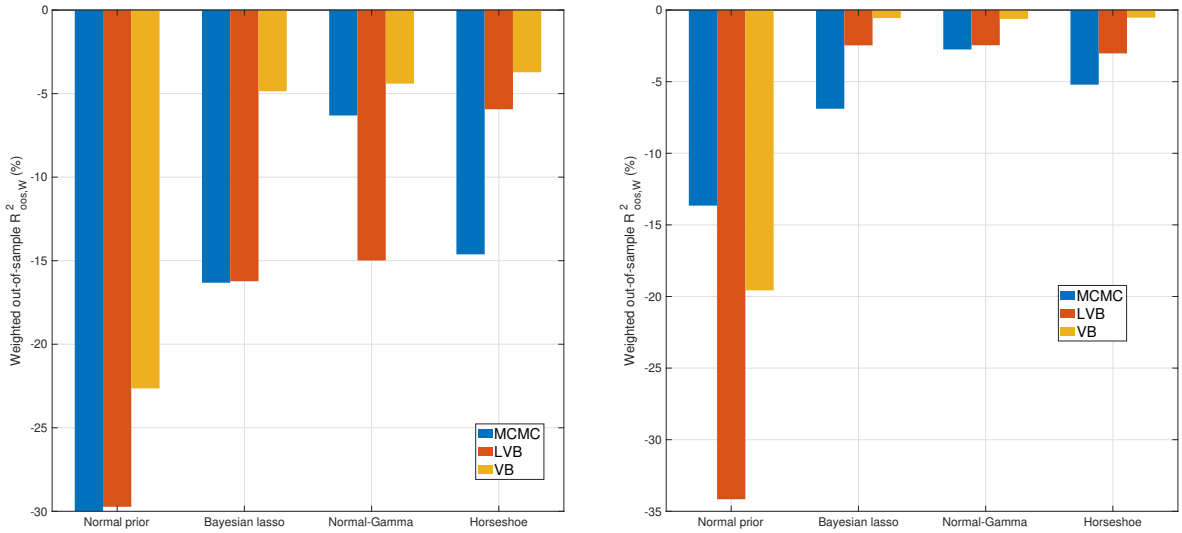
(a) $R^2_{W,oos}$ across 49 industries

(b) $R^2_{W,oos}$ across 30 industries

Figure E.6: Weighted out-of-sample R^2_{oos}. This figure plots the out-of-sample $R^2_{W,oos}$ where the aggregation is based on the average market capitalization of firms within a given industry. The left (right) panel shows the results for the cross section of 49 (30) industry portfolios.

As for the individual industry forecasts, the naive rolling mean forecast represents a highly challenging benchmark to beat, once individual industry forecasts are aggregated. Yet, our VB approach outperforms both the MCMC and LVB irrespective of the shrinkage prior specification. This is more evident for the 30 industry classification, where the $R^2_{W,oos}$ for the VB approach is essentially zero, whereas the LVB, which ranks second, is as low as -3%, for monthly observations. At a more general level, a sparse multivariate regression model outperforms a model with a normal prior, as indicated by a large and negative weighted $R^2_{W,oos}$ obtained by the latter.

Figure E.7 reports the results for a different aggregation scheme where the wights w_{it} in $R^2_{W,oos}$ represents the inverse of the sample volatility of the asset. That is, the weight of each asset's forecast errors depends on its volatility. All things equal, there will be less (more) penalty for the forecast errors of an asset with large (small) returns volatility, but also more reward when the forecast is accurate. Similar to the size-based aggregation, the naive rolling mean forecast turns out to be a highly challenging benchmark to beat. Yet, our VB approach outperforms both the MCMC and the LVB approach irrespective of the shrinkage prior specification. This is more evident for the 30 industry classification, where the $R^2_{W,oos}$ for the VB approach is essentially zero, whereas the LVB, which ranks second, has $R^2_{W,oos}$ as low as -3% for monthly returns.



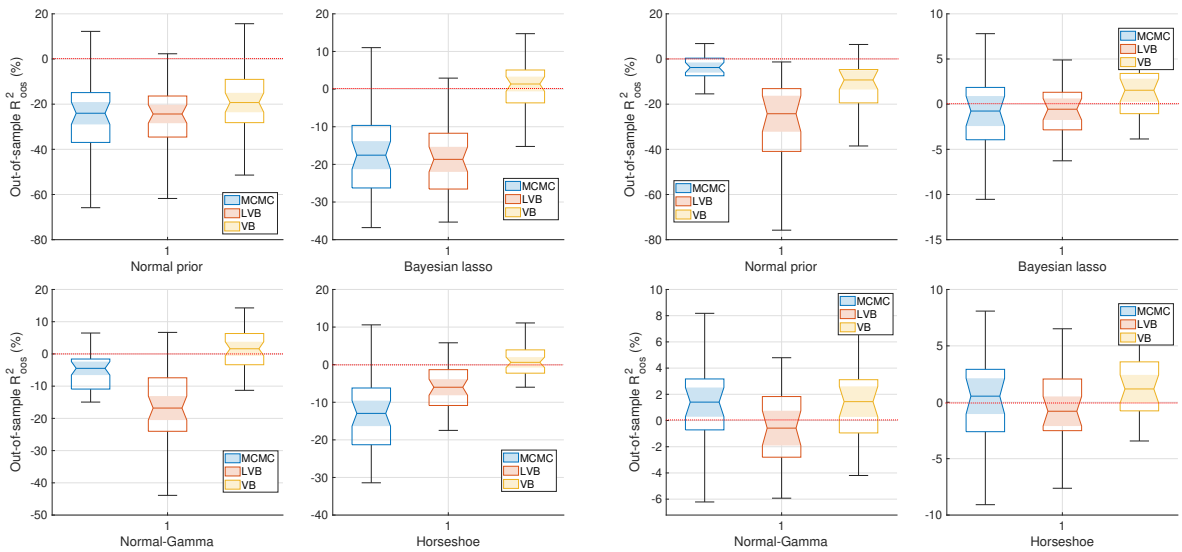
(a) $R^2_{W,oos}$ across 49 industries

(b) $R^2_{W,oos}$ across 30 industries

Figure E.7: Weighted out-of-sample R^2_{oos}. This figure plots the out-of-sample $R^2_{W,oos}$ where the aggregation is based on the inverse of the sample volatility of the industry portfolio returns.

E.2 Predictability over the business cycle

The spread in the cumulative square prediction errors shown in Figure 6 suggests that the forecasting performance across models can be correlated with the aggregate business cycle. We now delve further into the analysis of the forecasting performance in recession periods. More precisely, we split the data into recession and expansionary periods using the NBER dates of peaks and troughs. This information is used *ex-post* and is not used at any time in the estimation of the predictive models. Then, we compute the corresponding R^2_{oos} for the recession periods only. Figure E.8 reports the results.



(a) $R^2_{i,oos}$ across 49 industries

(b) $R^2_{i,oos}$ across 30 industries

Figure E.8: Cross-industry returns predictability during recession periods. The left (right) four panels show the cross-sectional distribution of the R^2_{oos} for the 49 (30) industry portfolios. The red horizontal line separates the $R^2_{oos} > 0$ from $R^2_{oos} < 0$.

The predictive ability of all prior specifications, including the non-informative normal prior, substantially increases for both the cross section of 49 and 30 industry portfolios. Nevertheless, with the only exception of the normal-gamma for the 30 industry portfolios, our `VB` estimation approach generate the highest *median* R^2_{oos} across industries. The outperformance of the `VB` approach is more evident for the cross section of 49 industry portfolios. The `VB` approach is the only that delivers a positive median R^2_{oos} irrespective of the shrinkage prior specification. The performance gap across methods during recessions is more tight for the 30 industry portfolio classification. Nevertheless, the median out-of-sample R^2_{oos} is higher than both the `MCMC` and `LVB` for both the Bayesian adaptive lasso and horseshoe priors.